

2014/15 Knowledge Sharing Program with Belize:

Formulation of a National STI Strategy and Action Plan and Establishment of a Belize-Korea Science, Technology and Innovation (STI) Institute



MINISTRY OF STRATEGY
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KDI Korea Development Institute

2014/15 Knowledge Sharing Program with Belize

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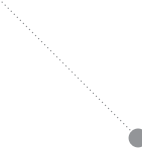
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Institute





Preface

In the 21st century, knowledge is one of the key determinants of a country's level of socio-economic development. Based on this recognition, Korea's Knowledge Sharing Program (KSP) was launched in 2004 by the Ministry of Strategy and Finance (MOSF) and the Korea Development Institute (KDI).

KSP aims to share Korea's experience and knowledge with the partner countries to achieve mutual prosperity and cooperative partnership. Former high-ranking government officials are directly involved in the policy consultation to share their intimate knowledge of development challenges, and to complement the analytical work of policy experts and specialists who have extensive experience in their fields. The government officials and practitioners effectively pair up with their counterparts in the development partner countries to work jointly on pressing policy challenges and share development knowledge in the process. The program includes policy research, consultation and capacity-building activities, all in all to provide comprehensive and tailor-made assistance to the development partner countries in building a stable foundation and fostering capabilities to pursue self-sustainable growth.

In 2014, policy consultation and capacity building workshop were carried out with 29 partner countries covering over 100 research agendas. As a new partner country, Kyrgyz Republic, El Salvador, Guatemala, Cuba were selected in consideration of the country's policy demand, growth potential, and strategic economic partnership.

The 2014/15 Knowledge Sharing Program with Belize was carried out with the aim of exchanging socio-economic development experience of two countries for improving Belize's policy making capacity and achieving her socio-economic development. Under the MOU signed between the Ministry of Strategy and Finance of Korea and the Ministry of Energy, Science & Technology and Public Utilities of Belize, the joint research and seminars were conducted in order to support the "Formulation of a National STI Strategy and Action Plan and Establishment of a Belize-Korea Science, Technology and Innovation (STI) Institute."

I would like to take this opportunity to express my sincere gratitude to Senior Advisor Dr. Myung Ja Kim, Research Program Manager Prof. Doogab Yi, as well as the project consultants including Prof. Hyungsub Choi, Mr. Gertrudes Velasquez, and Mr. Ward McGregor for their immense efforts in successfully completing the 2014/15 KSP with Belize. I am also grateful to Executive Director Dr. Si Wook

Lee, Former Executive Director Dr. Hong Tack Chun, Program Director Dr. Byung Koo Cho, and Program Officer Ms. Jin Ha Yoo, and all members of the Center for International Development, KDI for their hard work and dedication to this program. Lastly, I extend my warmest thanks to the Belizean counterparts, the Ministry of Energy, Science & Technology and Public Utilities and other related agencies, program coordinators, and participants for showing active cooperation and great support.

In your hands is the publication of the results of the 2014/15 KSP with Belize. I believe that KSP will serve as a valuable opportunity to further elevate mutual economic cooperation of Belize and Korea to a new level. I sincerely hope the final research results on the selected areas could be fully utilized to support Belize in achieving economic development goal in the near future.

Joon-Kyung Kim
President
Korea Development Institute

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2014/15 KSP with Belize

Jin Ha Yoo (Program Officer, Korea Development Institute)

Korea Development Institute (KDI), in collaboration with the Ministry of Strategy and Finance (MOSF) of the Republic of Korea, has been implementing the Knowledge Sharing Program (KSP) with numerous partner countries since 2004 with the overarching goal to assist the enhancement of the national competitiveness of its partner countries by sharing Korea's development experience.

Located adjacent to Mexico and Guatemala, Belize is a Central American country with a population around 340,000 persons and land area around 22,966. (CIA Word Factbook 2015) Considering the small size, it is evident that the Belizean government's main concern for economic growth are acquiring adequate workforce with proper set of skills and identifying sizable market beyond her national boundaries. With such concern, Belizean government is eager to enhance its STI (Science, Technology and Innovation) capability to utilize it as a driver of her economic success.

KSP with Belize was initiated in 2013 based on the strong will of Belizean government. The second year of KSP was conducted under the theme of "Formulation of a National STI Strategy and Action Plan and Establishment of a Belize-Korea Science, Technology and Innovation (STI) Institute." After careful consideration, two subtopics were selected under the theme; the following table is a brief overview of the 2014/15 KSP with Belize's topics and respected Korean researchers.

〈Table 1〉 2014/15 KSP Topics and Researchers

Formulation of a National STI Strategy and Action Plan and Establishment of a Belize-Korea Science, Technology and Innovation (STI) Institute	
Senior Advisor: Myung Ja Kim (Former Minister of Environment) Program Director: Byung Koo Cho (KDI) Research Program Manager: Doogab Yi (Seoul National University)	
Research Topics	Researchers
Formulation of Belize STI Strategy and Action Plan	Hyungsub Choi (Seoul National University of Science & Technology)
Establishment of a Belize-Korea Science, Technology and Innovation (STI) Institute	Doogab Yi (Seoul National University)

For the first stage of the 2014/15 KSP with Belize, the High Level Demand Survey and Pilot Study visit was conducted in November to identify research themes regarding the proposed topics, and to discuss about the implementation cycle and procedures. At the series of meetings, the Korean delegation collected necessary information and data and took advice from experts and relevant organizations. The Additional Pilot Study and Local Reporting Workshop visit was conducted with a purpose to review current research findings and direction. During the visit, the Korean delegation gathered more in-depth information and received feedback from the Belizean experts.

For the next stage, Belizean delegation, headed by Minister Grant of Energy and Minister Hulse of Labour, visited Korea at the end of April to participate in Interim Reporting and Policy Practitioners' Workshop. At the Interim Reporting session, both Korean researchers and Belizean local consultants shared their interim research outcomes and the participants had rigorous discussion to enhance the quality of policy recommendations for the given topics. Furthermore, the Belizean delegation visited various institutions including KIST, Ministry of Science, ICT, and Future Planning, Seoul National University and others where they first-handedly experienced Korea's economic development.

As the final stage of 2014/15 KSP with Belize, Korean delegation headed by Former Minister of Environment, Dr. Myung Ja Kim visited Belize City from July 18th to 24th to conduct the Senior Policy Dialogue and Final Reporting Workshop to share final research outcome and policy recommendations. During the mission, the delegation presented their findings to approximately 100 stakeholders including high-level officials, professors, researchers, and other interested parties.

Executive Summary

Doogab Yi (Seoul National University)

The Belizean economy is small, open and based on commodities, as well as dominated by non-technology-intensive activities. The service industry is the biggest contributor to GDP and employment, but agriculture accounts for most exports. Output of agricultural commodities and tourism accounts for almost 60 percent of exported goods and services. Realizing its heavy reliance on outdated views of socioeconomic development and production and management systems, the Belizean government devised a strategy to provide a level playing field for all businesses and entrepreneurs, marked by the use of appropriate technology to increase productivity and competitiveness in an environmentally sustainable way.

Development of sci-tech initiatives applied to strategic economic areas can serve as the driver of a resilient Belizean economy. The Ministry of Energy, Science and Technology and Public Utilities' (MESTPU) Strategic Plan 2012-17 describe a "framework that supports the economic growth and competitiveness of Belize in the global market through the promotion of energy and food security, improved access to potable water, and utilization of science and technology to advance the wellness and livelihoods of the population to lead longer, wealthier and more productive lives."

The 2014/2015 Knowledge Sharing Program (KSP) with Belize was carried out under an MOU signed by the Belizean Ministry of Energy, Science and Technology and Public Utilities. The main purpose of the program is to share experiences and ideas in improving the policymaking capacity of Korea and Belize in science,

technology and innovation (STI). The report's primary objective is to promote greater understanding of the Korean experience focusing on best practices in the development of STI to help Belize improve its agencies responsible for strategic planning and implementation of program activities within STI.

Chapter 1. Formulation of Belize STI Strategy and Action Plan

The goal of this report is to propose a national science, technology, and innovation (STI) strategy and action plan for Belize, in order to realize the nation's growth potential. In recent years, Belize is paying increasing attention to the role STI could play in fostering sustainable economic development. For that purpose, the country has developed strategic plans to lay out the basic parameters into the future. What is missing is a more detailed roadmap to realize the broad vision. To accomplish this objective, the report first analyzed STI's status in Belize covering its capacities in research, education and innovation.

The first step in designing the roadmap is to determine the strategic focus of action, which should be determined by the relevant authorities in Belize. As a short-cut, this report will assume that the agricultural sector has the greatest growth potential when integrated with STI. The tourism sector, which is currently the largest contributor to the Belizean economy, is taken as a potential revenue source for kick-starting the STI Institute and overall system. Based on these assumptions, we will propose an STI strategy and deduce concrete action plans adequate for Belize's current conditions. Toward this goal, this report will review the relevant theories of STI policy; analyze the current status of Belizean efforts to establish the national innovation system (NIS); provide an overview of the South Korean STI experiences, focusing on the relevant elements.

South Korea since the 1960s provides a rare example of a late-developing country succeeding in overcoming the development barrier, the case of South Korea has gathered significant attention in recent years. From the rough beginnings in the early 1960s, South Korea established numerous institutions for STI, strengthened its educational system, and forged links among different sectors within the emerging NIS. Extracting lessons from this experience will require making interpretations about the South Korean past, which will be provided in a systematic fashion. Apart from institution-building, we will emphasize the importance of scientific culture throughout the broader society. Taking the Saemaul Movement (New Community Movement) as an example, we will describe how South Korea sought to disseminate the scientific way of life throughout society, especially in the rural communities. Scientists and engineers were mobilized to provide technology guidance. The slogan of "scientization of agriculture" was made visible in the field as the large group of professional scientists and engineers roamed the countryside.

Based on the discussions above, this report will make recommendations for STI strategy and roadmap for Belize. The ultimate vision of the STI strategic plan is to achieve “sustainable economic growth based on science and technology innovation.” In order to foster decision-making on STI issues, we suggest that the National STI Committee be established at the earliest convenience, comprised of relevant stakeholders, answerable to the Office of the Prime Minister. This committee will serve as the highest authority in matters related to STI. The STI Committee will seek funding for the STI Institute, both through overseas ODA funding and domestic sources; it will also take the lead in enacting the “Basic Law for the Promotion of STI” and the STI Comprehensive Long-Term Plan. Most important, the committee will be responsible for prioritizing the strategic industries, on which Belize will focus within various timeframes.

Chapter 2. Establishment of a Belize-Korea STI Institute

The second part of the report aims to develop a feasible framework for the Korean-Belizean STI institute. In order to accomplish this, greater understanding of the Korean experience focused on best practices in development of STI would help Belize improve its agencies responsible for strategic planning and implementation of program activities within STI areas. This report examines the success of KIST, Korea’s first government-funded research institute (GRI) that led the country’s STI-based economic development, and explores several factors crucial to KIST’s development as a prominent STI think tank in Korea. The conclusion will be used as a model for the proposed Belizean STI institute based on an analysis of Belize’s STI capability and related needs.

The Belizean STI institute will play a key role in the implementation of The STI Strategy and Action Plan for Belize. The plan identifies the institute as the principal actor in the implementation of actions in several strategic objectives. The institute aims to contribute to the country’s national and provincial development by serving as a strategic center of research knowledge, a hub for research activities, a center for STI talent and a site for building infrastructure for research and education.

The development of a national STI institute has positive implications for the development of a knowledge-based society and innovation-based economy. So a necessity-based approach in planning institute in Belize is crucial. We took three approaches (that are not exclusive to each other). First, we determined if Belize needs a strategic focus in developing an STI institute. Second, we assessed the feasibility of Belize having a training center to help the country get the talent needed. Third, we considered ICT infrastructure as having a transformative impact on various sectors in Belize such as education, tourism and other private industries.

This report proposes a feasible model for the institute. For this, we conducted a thorough assessment of the country's STI capability and related needs; evaluated the feasibility of establishing an STI institute in Belize; and examined the legal and institutional mechanisms for the institute. Next we proposed three approaches to provide a general framework for the institute. Our three models, which are tailored for our assessment of Belize's STI capability and needs, are based on a greater understanding of the Korean experience focused on best practices in the development of STI, especially those from KIST. Through these three models, we propose key policy recommendations for the proper establishment and sustainability of the Belizean STI institute.

2014/15 Knowledge Sharing Program with Belize:
Formulation of a National STI Strategy and Action
Plan and Establishment of a Belize-Korea Science,
Technology and Innovation (STI) Institute

Chapter 1

Formulation of Belize STI Strategy and Action Plan

1. Introduction
2. The Necessity of a National STI Strategy
3. Analysis of Belize's STI Status
4. Analysis of Key Industrial Sector and National Strategic Plans
5. The Korean STI Experience
6. STI Strategy & Roadmap for Belize
7. Conclusion

Formulation of Belize STI Strategy and Action Plan

Hyungsub Choi (Seoul National University of Science and Technology)

Gertrudes Velasquez (University of Belize, Belize)

Summary

The goal of this report is to propose a national science, technology, and innovation (STI) strategy and action plan for Belize, in order to realize the nation's growth potential. In recent years, the Belizean government has made the decision to embark on the endeavor, spearheaded by the initiatives of the Ministry of Energy, Science, Technology, and Public Utilities (MESTPU) aiming to utilize STI for economic development. In order to realize this broad mandate, we will propose an STI strategy and deduce concrete action plans adequate for Belize's current conditions. Toward this goal, this report will review the relevant theories of STI policy; analyze the current status of Belizean efforts to establish the national innovation system (NIS); provide an overview of the South Korean STI experiences, focusing on the relevant elements.

The Necessity of a National STI Strategy

The first step toward designing a national STI strategy would be to identify the basic components of the national innovation system (NIS), and gauge how these may or may not be applied to the context of latecomer countries. The NIS should be seen as an approach, not a theory. The NIS does not tell you how to foster technological innovation; rather it provides a vantage point from which one can observe the various players and institutions that play important roles in fostering technological innovation within a country. Policy analysts who propose the NIS approach largely

agree that the three key components of the modern innovation system are the industry, university, and government sectors, each playing critical roles in fostering technological innovation.

One of the problems in applying the NIS approach to latecomer countries is that some of the institutions assumed in the “triple helix” structure are weak or sometimes virtually non-existent. Therefore, the NIS approach should be seen only as a template for designing the STI strategies. As a template, one can take the three major components of the NIS as an aim to attain in the long-term. But the detailed strategies to arrive at the long-term aim could take diverse pathways depending on the specific conditions and past trajectories of these institutions. This means paying attention to the broader environment surrounding the NIS proper, such as labor markets, the education and training system, and their relation to systems of corporate governance. Being reflective of the pitfalls of adopting the language of NIS would allow us to utilize the approach as a useful template in framing the strategies of STI development.

Analysis of Key Industrial Sector and National Strategic Plans

In recent years, Belize is paying increasing attention to the role STI could play in fostering sustainable economic development. For that purpose, the country has developed strategic plans to lay out the basic parameters into the future. What is missing is a more detailed roadmap to realize the broad vision. The first step in designing the roadmap is to determine the strategic focus of action, which should be determined by the relevant authorities in Belize. As a short-cut, this report will assume that the agricultural sector has the greatest growth potential when integrated with STI. The tourism sector, which is currently the largest contributor to the Belizean economy, is taken as a potential revenue source for kick-starting the STI Institute and overall system.

The South Korean STI Experience

This section describes the development experience of South Korea since the 1960s to extract lessons for latecomer countries. As a rare example of a late-developing country succeeding in overcoming the development barrier, the case of South Korea has gathered significant attention in recent years. From the rough beginnings in the early 1960s, South Korea established numerous institutions for STI, strengthened its educational system, and forged links among different sectors within the emerging NIS. Extracting lessons from this experience will require making interpretations about the South Korean past, which will be provided in a systematic fashion.

Apart from institution-building, we will emphasize the importance of scientific culture throughout the broader society. Taking the Saemaul Movement (New Community Movement) as an example, we will describe how South Korea sought to disseminate the scientific way of life throughout society, especially in the rural communities. Scientists and engineers were mobilized to provide technology guidance. The slogan of “scientization of agriculture” was made visible in the field as the large group of professional scientists and engineers roamed the countryside.

STI Strategy and Roadmap for Belize

Based on the discussions above, this report will make recommendations for STI strategy and roadmap for Belize. The ultimate vision of the STI strategic plan is to achieve “sustainable economic growth based on science and technology innovation.” In order to foster decision-making on STI issues, we suggest that the National STI Committee be established at the earliest convenience, comprised of relevant stakeholders, answerable to the Office of the Prime Minister. This committee will serve as the highest authority in matters related to STI. The STI Committee will seek funding for the STI Institute, both through overseas ODA funding and domestic sources; it will also take the lead in enacting the “Basic Law for the Promotion of STI” and the STI Comprehensive Long-Term Plan. Most important, the committee will be responsible for prioritizing the strategic industries, on which Belize will focus within various timeframes.

1. Introduction

A brief overview outlined above shows that Belize is relatively advanced and possesses considerably more resources than other less developed countries (LDCs), but has yet to acquire the workforce and growth engines required to ensure sustainable growth. The main challenges in boosting economic development in Belize are: 1) to identify export markets to absorb its products and 2) to acquire an adequate workforce with the proper technical skills. In short, size is the major constraint on continued growth in Belize.

As many observers and analysts point out, the key engine of Belizean growth since the 20th century has been technological innovation (Abramovitz and David, 2000). The world’s most successful economies are those that have learned the arts of fostering, utilizing and disseminating scientific and technological knowledge, though full mastery remains largely elusive. Development and utilization of new technologies and the associated institutional framework to support these activities are crucial foundations for the continued well-being of a country. Throughout the past century, however, technological innovation has been concentrated in a select

few nations, mostly in the West, but gradually expanding to a few countries in Asia. It is safe to say Belize is not one of them. The challenge, therefore, is to create the capability and institutions to allow Belize to utilize science and technology as a growth engine. This tall order has been unsuccessfully attempted by many countries despite their diverse and serious efforts.

The goal of this project is to establish a national science, technology, and innovation (STI) strategy and action plan for Belize to realize the nation's full growth potential. In recent years, Belmopan has embarked on this endeavor with initiatives spearheaded by the Ministry of Energy, Science, Technology, and Public Utilities (MESTPU). In 2012, MESTPU announced Strategic Plan 2012-2017, outlining a five-year plan to implement the nation's STI options. The purpose of the plan is to "utilize STI to act as a catalyst to reduce poverty, promote sustainable livelihoods, and improve the quality of life of the population" (p. 1, Part II). In this project, we will propose an STI strategy and deduce concrete action plans tailored to Belize to realize its government's vision of the future.

The remainder of this report will cover a range of topics. Chapter 2 will propose an appropriate macro-level strategy for Belize based on recent discussions of the theoretical foundations in STI policy literature. Chapter 3, based on the information and able assistance of Mr. Ward McGregor, will analyze the status of Belize and outline several implications in the implementation of STI policy, focusing on agriculture and tourism as strategic targets. This latter portion of this chapter will be based on a close reading of MESTPU's Strategic Plan 2012-2017, as well as other planning documents prepared by Belizean ministries and institutions. These will be supplemented by the KSP data gathered in a visit to Belize in February 2015. Chapter 4 introduces the development experience of Korea, focusing on aspects Belize could benchmark. Given the inherent complexities of STI policy and strategy, this section will cover a broad spectrum of issues in the Korean economy, society and industry. In particular, the focus will go on the experiences of technical personnel, education system and industry-academia-government cooperation. Based on these discussions, Chapter 5 will propose a short-, medium- and long-term strategy for Belizean STI, as well as concrete steps to implement these broad goals.

2. The Necessity of a National STI Strategy

How can a nation jump-start its scientific and technological capability? If history serves as a guide, the task is highly difficult, to say the least. A brief historical survey of countries achieving this feat would immediately show that science and technology development in a given society requires not only the long-term nurturing of its education system and research institutions, but also shifts in the broader social and

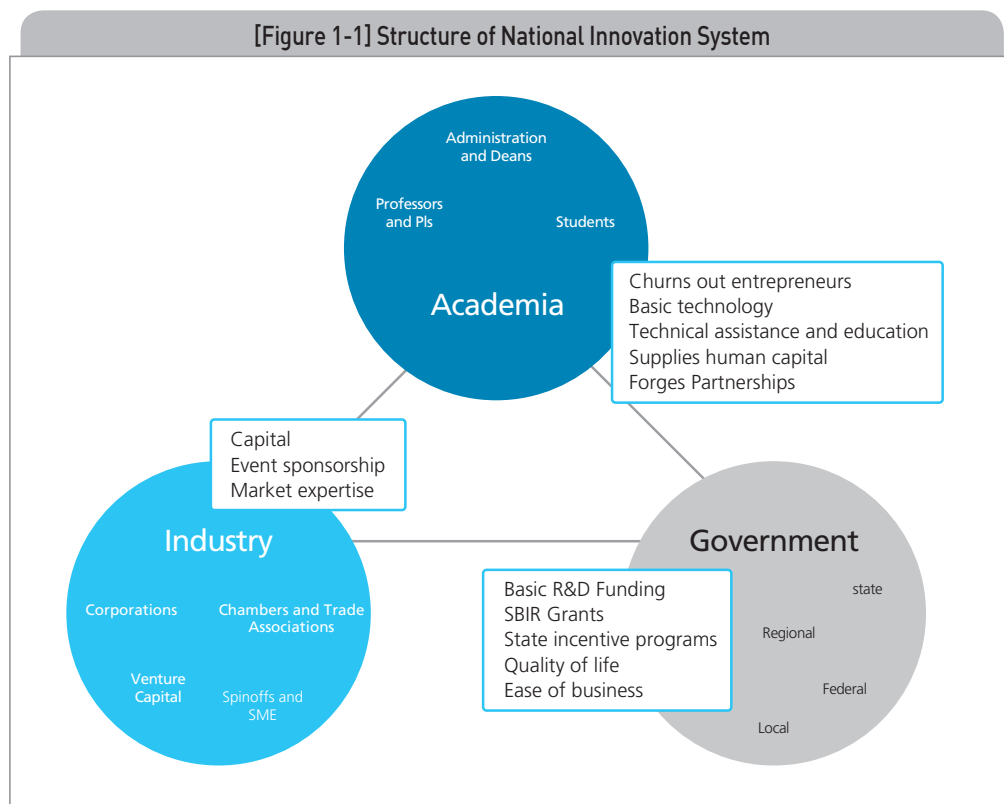
cultural contexts. The process also has much to do with contingent and/or external factors over which a country has little control, such as migration of skilled personnel. For example, the U.S. through the 19th century was a backwater in scientific and technological activities compared to countries in Europe (Mowery and Rosenberg, 1999). Japan, too, was highly underdeveloped until the late 19th century, when it embarked on a campaign to modernize itself after the Meiji Restoration (Morris-Suzuki, 1994). Both the U.S. and Japan spent a long period establishing new schools, nurturing research institutions and fostering a culture of scientific thought. Through these efforts, the two countries rose to prominence on the global science and technology scene in the second half of the 20th century.

This, of course, is not to say both global superpowers should serve as role models for today's developing countries in charting their trajectories with an eye toward the future. Indeed, numerous examples show good-faith efforts to foster a nation's scientific and technological capability that yielded ambiguous results, if not complete failure. The case of Korea, to be examined in closer detail later in this report, is perhaps that rare exception of an LDC transforming itself into a developed country largely through technological innovation. Nevertheless, the first step toward designing a national STI strategy is to identify the theoretical components of the national innovation system, and gauge how these might or might not be applied to countries late to industrialization. Therefore, this chapter will take a sweeping overview of the basic theoretical structure of the national innovation system, accounting for major pitfalls that such countries aim to avoid in implementing institutions for fostering STI.

2.1. National Innovation System

It needs to be emphasized at the outset that the national innovation system (NIS) is an approach, not a theory. It does not tell us how to foster technological innovation; rather it provides a vantage point to observe the various players and institutions that play important roles in fostering technological innovation within a country. The operative term here is system. The standard dictionary definition of system is "a set of interacting or interdependent components forming an integrated whole." Underlying the NIS approach is the assumption that the components within the system are constantly interacting with each other. In other words, change in one component affects every other component due to their interdependence. Adopting the systems approach in STI policy means to take a holistic view toward the complex web of interactions among the components. Though the basic agent for formulating STI strategy is frequently the central government, the public sector does not always have the wherewithal to dictate or direct driver players in a unilateral fashion (Nelson, 1993; Freeman, 1995).

Policy analysts who propose the NIS approach largely agree that the three key components of the modern innovation system are industry, academia and government. The industrial sector is the major player in the national economy and seeks to improve its products and processes, occasionally through science and technology. The role of the university, or academia, is to train skilled personnel (not only scientists and engineers but also entrepreneurs) required for innovative activities, as well as produce scientific and technological knowledge through research and development (R&D). At times, the academic sector gets closely involved with industry through academic-industrial partnerships or industrial extension services. The government is responsible for coordinating the overall system. At times, the government establishes public research institutes (PRIs) that conduct R&D deemed critical for the nation and bridge the diverging concerns of industry and academia. The three sectors form the “triple helix” (Etzkowitz and Leydesdorff, 2000; Leydesdorff and Etzkowitz, 1998; Etzkowitz, 2003). (Refer to Figure 1-1)



As such, leading scholars in STI policy define the NIS as the “network of institutions in the public and private sectors whose activities and interactions initiate, import, modify and diffuse new technologies (Freeman, 1987).” For the NIS

to successfully promote STI-based development, its three components must form organic relationships with one another. Furthermore, the government should play the critical role of creating a favorable policy environment under which the industrial and academic sectors can operate effectively. In other words, the NIS is “the system of interacting private and public firms (either large or small), universities, and government agencies aiming at the production of science and technology within national borders. Interaction among these units may be technical, commercial, legal, social, and financial, in as much as the goal of the interaction is the development, protection, financing or regulation of new science and technology (Niosi *et al*, 1993).”

2.2. NIS for Developing Countries

Can the NIS approach be applied to countries late to industrialization such as Belize? To answer this question, consider the context from which the NIS emerged. One of the earliest proponents of the approach was the American economist Richard R. Nelson (1993), who pioneered evolutionary economics as a critique against neoclassical economics. The basic tenet of evolutionary economics is its emphasis on history and institutions. Rather than focus on supply-demand relationships and the equilibrium point, evolutionary economics focuses on the process and trajectory of economic change (Nelson and Winter, 2009). In short, the NIS approach says “the economic structure and the current institutional setup, both with historical roots, need to be analysed and understood in order to set policy priorities (Lundvall and Lema, 2015).”

One of the problems in applying the NIS approach to countries late to industrialization is that certain institutions assumed to be in the “triple helix” structure are weak or sometimes virtually non-existent. This is due to the development of NIS theories developed in more advanced economies with a long tradition of universities, industrialization and the public sector. This poses a significant challenge for developing countries over where to begin. Should a nation start with strengthening the industrial sector or instead invest more in the academic sector? Or is it best to begin with a state-funded R&D institution to spark interest within the industrial sector in the power of science and technology for driving economic growth? As discussed above, the answer to this question depends on the historical trajectory of the specific locale.

The NIS approach, then, should be seen only as a template for designing STI strategies in developing countries. As a template, the three major components of the NIS can serve as aims to attain over the long term or ideal types. But detailed strategies to arrive at the long-term aims could take diverse paths depending on the specific conditions and past trajectories of these institutions. Moreover, if the aim

is to establish or strengthen the components in the triple helix, closer attention to the broader environment surrounding them is required. This is perhaps the reason Lundvall and Lema (2015) urged a “broadening” of the innovation system concept. In recommending development strategies for African nations, they argue that the NIS approach has so far ignored the “broader set of institutions shaping competence building in the economy, such as labour markets, the education and training system, and their relation to systems of corporate governance.” In addition, they add that “nor did they consider the broader connections between these institutional subsystems and national political cultures and welfare regimes.”

Thus, a certain pitfall exists for countries late to industrialization in adopting the language of STI analysts in more advanced economies. However, if one is reflective of the pitfalls and avoids falling into the “deficit model” of attempting to fill in the boxes within the triple helix structure, the NIS approach can serve as a useful template in framing strategies for STI development strategies.

2.3. The Evolution of NIS

As touched on in the previous discussion on evolutionary economics, the NIS does not remain static but undergoes dynamic evolution over time. In successful developing economies countries, government intervention plays an important role in catalyzing such a transformation. Several policy measures governments can take could spark the evolution of NIS in positive directions. These measures can be categorized as intervening in (1) personnel management, (2) institution building and (3) environment creation. These categories are not mutually exclusive; they overlap with each other.

First, the government’s role in acquiring the proper personnel with the appropriate scientific and technical expertise is critical. The central government can take a number of measures to achieve this goal. One is to retool the national education system to domestically produce a workforce possessing the required knowledge and skills. The other is to create opportunities and design incentive structures for skilled personnel from outside the country, either overseas-based nationals or foreigners, to relocate. Either way, the effective operation of the NIS depends on the availability of people with the necessary skills and experience.

Second, the government can establish new institutions or revamp existing ones to play important roles within the NIS. The central government can play a direct role in establishing PRIs to meet current and projected industrial R&D needs. This can be seen in the experience of Korea in 1966, when Korea Institute for Science and Technology (KIST) was established. The case of KIST shows that the initial demand for R&D might not come from the industrial sector, especially in the early stages of NIS

development. Therefore, the government sometimes needs to educate the industrial sector and induce demand for technological innovation. Also, the government has a particularly large role to play in fostering academia. Universities are key components of an NIS, since they play the dual role of training skilled experts and conducting long-range R&D, both of which are critical to the long-term success of tech-based development.

Third, the government should intervene in the evolutionary process of NIS to create an environment conducive to technological innovation giving rise to a virtuous cycle. This would include measures such as modifying laws and regulations to incentivize technological innovation and creating new opportunities for technological learning through forming links with the overseas scientific and technological community. Moreover, the government must look beyond the narrow concept of NIS and try to nurture a cultural and societal attitude that embraces the power of science and technology in national development. Without a broader social, cultural and social environment that places value in science and technology, all attempts to establish an NIS and strengthen STI are bound to fail. In short, a combination of top-down and bottom-up approaches is necessary for building a strong national system for technological innovation.

The evolution of NIS by no means follows a predetermined trajectory. Yet a set of common characteristics of successful NISs can be derived, such as the existence and effective operation of key institutions, division of labor and close relations among key NIS components, and the interconnectedness of the national system with the outside world. Of particular importance for countries late to industrialization are the social, political and legal environments surrounding major NIS components. The external environment largely determines the input (skilled personnel and effective institutions), which ensures the smooth evolution of an NIS into maturity. This would, in turn, lead to positive output in sustainable economic development.

2.4. Global Circulation of Knowledge and Personnel

Most discussions of the innovation system at the national level have focused on the nation-state. In recent years, however, scholars have said innovation systems are not closed systems but open to the outside world. The immediate context to this perspective is the rapid and pervasive advance of globalization since the 1980s. In other words, maintaining links and connections with the outside world is just as important as acquiring domestic resources and modifying the legal environment. With the recent development of communication and transportation technologies, the capabilities to identify and acquire resources from around the world are critical to the enhancement of national competitiveness.

The notion of “open innovation” was originally coined to describe and explain the importance of global networks in the circulation of knowledge, or more specifically, intellectual property (Chesbrough, 2006). The concept, however, also applies to the circulation of technical personnel. For example, policy expert Anna Lee Saxenian of the University of California at Berkeley argues that the rise of high-tech regions in India, China and Taiwan is due to the global circulation of immigrant engineers from Silicon Valley. Gaining training and experience in advanced industrial economies, these engineers serve as agents disseminating technological knowledge and culture in their home countries. In this way, Saxenian’s analysis turns the rhetoric of brain drain on its head, emphasizing the positive impact of immigrant engineers as they circulate between Silicon Valley and Bangalore or Hsinchu (Saxenian, 2007).

So the “open innovation” perspective provides important implications for STI policy in developing countries. Though Saxenian’s perspective might not be directly applied to the context of countries late to industrialization, it would be a narrow vision to focus exclusively on developing technical personnel within the country. Rather, the more beneficial course of action is to simultaneously devise strategies to tap top talent living outside the country (Mountford, 1997). The targets can include those who left a country to study and find employment abroad, as well as foreigners who might be interested in new opportunities to further their careers. Seen from this angle, the brain drain is not necessarily a bad thing; if the “drained brains” can be utilized as extensions of the nation’s human resource network, they can serve as valuable points of contact for identifying and procuring highly skilled technical personnel.

3. Analysis of Belize’s STI Status

3.1. Status of Science & Technology Research Capacity in Belize

While Belize has a level of engagement in basic research, engagement in applied research and experimental development is notably absent. A review of UNESCO’s country profile for Belize in research and experimental development showed the unavailability of data in all fields related to science and technology¹⁾ (UNESCO, 2010). While Belize’s Ministry of Energy, Science and Technology, and Public Utilities (METSPU) is tasked with facilitating R&D, the lack of a clear and comprehensive national research agenda has resulted in segmented pools of research activities in specific areas that do not necessarily align with national policies for development

1) UNESCO (2015). Country Profile. Retrieved from <http://www.uis.unesco.org/das/Country/Science?code=580®ioncode=null&SPSLanguage=EN>

and growth²⁾ (UNESCO, 2010). In addition, inadequate funding for R&D has led to a stagnant research culture.

According to the Approved Estimates of Revenue and Expenditure for Fiscal 2013/2014, 0.06 percent of GDP was invested in MESTPU (Ministry of Finance, 2015).³⁾ This is very much in line with most R&D investment in the Caribbean region, whose average has remained at 0.1 percent of GDP over the past decade. For example, Trinidad and Tobago spent 0.06 percent of GDP on R&D in 2007 and an average of 0.12 percent from 2000 to 2005.⁴⁾

In March 2010, the Prime Minister's Council of Science Advisors (PM-CSA) was commissioned to advise the Belizean government on formulating and implementing an effective strategy for national economic advancement through STI. Its mission is to further Belize's development through practical and environmentally sustainable applications of science, technology, engineering and mathematics (STEM) to create economic growth, strengthen education, enhance health and fortify Belize's national security. The council consists of domestic and international professionals in medical services, climate change, physics and business.

In 2012, the Belizean government established MESTPU. The ministry consists of three units: the Department of Geology and Petroleum, the Energy Unit and the Science and Technology Unit. The latter is responsible for the promotion of science and technology in Belize and plays a key role in Belize's efforts to achieve its Millennium Development Goal Target 8F: to make available the benefits of new technologies, especially those from information and communications, in cooperation with the private sector. Its main areas include the Information and Communication Technology Road Show held in March 2014, the Student Program for Innovation in Science and Engineering (SPISE) and the STEM program with the long-term goal being to "diversify the economies of the region by stimulating more technology-based entrepreneurship within the region (MESTPU, 2015)."

To respond to Belize's developmental goals, MESTPU has recognized the need to build its institutional capacity by strengthening skills, competencies, abilities, operations and services through the establishment of an STI institute (STI; MESTPU Strategic Plan 2012-17⁵⁾). Highly qualified researchers, professionals and technicians

2) UNESCO (2010). UNESCO Science Report 2010: The Current Status of Science Around the World. Retrieved from <http://www.uis.unesco.org/Library/Documents/UNNECOSR10-eng.pdf>

3) Ministry of Finance (2015). Retrieved from <http://www.cdn.gov.bz/mof.gov.bz/images/stories/docs/APPROVED%20BUDGET%202013-2014.pdf>

4) UNESCO (2010). UNESCO Science Report 2010: The Current Status of Science around the World. Retrieved from <http://unesdoc.unesco.org/images/0018/001899/189958e.pdf>

5) Government of Belize. Ministry of Energy, Science & Technology and Public Utilities Strategic Plan 2012 - 2017. Retrieve from <http://estpu.gov.bz/images/media/mestpu%20strategic%20plan%20-%202012-2017.pdf>

are needed to maintain and expand the country's STI capacity and ensure that appropriate incentives are offered to retain the human capital developed. Other tasks include tackling a number of weaknesses such as insufficient working conditions (including lack of instrumentation and technical support) and lack of access to high-level research networks and remuneration.

In addition to MESTPU's mandate to promote R&D, several other ministries and government bodies as well as other organizations conduct basic research in Belize. The Belizean Ministry of Education's Policy and Planning Unit is primarily responsible for gathering and analyzing data on students, teachers and scholastic performance. The Statistical Institute of Belize (SIB) is the country's official source of data on the population, economy and social conditions. The Environmental Research Institute (ERI), a semi-autonomous department of the University of Belize established in 2010, explores the large gap in domestic capacity for research in and monitoring of natural resources at the national level, including the longest barrier reef in the Western Hemisphere. ERI thus supports the country's most important industries of tourism and agriculture, and guided by the National Environmental and Natural Resources Management Research Agenda, focuses on producing results directly relevant and applicable to the sustainable management of Belize's natural resources and building domestic capacity. ERI partners with both domestic and international non-governmental organizations and conservation groups to accomplish its mandate⁶⁾ (UNCSD, 2012).

The Caribbean Agricultural Research and Development Institute (CARDI) is the sole R&D body among Caribbean Community (CARICOM) member countries for the R&D needs of the region's agricultural sector, as identified in national plans and policies. The think tank offers and extends the application of new technologies in production, processing, storage and distribution of the agricultural products of member states. Notable achievements include the implementation of integrated farming systems to beneficiary farmers, production of open pollinated corn (white and yellow types), soybean, cowpea and other grain legumes for commercial production, production of a manual on hot pepper production in the Caribbean and post-harvest submitted to the Food and Agriculture Organization (FAO) for distribution among Caribbean countries. It also implemented the Caribbean Agrometeorological Initiatives in collaboration with the Caribbean Institute for Meteorology and Hydrology (CIMH) to develop an effective system for forecasting pests and diseases⁷⁾ (CARDI, 2015). The Caribbean Community Climate Change Center coordinates the Caribbean region's response to climate change by working on effective solutions and projects to combat the environmental impact of climate change and global warming. The center serves

6) United Nations Council for Sustainable Development (2012). Belize National Sustainable Development Report. Retrieved from <https://sustainabledevelopment.un.org/content/documents/1012belize.pdf>

7) CARDI (2015). Country Profile. Retrieved from <http://www.cardi.org/country-offices/belize/>

as a database on regional climate change. It also provides policy guidelines on global warming and climate change to member states of the Caribbean Community (CARICOM)⁸⁾ (CCCCC, 2015).

While many organizations such as ERI, CARDI and CCCCC are actively engaged in R&D, academia's participation in research has been sporadic and, to a large extent, absent.

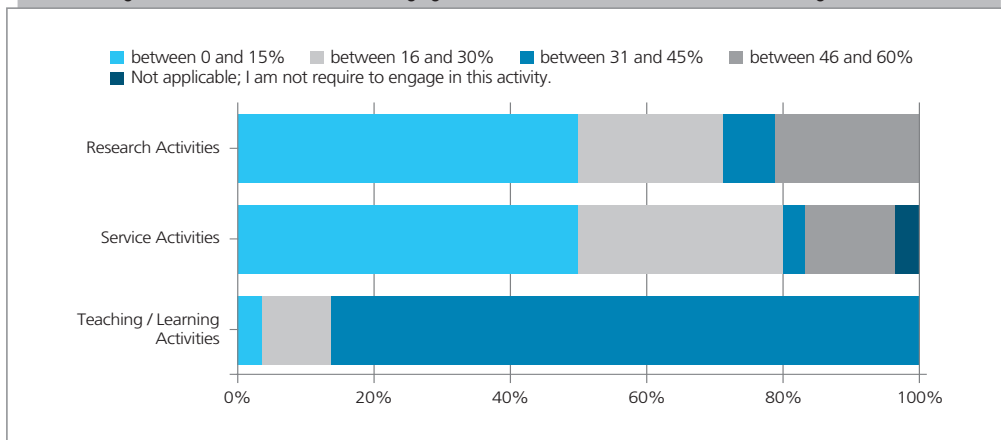
Intermittent efforts to develop a research culture at the University of Belize have resulted in minimal faculty participation in research in their areas of expertise. While the university maintains in its mission statement its commitment to excellence in higher education, research and service for national development, it has primarily taken on a teaching role. Among full-time sci-tech faculty, approximately 90 percents aid they spend between 46 and 60 percent of their academic commitment to the university through teaching or learning activities. Additionally, 20 percents of faculty were not required to engage in research, and an additional 50 percent spent no more than 15 percent of their academic commitment on research (Figure 1-2).

A qualitative analysis of reasons for low research activity found heavy teaching load, inadequate financial and physical resources, and lack of a university research framework⁹⁾ (Velasquez, 2014). Limited access to a comprehensive database of domestic and regional sci-tech journals has also meant limited regional cooperation and collaboration in sharing knowledge among researchers, teachers and educators. Collaboration in ongoing in-country research done by foreign universities has been minimal, despite the university's relations with North American counterparts through the Consortium of Belize Educational Cooperation (COBEC) and memorandums of understanding with other universities abroad.

8) CCCCC (2015). The Caribbean Community Climate Change Center. Retrieved from <http://www.caribbeanclimate.bz/>

9) Velasquez, G. (2014). *UB FST Faculty Needs Assessment Survey*. University of Belize.

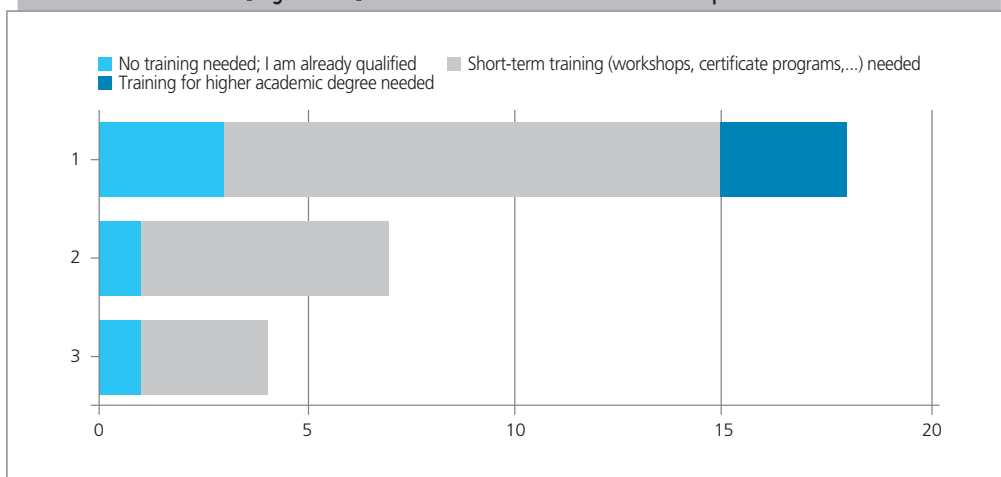
[Figure 1-2] UB FST Level of Engagement in Research, Service & Teaching Activities



Source: Velasquez, 2014.

In addition, the majority of University of Belize faculty of science and technology (UB FST) said some form of short-term training or graduate education was needed for conducting research (Figure 1-3).

[Figure 1-3] UB FST Needs for Research Participation

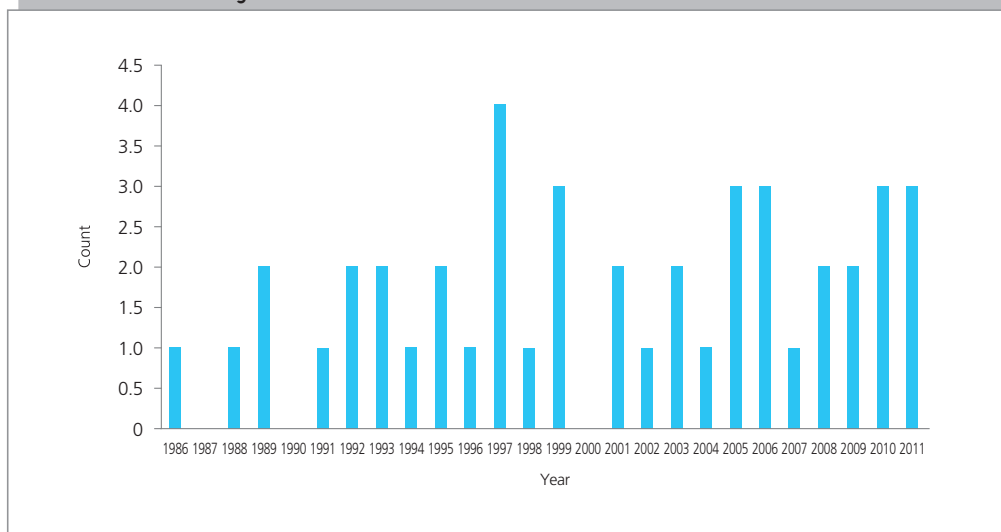


Source: Velasquez, 2014.

[Figure 1-4] presents the number of scientific and technical journal articles published in peer-reviewed journals between 1986 and 2014; data is unavailable for those published from 2011 to 2014.¹⁰⁾

10) Retrieved from http://www.theglobaleconomy.com/Belize/Science_articles/

[Figure 1-4] Published Scientific & Technical Journal Articles



Notes: The number of sci-tech articles published in Belize is a measure of “scientific output.”

Most scientific breakthroughs are described in journal articles that go through editorial review by peer scientists before being published. The quantity (and quality) of these articles indicates the activity level of scientific life in Belize.

Source: National Science Foundation

World Bank definition: Scientific and technical journal articles refer to the number of scientific and engineering articles published in the following fields: physics, biology, chemistry, mathematics, clinical medicine, biomedical research, engineering and technology, and earth and space sciences.

3.2. Status of Science & Technology Education Capacity in Belize

In recent years, Belize has invested more than 25 percent of its annual government budget in education. Investment in physical resources, incentives to keep students in school and teacher training at the primary and secondary levels have improved access to and gender equality in education.¹¹⁾ Of particular concern, however, is the quality of output from the education system.

A review of the Belizean education system shows a system with a heavy academic bias and low emphasis on technical and vocational training. This has resulted in underutilization of TVET educational resources. Passing standards in primary and secondary-level mathematics at the national and regional levels are low. Coupled with this is lower enrolment in mathematics, natural sciences and information

11) UNICEF (2011). *Situational analysis of children and women in Belize 2011: An ecological review*.

technology at the higher levels of education. In addition, feedback from stakeholders in UB FST Focus Group interviews, conducted as part of the university's curriculum review process, found that university graduates were well prepared with knowledge and information but short on skills and practical implementation of knowledge¹²⁾ (University of Belize, 2012).

3.2.1. Primary School Education

Education in Belize follows the British model. Under Belizean law, primary school attendance is compulsory and consists of eight years of schooling. Secondary school requires four years and undergraduate college programs are take four years. Upon completion of primary school, students take primary school examinations (PSE) to assess achievement in English, mathematics, science and social studies in accordance with the national curriculum for primary school set by the Ministry of Education, Youth and Sports (MoEYS). In high school, usually after their first two years, students select a science- or business-related curriculum for their remaining two years. Upon graduation, they take tests for the Caribbean Examinations Council Secondary Education Certificate (CSEC) in various subjects.

MoEYS also provides technical and vocational training through its technical and vocational education and training (TVET) programs. Access to these programs could follow completion of high school for full-time National Vocational Qualification (NVQ) programs or from various stages of education for entry to pre-vocational programs designed to provide learning and knowledge needed to access NVQ programs. In addition to the TVET track for further studies, high school graduates may opt to enroll directly in junior colleges for two years in traditional study of an academic field, after which they can earn an associate's degree and transfer academic credits to a four-year university for studying an additional two years for a bachelor's degree. Upon completion of two years of study beyond CSEC, students can take the Caribbean Advanced Proficiency Examination (CAPE) in various subjects.

<Table 1-1> shows overall enrolment in the Belizean education system for the 2012-13 academic year.¹³⁾

12) University of Belize (2012). UB FST Focus Group Interviews.

13) Ministry of Education, Youths & Sports (2012). *Abstract of Statistics 2012-13*.

〈Table 1-1〉 Overall Enrolment in Belizean Education System 2012-13

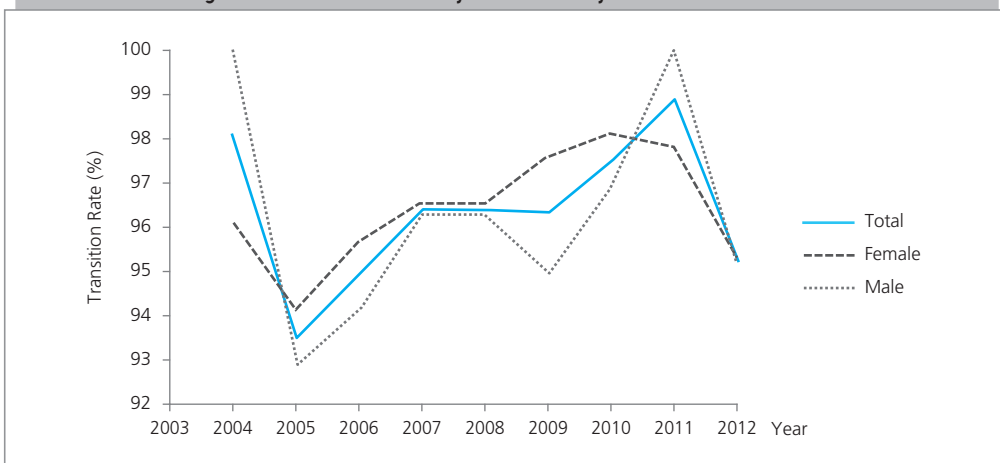
Education Level	Male	Female	Total
Preschool	3,764	3,636	7,400
Primary Education	35,287	33,525	68,812
Secondary Education	9,897	10,642	20,539
ITVET	542	177	719
Junior College	1,527	2,291	3,818
University	1,677	2,975	4,652
Total	52,694	53,246	105,940

Source: Ministry of Education, Youth & Sports.

3.2.2. High School Education

From 2004 to 2013, the national transition rate from primary to secondary school ranged from 93 to 99 percent (Figure 1-5), with a slightly lower rate for males (UNESCO, 2015). The net enrolment rate for secondary school from 2008 to 2013 steadily increased from 40.7 to 51.3 percent, with males having a lower net enrolment rate (37.4 vs. 48.2 percent) than females (44.1 vs. 54.1 percent). Average secondary repetition rates over the same period ranged from 6.5 to 9.1 percent, with higher rates for males. Dropout rates by academic year in the same period were highest in the first two years of high school (first year: 11.7-14.8 percent; second year: 8.4-10.3 percent). Completion rates for the same period ranged from 60.2 to 63 percent, with lower rates for males (53.4 to 58.9 percent) than females (63.1 to 69.5 percent)¹⁴⁾

[Figure 1-5] 2003-12 Primary-to-Secondary School Transition Rate



Source: Ministry of Education, Youth & Sports.

14) Ministry of Education, Youth & Sports (2012). *Abstract of Statistics 2012-2013*.

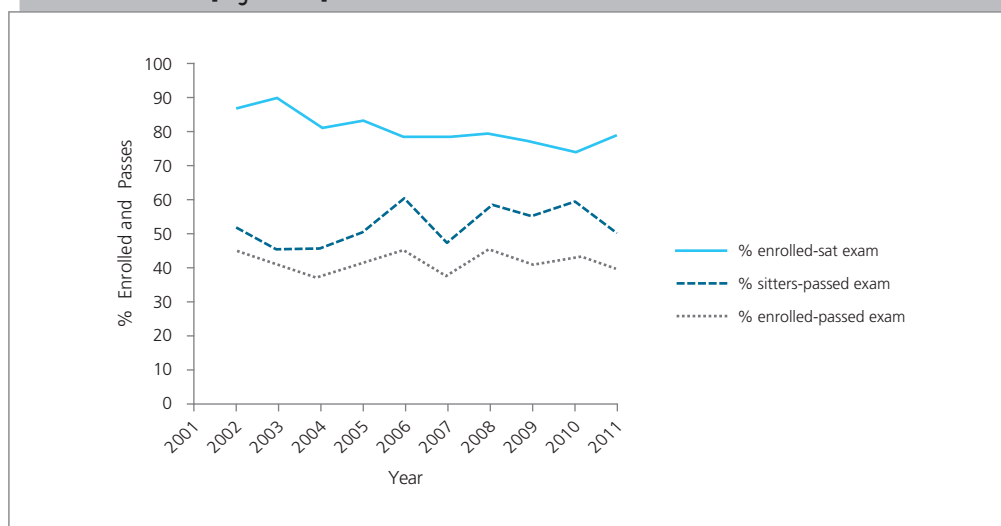
(Ministry of Education, Abstract of Statistics 2012-13).

Students in their first two years of high school follow a general mathematics and integrated science curriculum. High school mathematics and science education curricula in the junior and senior years focus on preparation for examinations of the Caribbean Examinations Council Secondary Education Certificate (CSEC). Courses consist of preparation for school-based assessments (SBAs), a major component of the CSEC examinations. The Ministry of Education, Youth and Sports rewards high performance on such tests by providing scholarships for tertiary education.

Students who passed with at least a satisfactory grade (three or higher) in CSEC mathematics comprised between 45 percent and 60 percent of the total [Figure 1-6]. In 2014, 55.1 percent passed the CSEC exam for mathematics, 77.9 percent passed that for IT, 70 to 80 percent passed those for natural sciences (biology, chemistry, physics and integrated science), and 46 to 73 percent passed those for technical studies (building technology, electrical and electronic technology, mechanical engineering technology and technical drawing). Refer to <Table 2-2>.

The Ministry of Education, Youth and Sports continues to provide training for secondary school teachers. In 2012, 25.5 percent of teachers were university graduates with professional training or a professional degree, 35.9 percent were university graduates and 4 percent had formal teacher training (Ministry of Education, 2012).

[Figure 1-6] 2001-11 Performance in CSEC Mathematics Test



Source: Ministry of Education, Youth & Sports.

〈Table 1-2〉 2014 CSEC Examination Results by Subject & Gender

Subject	No. of Takers	% Passed	% Male	% Female
Mathematics and Computing				
Mathematics	18,354	55.1	53.5	56.4
Additional Mathematics	2,265	79.2	76.1	82.3
Information Technology	3,252	77.9	76.4	79.6
Natural Sciences				
Biology	4,761	72.4	70.2	73.9
Chemistry	3,939	71.4	70.6	72
Physics	3,894	78.2	75.4	81.6
Integrated Science	3,381	70.6	65.3	75
Technical Studies				
Building Technology - Construction Building	58	46.6	42.3	83.3
Technology - Woods	193	73.1	73.2	71.4
Electrical & Electronic Technology	435	45.5	46	38.7
Mechanical Engineering Technology	365	71.5	70.6	78.9
Technical Drawing	1,690	73	71.7	80

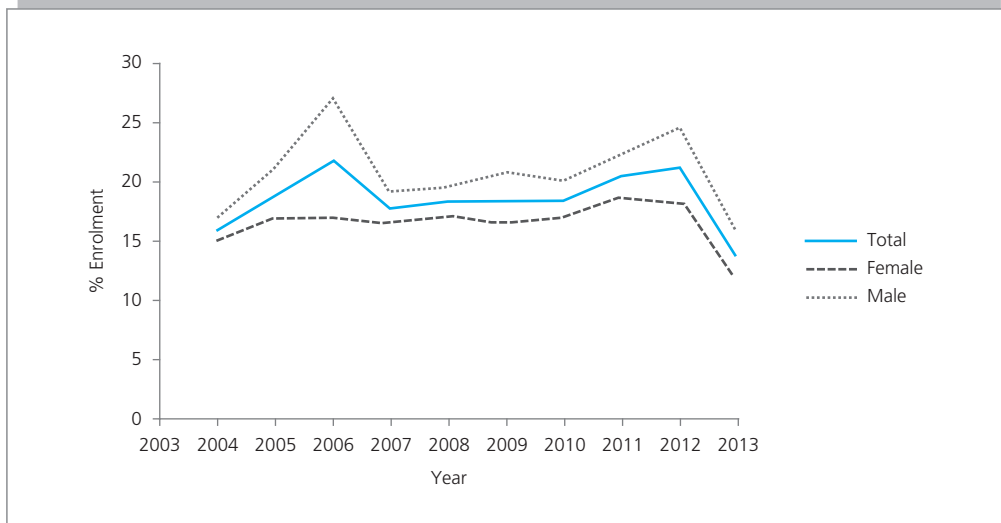
Source: Ministry of Education, Youth & Sports.

3.2.3. TVET Education

Enrolment in technical and vocational education and training (TVET) in Belize remains problematic. Despite substantial investment, rising enrolment is due partly to the opening of two centers; four centers established in 2001 have seen no increase in enrolment (Figure 1-7). This has led to underutilization of the six facilities. In two instances, this could be due to location of ITVET centers but the broader issue is the subsector's articulation with secondary education. Another problem is the prevalent and persistent view that TVET options are for students who are less academically inclined and thus opt for this as a last resort after failing to enter junior college or regular secondary school¹⁵⁾ (Ministry of Education, 2012).

15) Ministry of Education, Youth & Sports (2012). *Education Sector Strategy 2011-2016*.

[Figure 1-7] 2003-13 Enrolment Trends in Technical & Vocational Education & Training (TVET) as % of Overall Enrolment in Secondary Education



Source: UNESCO (2015). Country Profile

<http://www.uis.unesco.org/das/Country/?code=580®ioncode=null&SPSLanguage=EN>

3.2.4. Higher Education

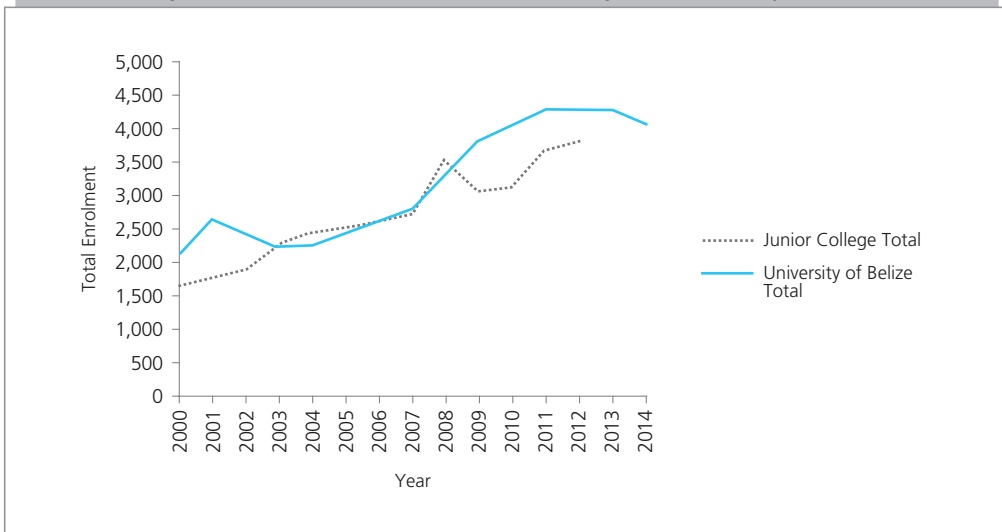
Enrolment has seen a steady increase at the tertiary level of education, with that of females outpacing the males'. Junior college enrolment has doubled over the past decade largely because of the addition of four junior colleges. Similarly, university enrolment has continued to expand annually, passing 4,000 in the 2013-14 academic year. The University of Belize accounts for the largest share of enrolment (4,040). It offers associate and bachelor's degrees in four departments: management and social science, science and technology, nursing, allied health and social work, and education and arts. The private Galen University has an enrolment of 550 students and offers associated and bachelor's degrees, with a few master's degree programs primarily in teacher education and social sciences (Figure 1-8).

These two universities make up 90 percent of Belize's university enrolment, with the rest either pursuing open and distance courses, largely with the University of the West Indies (UWI), or attending universities abroad. Females account for two-thirds of the student total. Despite this progress, gross enrolment rates estimated at 10-13 percent are at best less than half that of the regional average (27 percent) and on par with Central American neighbors Honduras and El Salvador (Ministry of Education, 2012). In addition, enrolment in associate and bachelor's programs in agriculture, mathematics, information technology, natural sciences (biology, chemistry, marine biology and natural resource management), and engineering (architecture,

building and civil engineering, and mechanical and electrical engineering) offered at the University of Belize favors male enrolment (32 percent higher on average) than female enrolment (Refer to Figure 1-9). The percentage of those passing the mathematics section of the Caribbean Advanced Proficiency Examination (CAPE) increased from 71.8 to 91.8 percent from 2006 to 2010 (Ministry of Education, 2012).

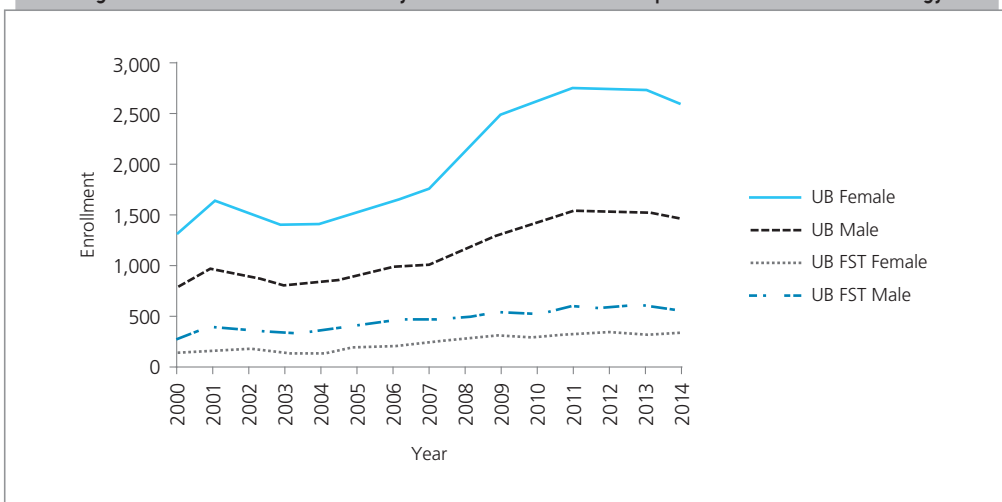
Among sci-tech faculty at the University of Belize, 19 percent had a degree in

[Figure 1-8] 2000-14 Enrolment in Junior Colleges and University of Belize



Source: Ministry of Education (2012); Velasquez (2014).

[Figure 1-9] 2000-14 Enrolment by Gender at UB and UB Dept. of Science and Technology



Source: Velasquez, 2014.

education and 77 percent a professional degree. <Table 1-3>, [Figure 1-10], and [Figure 1-11] provide data on academic qualifications and fields for full-time sci-tech faculty at the university:

<Table 1-3> UB FST Faculty Qualifications by Gender

Gender	Bachelor's (%)	Master's (%)	Doctorate (%)	Total (%)
Female	2 (3.9)	7 (13.7)	4 (7.8)	13 (25.5)
Male	4 (7.8)	27 (52.9)	7 (13.7)	38 (74.5)
Total	6 (11.7)	34 (66.6)	11 (21.6)	51

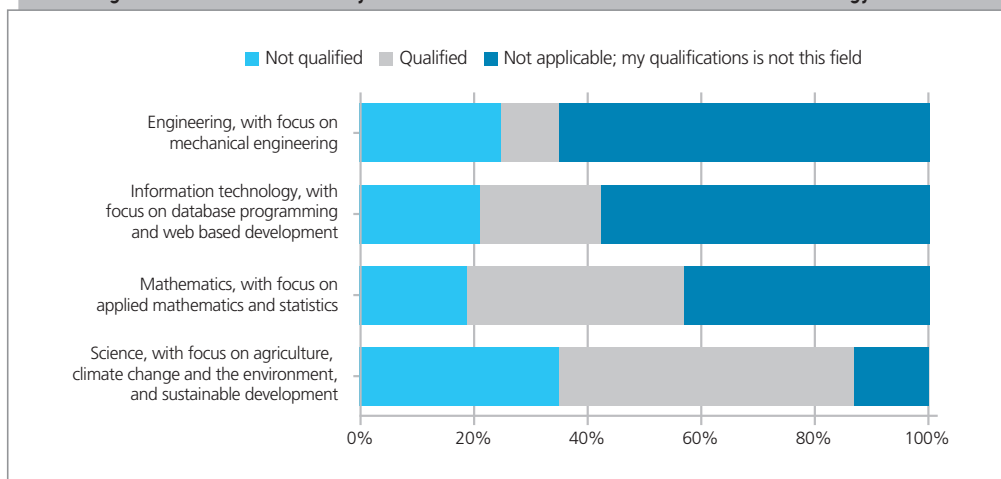
Source: Velasquez, 2014.

[Figure 1-10] UB FST Faculty's Areas of Expertise

#	Area of Expertise		Response
1	Agriculture		6
2	Architecture		3
3	Biology		10
4	Building Engineering		5
5	Civil Engineering		3
6	Chemistry		4
7	Electrical Engineering		3
8	Information Technology		5
9	Marine Science		3
10	Mathematics		6
11	Mechanical Engineering		3
12	Natural Resources Management		4
13	Physics		3
14	Other		0

Source: Velasquez, 2014.

[Figure 1-11] UB FST Faculty's Qualifications to teach in Science and Technology Fields



Source: Velasquez, 2014.

3.3. Status of Belize's Innovation Capacity

Through enacting laws, signing international treaties and setting up an office to protect intellectual property rights, Belize has created the proper conditions to encourage entrepreneurship and innovation. Despite being an example for the region, the Belizean population has, to a large extent, consumed the knowledge of other nations.

The Belizean legislature has enacted several primary laws on intellectual property rights (IPR), including the following:

1. Patents (Amendment Act, No. 40 of 2005)
2. Ancient Monument and Antiquity Act - Cap. 330 (2000)
3. Broadcasting and Television Act - Cap. 227 (2000)
4. Business Name Act - Cap. 247 (2000)
5. Cinematography Act - Cap. 241 (2000)
6. Copyright Act - Cap. 252 (2000)
7. Trademark Act - Cap. 257 (2000)
8. Industrial Design Act - Cap. 254 (2000)
9. Patent Act - Cap. 253 (2000)
10. Protection of Layout-Design (Topographies) of Integrated Circuit Act-Cap. 256 (2000)
11. Protection of New Plant Varieties Act - Cap. 255 (2000)

The following IPR-related laws have also been enacted:

1. Electronic Transaction Act - Cap 290:01 (2003)
2. Arbitration Act - Cap. 125 (2000)

3. Customs and Excise Duty Act - Cap. 48 (2000)
4. Freedom of Information Act - Cap. 13 (2000)
5. Imperial Law (Extension) Act - Cap. 2 (2000)
6. Reciprocal Enforcement of Judgment Act - Cap. 171 (2000)

Belize is a signatory to the following international IPR treaties:

1. Hague Agreement Concerning the International Registration of Industrial Designs (July 12, 2003)
2. Berne Convention for the Protection of Literary and Artistic Works (June 17, 2000)
3. Convention Establishing the World Intellectual Property Organization (June 17, 2000)
4. Paris Convention for the Protection of Industrial Property (June 17, 2000)
5. Patent Cooperation Treaty (June 17, 2000)

Belize is also a signatory to a number of IPR-related multilateral and bilateral treaties including the World Trade Organization (WTO) Agreement on Trade-related Aspects of Intellectual Property Rights (TRIPS Agreement) (1994) that took effect on January 1, 1995. Belize is deemed fully compliant with its obligations under TRIPS.¹⁶⁾

In addition, the Belize Intellectual Property Office (BELIPO), in conjunction with the World Intellectual Property Organization (WIPO), administers and maintains a database of IPR filings related to copyrights, patents, utility models, trademarks, industrial design, plant types and protection of integrated circuits.¹⁷⁾ BELIPO is automated to a degree and allows searches to be done online as well as procurement of filing forms. Through BELIPO, Belize is an example for IPR advancement in the Caribbean region.¹⁸⁾

A review of a number of IPR filings from 2000 to 2013¹⁹⁾ found the following:

1. Minimal number of patent, utility model and industrial filings (Figure 1-16, Figure 1-12 and Figure 1-13)
2. General increase in number of trademark applications (Figure 1-14)
3. Patent applications made primarily in sci-tech fields (Figure 1-15)
4. Majority of IPR filings made by non-residents or from abroad

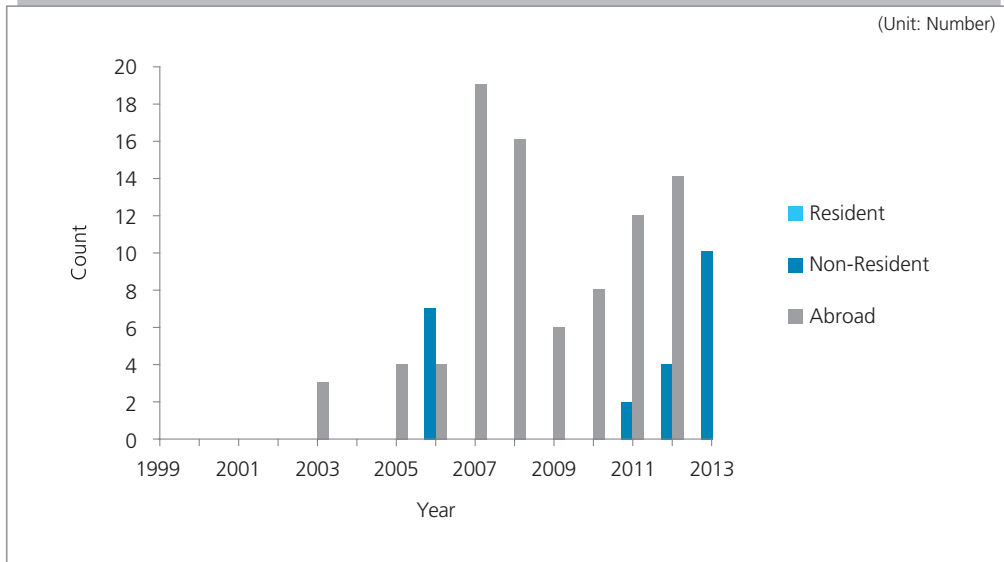
16) See Legislation Belize, THE WORLD LAW GUIDE, <http://www.lexadin.nl/wlg/legis/nofr/oeur/lxwebbz.htm> (last visited on March 31, 2015) (detailing Intellectual Property Laws in Belize).

17) Belize Intellectual Property Office (BELIPO). <http://www.belipo.bz/> (last visited on March 31, 2015).

18) Wilson, D.C. (2011), The Caribbean Intellectual Property Office (CARIPO): New, Useful, and Necessary. *Michigan State Journal of International Law*, Vol. 19:3

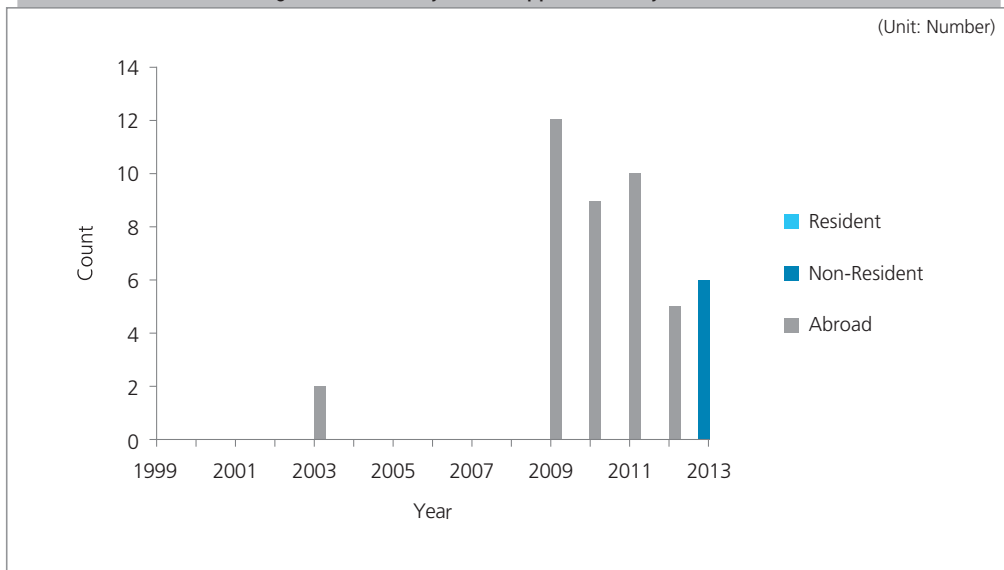
19) Retrieved from http://www.wipo.int/ipstats/en/statistics/country_profile/profile.jsp?code=BZ

[Figure 1-12] Patent Grants by Residence



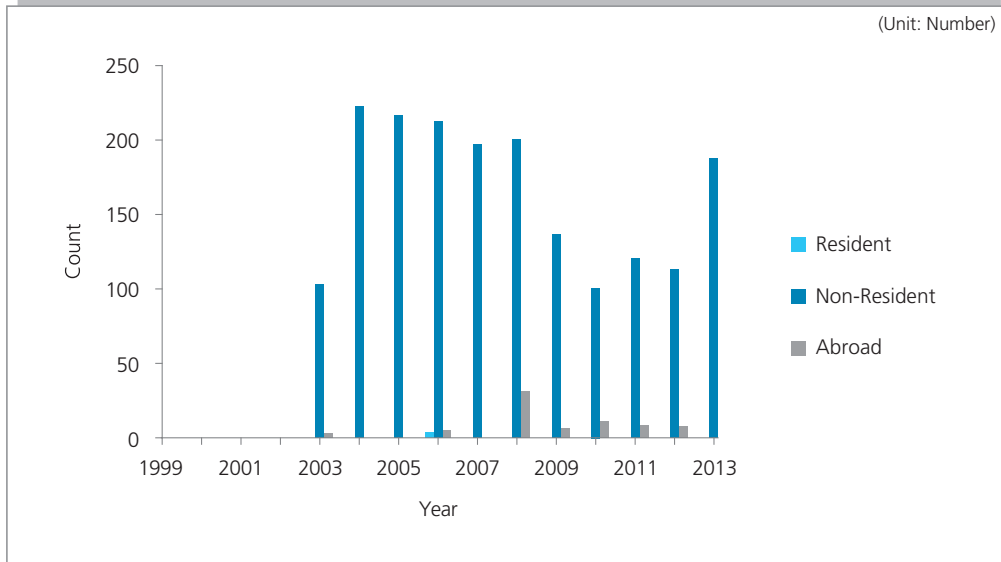
Source: WIPO.

[Figure 1-13] Utility Model Applications by Residence



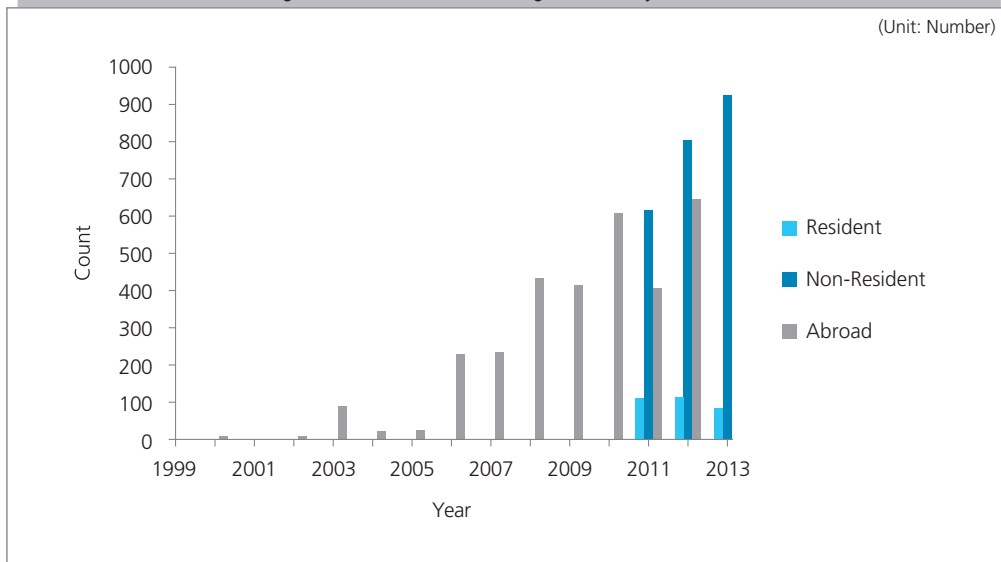
Source: WIPO.

[Figure 1-14] Industrial Design Registration by Residence



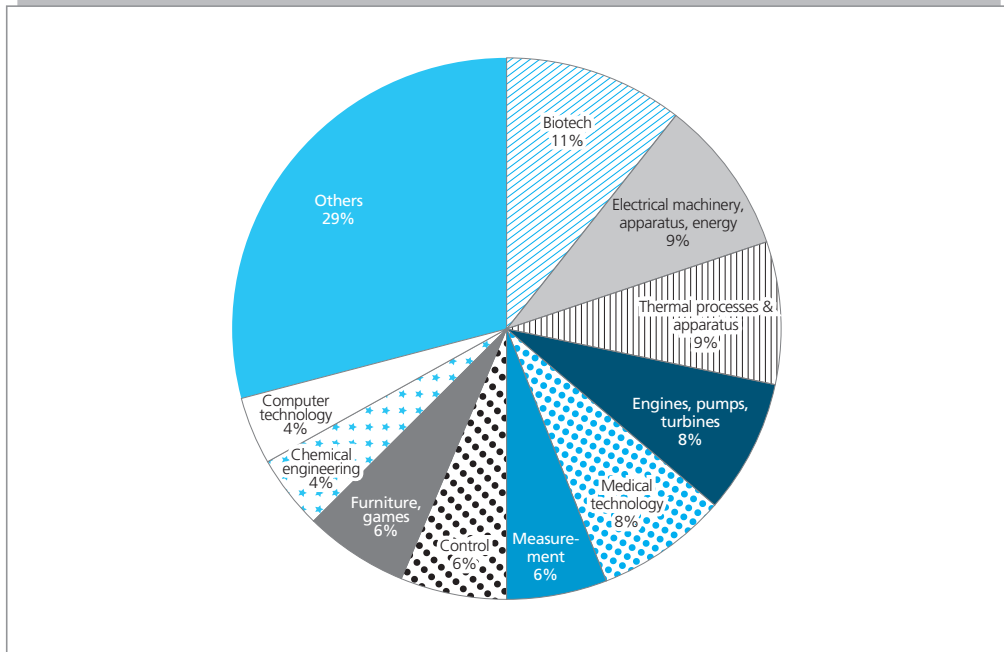
Source: WIPO.

[Figure 1-15] Trademark Registration by Residence



Source: WIPO.

[Figure 1-16] Patent Applications by Top Technology Fields (1999-2013)



Source: WIPO.

3.4. Status of STI Personnel: Study Abroad Statistics & Brain Drain

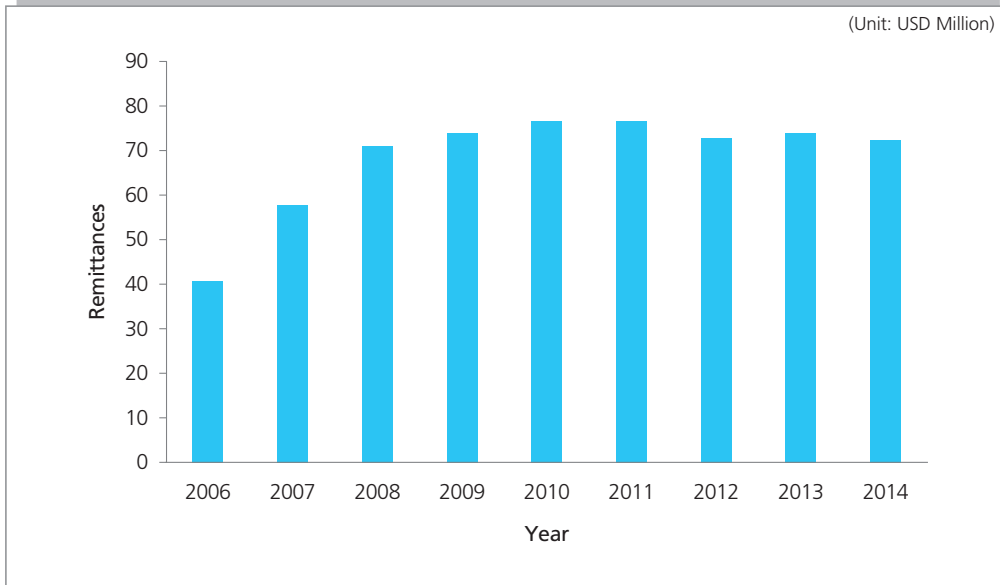
While the Belizean government has made substantial efforts to encourage Belizeans living abroad to return home and/or to invest in Belize, especially those with technical skills and expertise, the results of this campaign have been minimal. Few qualified Belizeans or persons of Belizean ancestry return to resettle in their motherland. Another challenge is to estimate the number of ethnic Belizeans living abroad and their qualifications as the country lacks an updated database of study abroad statistics and the brain drain.

The U.S. Census Bureau said 61,302 (margin of error: +/- 6,324) Belizeans and people of Belizean ancestry were living in the U.S. in 2013.²⁰⁾ Smaller numbers live in Canada and European countries. Personal remittances exceeded 70 million USD from 2008 to 2014, accounting for 4.58 (2013) to 5.88 percent (2009) of Belize's GDP.²¹⁾

20) U.S. Census Bureau. Retrieved from http://factfinder.census.gov/faces/tableservices/jsf/pages/productview.xhtml?pid=ACS_10_1YR_B04003&prodType=table. April 1, 2015.

21) The World Bank. Data retrieved from <http://data.worldbank.org/indicator/BX.TRF.PWKR.DT.GD.ZS>

[Figure 1-17] Personal Remittances in USD Million (2006-14)



Source: The Central Bank of Belize.

F. Docquier and A. Marfouk estimate that between 1965 and 2000, the following percentages of the Belizean labor force by education level immigrated to the U.S.: primary education 7 percent, secondary education 61 percent and tertiary education 73 percent.²²⁾ K. Nurse and J. Jones estimate that the migration rates of persons from Belize with secondary and tertiary education to member countries of the Organization for Economic Cooperation and Development (OECD) in the 1990s were 48.6 and 62.6 percent, respectively, and the corresponding figures for the 2000s were 49.2 and 51 percent.²³⁾ These statistics are noteworthy given that the Belizean government allots more than 25 percent of its national budget to education.

Belizeans pursuing studies abroad are an important factor behind the brain drain given their significant contribution to the skilled and professional labor force of countries where they study and gain further qualifications.²⁴⁾ Unfortunately, the Belizean Ministry of Education maintains an active list of scholarship opportunities for study abroad on its site,²⁵⁾ but a search for data on the students who are studying abroad yields no information .

22) Docquier, F., and Marfouk, A. (2005). *International Migration by Educational Attainment (1990-2000)*. Release 1.1. (Washington, D.C.: World Bank).

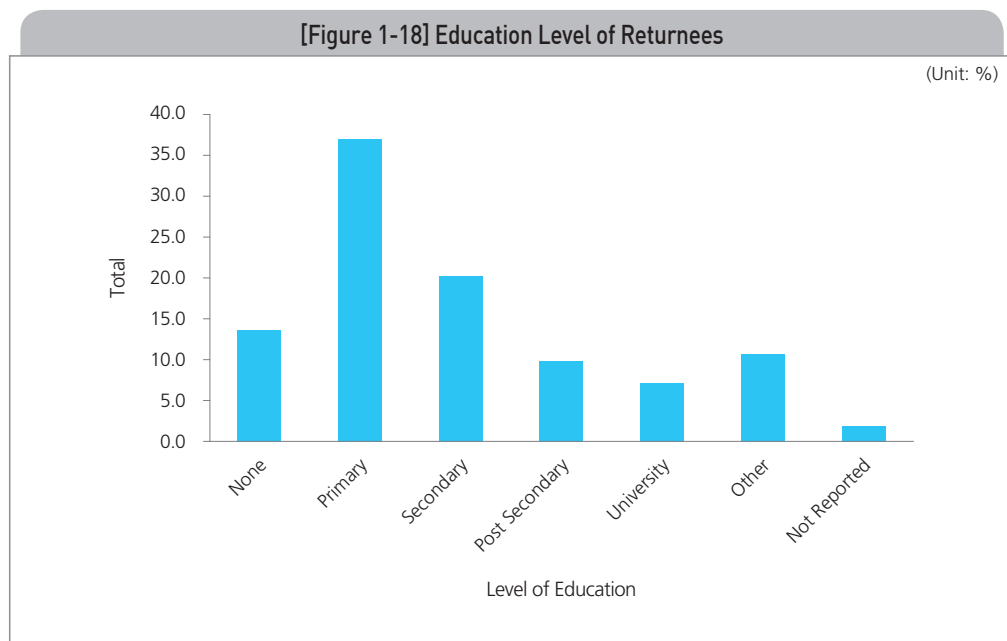
23) Nurse, K., and Jones, J. (2009). *Brain Drain and Caribbean-EU Labor Mobility*. Shridath Ramphal Center for International Trade Law, Policy and Services, UWI, Cave HILL, Barbados..

24) E. Thomas-Hope, E. (2001). *Skilled Labor Migration from Developing Countries: The Caribbean Case*. University of the West Indies.

25) Retrieved from <http://www.moe.gov.bz/index.php/scholarships>

In recognition of the contributions Belizeans abroad can make to the development of their motherland, the Belizean government, through the Ministry of Foreign Affairs and Foreign Trade, formally established in 2009 a desk specifically to engage the Belizean diaspora. In addition to the major contribution to the Belizean economy in the form of remittances, the government said the diaspora could serve as a potential source of investment capital, an untapped market for Belizean goods and services, and an underutilized channel for technology and expertise. Proposed initiatives included voluntary registration as a Belizean living abroad, a Belizean diaspora “medical brigade” to provide “an immediate embrace of high-value diaspora skills, knowledge and investment fund technology,” and “to deploy/ channel our intellectual and economical resources available within the diaspora; thus providing instant global capacity in partnership that benefits both participants and Belize.”²⁶⁾ Additionally, the government established the Diaspora Returnee Incentive Program (DRIP) to attract Belizeans abroad considering repatriation to invest in their homeland.

In contrast, the highest education level attained by approximately 70 percent of returnees who participated in the 2010 census carried out by the Statistical Institute of Belize was secondary education (Figure 1-18).



Source: Statistical Institute of Belize.

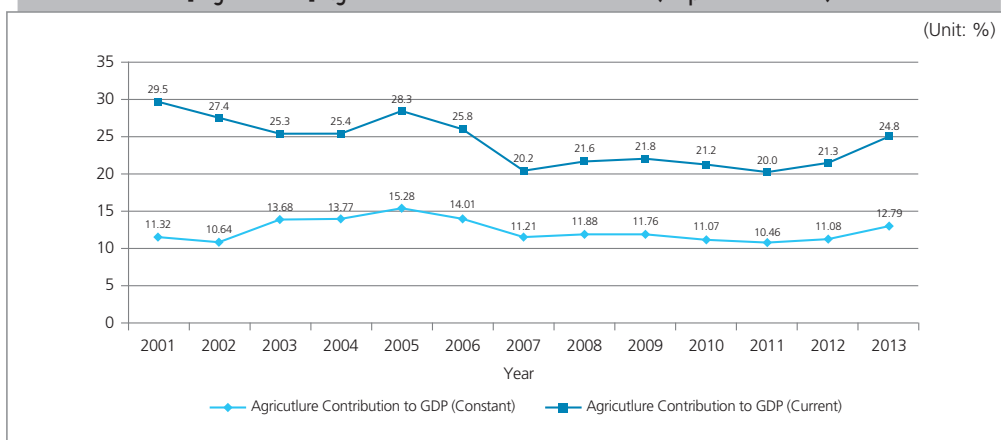
26) The Belizean Diaspora. Retrieved from http://www.diasporaofbelize.org/?page_id=16

4. Analysis of Key Industrial Sector and National Strategic Plans

As briefly noted in the introduction, Belize has been one of the better performing economies in the Caribbean region over the last few years (Caribbean Economic Review & Outlook, 2015). However, its small size and population have been and continue to be major constraints on further economic growth. The small population of 350,000 has and will pose significant challenges for Belize's STI-based growth in two ways. First, the human resource pool from which the country can draw skilled technical personnel is limited. This is another reason that makes the global circulation approach attractive to Belize, rather than sole reliance on domestic training. Second, the domestic market for Belizean industries is obviously limited. To achieve sustainable growth, the country must look outward to export markets with competitive products that can capture significant market share in the regional and global marketplace. In meeting these two challenges, STI should be effectively utilized to solve the problems facing the country.

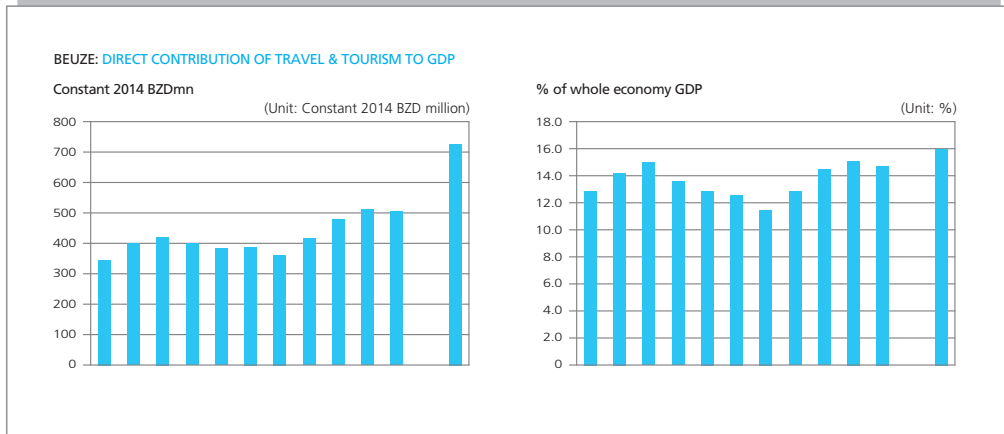
The two sectors that make the largest contributions to the Belizean economy are tourism and agriculture. In recent years, the agricultural sector has made a 140 million USD investment in the sugar industry, which partly accounts for the uptick in its economic contributions in 2012-13 (Figure 1-19). Also, the tourism sector has made impressive improvements due to the rapid growth of cruise and overnight tourism (Figure 1-20). Taken together, both sectors should comprise the "steeples of excellence" on which Belize should focus as major drivers of STI-based growth. Judging from interview results, agriculture, in particular, apparently warrants extra attention as the sector to benefit from additional STI activities in Belize. On the other hand, the recent growth of tourism could mean a potential revenue source

[Figure 1-19] Agriculture's Contribution to GDP (September 2014)



Source: Statistical Institute of Belize.

[Figure 1-20] Tourism's Contribution to GDP



Source: The World Travel & Tourism Council's Travel & Tourism Economic Impact 2015 Belize.

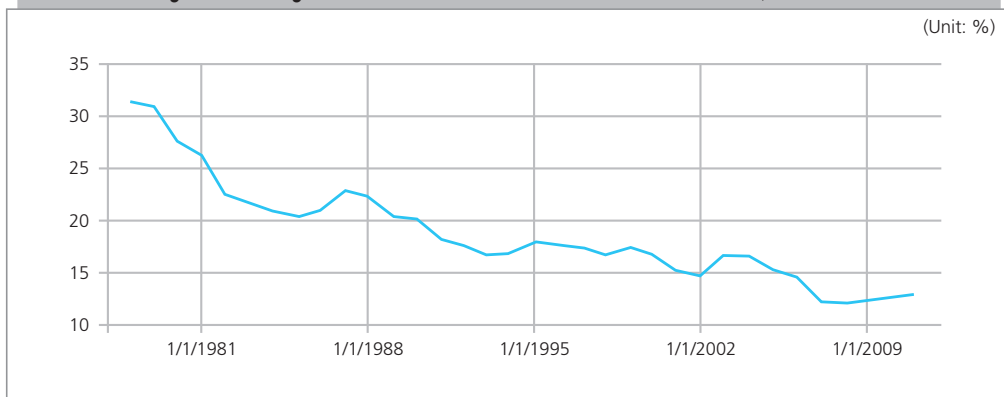
for securing a stable funding base for STI activities if appropriate tax legislation is enacted.

4.1. Analysis of Key Industrial Sectors in Belize

4.1.1. Agriculture

The agricultural sector is the second-largest contributor to Belize's GDP following the tourism industry. Since 1981, however, agriculture's status as the main contributor to GDP has seen a steady decline, as painfully evident in the following graph (Figure 1-21). According to Trading Economics, the sector's added value contribution to Belize's GDP was 13.02 percent in 2011. The dominant export commodities

[Figure 1-21] Agriculture's Decline in Added Value as % of GDP, 1981-2011



Source: Trading Economics.

are sugarcane and citrus, but several other commodities are beginning to assert themselves. Marine products, for example, experienced startling growth of 40 percent in 2013-14, largely due to a stellar increase in shrimp production.

These numbers indicate substantial untapped potential in Belize's agricultural sector that can be triggered through STI activities. Indeed, those in the sector are quite eager to provide a list of STI needs. Examples both from information surveys and interviews are as follows. The variety of needs identified by industry participants shows that real needs can be met, though prioritization is necessary to identify the most technical areas with the potential for high rates of return.

4.1.1.1. Science: genetics, bioinformatics & biology

A. Biotechnology

- 1) Tissue culture to acquire disease-free planting material & true-to-type materials
- 2) Planting materials resilient to extreme agro-climatic conditions such as drought & water logging
- 3) Research in eco-friendly bio-pesticides that can be developed from domestic materials
- 4) Soil chemistry & soil amendments to maintain fertility of soil like bio-fertilizer

B. Genetics

- 1) Embryo transplant to improve genetic stock in livestock
- 2) Plant breeding for purification of grain stock material

C. Food Technology

- 1) Expertise in product development
- 2) Product differentiation
- 3) Processing technology for branding & packaging
- 4) Quality certification technologies & services

D. Agro-meteorology & Hydro-meteorology

- 1) Best practices for precision forecasting of weather conditions – three to five day forecasts to help application of land preparation, fertilizer application & harvesting
- 2) Water demand

4.1.1.2. Technology: advances in agriculture, product development & ICT

A. Precision agriculture

- 1) Land leveling for irrigation
- 2) Geo-referencing agricultural areas & use of GIS
- 3) High resolution topographical mapping for planning & development
- 4) Technology that increases labor productivity like small, efficient and effective machines
- 5) Best practices for removal of excess moisture around plant root zones
- 6) Improving processes of production & post-production using technology & innovation

B. ICT

- 1) Development of phone applications for dissemination of agricultural information
- 2) Apps to collect data & statistics

4.1.1.3. Innovation: new products & services & developing Belize's competitive advantage

- 1) Training in problem solving
- 2) Training in organization & clustering to achieve comparative advantage
- 3) Learning to utilize available domestic materials for best results (e.g. building materials)
- 4) Use of solar, wind and hydroelectric energy to supply power in rural farming communities
- 5) Lifting & storing water for irrigation purposes
- 6) Protective covered structure for vegetable production
- 7) Aquaponics for backyard gardens

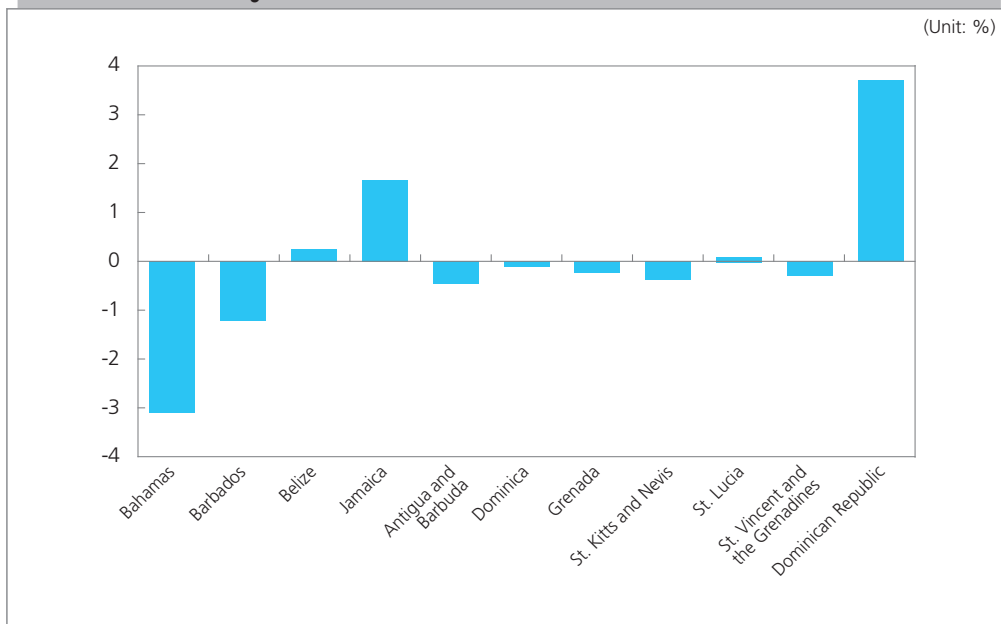
4.1.2. Tourism & Services

Tourism in Belize is one of its economy's most vibrant and fastest growing sectors. The industry's contribution to GDP, inclusive of all value-added products, foreign and domestic investments, supply chain and food products, was 39.2 percent, making tourism the top contributor to the Belizean economy. The sector is expected to grow 0.6 percent in 2015 (World Travel & Tourism Council, 2015). In particular, the number of cruise ship disembarkations saw healthy growth from 40,000 in 2001 to 600,000 in 2013, having the potential to be a major source of tax revenue with the appropriate legislation (Central Bank of Belize; Belize Tourism Board). The growth of the tourism

sector leads to secondary economic impact such as employment and consumption of subsidiary goods. According to the Central Bank of Belize's Annual Economic Report 2013, the unemployment rate fell 2.3 percentage points to 13 percent from 2012 due in part to job creation resulting from a resilient tourist industry (Central Bank of Belize, 2013). This growth must be managed responsibly lest the country face the danger of harming the very essence that attracts multitudes of tourists to its shores. As outlined in the Horizon 2030 Framework, "care for the natural environment" is one of the central themes of long-term Belizean development, and this is important not only for nature and the environment, but also for the healthy growth of the country's eco-tourism.

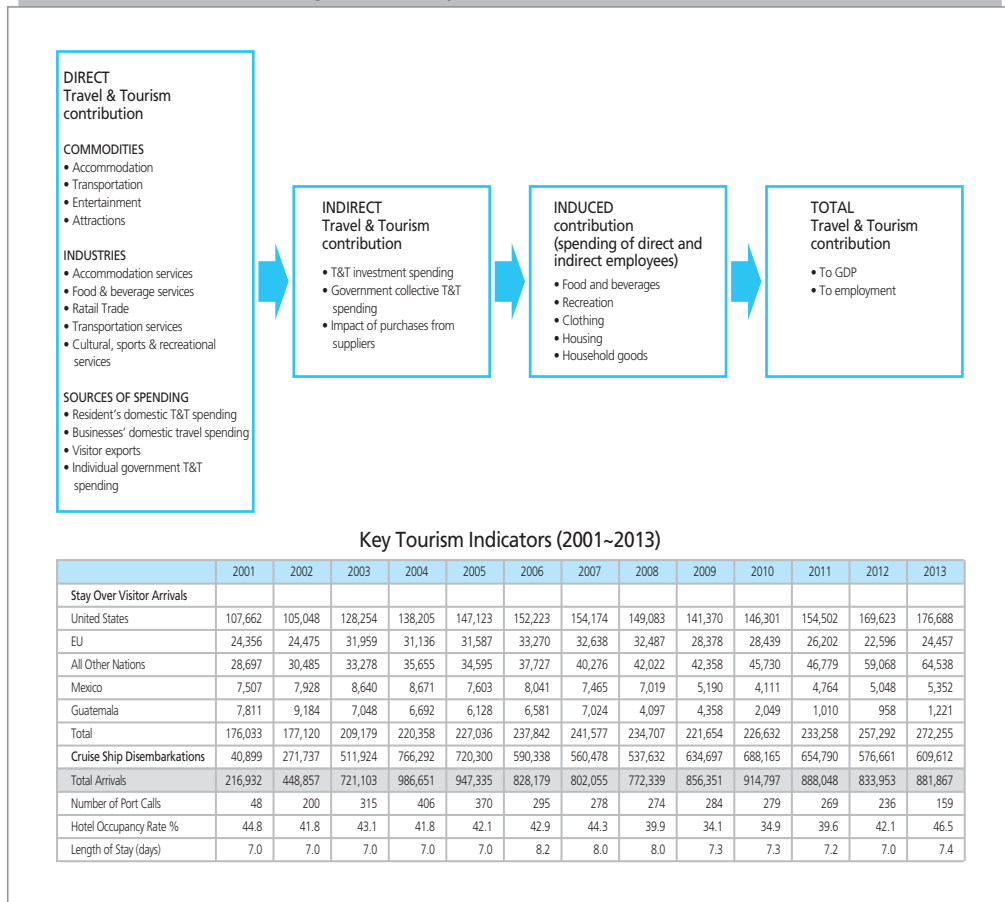
Travel and tourism are important economic activities in most Caribbean countries. The tourism sector in Belize has proven quite resilient throughout the recent economic downturn caused by the 2008 financial crisis. As shown in [Figure 1-22], the tourism market's share in the region has experienced decline, with the exception of Belize, the Dominican Republic and Jamaica. This poses a great opportunity for tourism growth and consequently increases the direct and indirect impact on the Belizean economy. The synergies between tourism and other sectors, especially agriculture, are clearly illustrated in [Figure 1-23]. In other words, when foreign tourists visit and spend time in Belize, they consume additional commodities such as food and drinks, which have supplementary economic impact.

[Figure 1-22] Share of Caribbean Tourism Market 2007-13



Source: IMF WP/14/229.

[Figure 1-23] Key Tourism Indicators 2001-13



Source: Central Bank of Belize & Belize Tourism Board.

4.2. Analysis of National Strategic Plans

4.2.1. Horizon 2030 Framework

Horizon 2030 Framework is a long-term national development plan prepared in 2011 by Barnett & Co., Ltd. The report is based on extensive consultations with key stakeholders in Belize who outlined a vision for the nation's development. Their vision covers the entire scope of Belizean national development, and as such is broader than the intended coverage of this report. But several prominent themes distilled in the *Horizon 2030 Framework* provide important guidelines for establishing Belize's STI strategy. The seven thematic areas are:

1. Democratic governance as a foundation for development
2. Crime, public safety & access to justice

3. Education for development – education for life
4. Building a resilient economy
5. Focus on key productive sectors
6. Keeping citizens healthy throughout their lifecycles
7. Care for the natural environment

Based on the themes outlined above, the Horizon 2030 team makes numerous recommendations on strategic goals to achieve the national vision, as well as identify suitable indicators to track progress. Certain indicators are directly relevant to establishing Belize's STI strategy such as the "number of students engaged in study of agriculture at secondary and tertiary levels," "availability of extension services to farmers" and "number of small businesses." Overall, the project aims for higher investment in science and technology education to promote innovation, building a vibrant agricultural sector with increased exports and governmental support for agriculture to raise productivity.

4.2.2. MESTPU Strategic Plan 2012-2017

The MESTPU Strategic Plan 2012-2017 was prepared in 2012 with the goal to "integrate energy, science and technology into national development planning and decision making to catalyze sustainable development." The plan is divided into two parts. Part I is the Belize National Sustainable Energy Strategy, with an eye toward medium- and long-term plans until 2033. Part II is the Science, Technology & Innovation (STI) Strategic Options for Implementation (2012-17). The latter will have a direct bearing on this project, so this paper's focus is on Part II.

Strategic Plan 2012-2017 proposes five strategic goals:

1. Introduce an STI capacity building & institutional strengthening program
2. Promote STI as important powerful ingredients to foster the economic growth and global competitiveness of Belize
3. Bolster development of microenterprise
4. Push for advancement of ICT (information and communication technologies)
5. Build a sustainable MESTPU

Within the framework of the goals outlined above, MESTPU has proposed the development of STI policy and action plan, as well construction of the Belize Center for Excellence in STI. Furthermore, the plan lists a set of priority projects that includes:

1. Rice, Corn & Beans Post-Harvest Technology – Introduction of Solar Drying Technology
2. Gasification Technology – Introduction of Bio-energy (Butane Substitution) for Commercial Crop Drying
3. Bio fuel (Diesel Substitution) Technology

4. LED Lighting & Solar Cooling to Improve Energy Efficiency & Conservation
5. Anaerobic Fermentation Technology
6. Water Purification Systems for Rural School without Potable Water
7. Hygiene & Sanitation Systems with Low Water Requirements for Rural Schools
8. Ocean Thermal Energy Conversion (OTEC) Technology

The selection of these topics reflects the primary concerns of MESTPU. First, the ministry's main attention is on energy issues, especially substituting fossil fuel with renewable energy. Second, these topics show the importance of technologies required for the agricultural sector. (This confirms what KSP members during their visit heard from their Belizean counterparts that agriculture should be the priority of the new STI institute.)

4.3. Simple SWOT Analysis

Based on the brief information outlined above as well as more detailed discussions of Belize in other parts of this report, a simple strengths-weaknesses-opportunities-threats (SWOT) analysis of the country's status is possible. The items in the strengths and weaknesses boxes in the diagram are factors internal to Belize; those in the opportunities and threats boxes are factors external to the country.

〈Table 1-4〉 SWOT Analysis	
Strengths	Weaknesses
• Natural & environmental resources • Active & forward-looking political leadership	• Small population that hinders economies of scale • Little experience in STI activities • Lack of popular understanding of STI
Opportunities	Threats
• Increasing interest in eco-tourism worldwide (↑ revenue) • Real & perceived need for STI in industry (esp. agriculture)	• Global market fluctuations • Lack of interest in technical personnel to participate in Belizean development

Two key strengths of Belize are the availability of natural and environmental resources and an active and forward-looking political leadership. The country is widely known for its rich deposits of natural resources. It is also a leading producer of agricultural products such as citrus, sugarcane and cacao; its fisheries production (i.e. shrimp) forms a major industry. Thus, designing a strategy should incorporate use of these resources and maximization of added value. Also, Belize's environmental resources are also well-known abroad. The cayes and atolls adorning the coast are

preferred tourist destinations in the region. Part of the undeveloped jungle area has the potential to attract tourists for its environmentally-friendly appeal.

On the other hand, the Ministry of Energy, Science and Technology, and Public Utilities' initiative to develop a national strategic plan for STI reflects the availability of active and forward-looking political leadership in Belize. Apart from the ministry, officials from relevant ministries are keenly aware of the importance of STI in ensuring further development. If the capability of the political leadership can be aligned to push the nation toward a particular direction, it will serve as a key driving force that advances the Belizean economy to the next stage of development.

Belize, however, has several weaknesses as well. The first is its small population, which inherently hinders exploiting economies of scale in the domestic market. Therefore, the country must direct its attention toward the regional and global markets if it wishes to achieve sustainable growth and development. Select portions of the Belizean economy are following this principle, though their efforts might be focused in a strategic fashion. Other weaknesses include lack of experience in dealing with STI activities and the absence of popular understanding in the area. For the Belizean people to understand the power of science and technology in fostering national development is critical, and this is possible through strengthening primary and secondary education and public campaigns explaining the mechanisms through which STI can play a leading role in economic development. The country's lack of experience in and understanding of STI should be seen as an important weakness, but should also be seen as a gateway through which all nations must pass to reach the stage of STI-driven development. There is no other way to gain experience other than experiencing it.

Opportunities for STI-driven development are abundant for Belize. One of the leading opportunities is worldwide interest in ecotourism, a sector that Belize should skillfully exploit. Already, indications show that tourism-related revenue is on the rise, as reflected in the number of cruise ship embarkations, and this could serve as an important financial source to support STI activities. As discussed below, a significant portion of this revenue should be earmarked for STI to ensure sustainability of STI activities. On the other hand, industry participants have a fairly good idea of their technical needs, especially in agriculture, as was made clear in many interviews and consultant reports in Belize. This means the agricultural sector has high growth potential if integrated with intense STI activities in the area. It is important to identify and prioritize technological needs, so that the most high-impact items can receive particular attention in the beginning stages of STI investment.

Finally, Belize must watch out for several external threats, the most critical being global market fluctuations. This is something Belize as a nation has little control

over, so keeping close tabs on the ebb and flow of the regional and global markets is always helpful, especially in strategic areas. This kind of information surveying should be part and parcel of the new STI institute's operations.

5. The Korean STI Experience

Over the last half century or so, the Republic of Korea has transformed itself from an agriculture-oriented economy with per capita GDP of less than 100 USD to one dominated by high technology such as semiconductors, automobiles and mobile phones, along with per capita income of almost 30,000 USD. Its spectacular growth in the second half of the 20th century has earned the country the nickname “Miracle on the Han River (Amsden, 1992).” What can countries late to industrialization learn from the development experience of Korea? Is there a Korean model extractable from the Korean development experience? As a rare example of a late-developing country overcoming the development barrier, Korea's success story has gathered significant attention in recent years. But other than the “developmentalism” argument of emphasizing the central role of government in economic planning and choosing winners, analyses have lacked sufficient detail to construct a model that latecomers to industrialization can emulate (Woo-Cumings, 1999; Kohli, 1994; Öniş, 1991).

To extract the right “lessons” from historical experience, a cautious approach is recommended. What worked in one place might not work in another, given differences in social, economic, cultural and geopolitical contexts of each locale. Indeed, Korea's rapid development was possible only within a specific set of internal and external circumstances, including the Cold War and U.S. involvement in East Asia, the division of the Korean peninsula and the ensuing bloody civil war (1950-53), and an elongated period of dictatorial rule (1961-1987) (Brazinsky, 2005; Brazinsky, 2007). This means developing countries looking to emulate the Korean experience will soon discover that emulation is by no means straightforward, and perhaps even impossible. A “stage theory” perspective on development, or a simple catch-up attitude without considering the historical trajectories of the “model” and the emulator, could be detrimental (Latham, 2000).

Nevertheless, with obligatory caution, the Korean experience can be used as a template for a successful case in building a nation's capacity for STI. Seen from the NIS perspective, Korea has successfully nurtured an innovation system comprised of a vibrant private industry, research-oriented universities and an array of specialized PRIs—the three components of an NIS as discussed in Chapter 2. Compared to what the nation had in place in 1960, the transformation is quite remarkable. When Korea embarked on a strong development program in the early 1960s, it had a

mediocre industrial sector, a negligible academic sector and a small number of PRIs with limited capabilities, mostly remnants from Japanese colonial rule. Today, the components within the “triple helix” of the Korean NIS are closely interconnected and coordinated through various public-private cooperative bodies (Kim, 2000). To understand the process Korea used to accomplish this transformation, closer attention is needed to the evolutionary process of industrial shift and government policies.

5.1. The Evolution of the Korean NIS

5.1.1. The Beginning of Korean NIS

As a nation’s capacity in technological innovation grows, it is perhaps inevitable that the economic focus shifts to a more technologically sophisticated sector. Then it follows that the innovation system will have to shift in a corresponding manner to provide the knowledge, expertise and skills required to undertake the new task. The result will be a gradual evolution of the NIS from simple to complex, and adding more components to the system as needs arise. This was certainly true for the Korean development experience in the second half of the 20th century. In other words, industrialization was rapid but did not happen overnight.

Korea has a long history as a unified nation, but many would agree that conscious efforts to drive national economic development began only in the early 1960s. The policy drive led by President Park Chung-hee, for better or worse, provided the critical push urgently needed at the time. When the president came to power in 1961, however, he took over a country still struggling to recover from the rubble of the Korean War (1950-53). Korea was still heavily dependent on developmental aid, mostly from the U.S. Korea had no industry to speak of and overall productivity was dismal. Few Koreans at the time had any idea how to begin the process of reconstruction.

In envisioning the nation’s long-term development, Park and his advisers adopted the strategy of export-oriented growth. The basic strategy was to focus on a select few sectors deemed to have comparative advantage in the global market, mass produce items by relying on imported technology and cheap labor, and export the products. This strategy made sense given the relatively small size of the Korean economy, and consequently, the paucity of its domestic market at the time. Adopting the export-oriented strategy would ensure the accumulation of foreign exchange critical for overall economic stability. Though it seems to make perfect sense in retrospect, this strategy was not the obvious choice in the early 1960s for an underdeveloped country such as Korea. Another option was the import-substitution strategy, or developing domestic industries necessary for the sustained operation

of the economy. The latter strategy, also known as autarky, was deemed at least as viable as the export-oriented option among key policymakers at the time (Kimiya, 2008).

So what accounts for Korea's strategic choice? The country's developmental direction in the early 1960s reflected the geopolitical context within which the nation found itself at the time. At the height of the Cold War, the U.S. remained fixated on containing the expansion of communism. Korea was considered one of the key barriers of the "free world" against communist expansion in East Asia. One way to maintain Korea's position was to integrate its economy as strongly as possible with those of American allies, especially Japan. Seen from the perspective of U.S. leaders, the autarky policy in Korea was something to be avoided at all costs. The flipside of this was that the Korean strategy of pushing an export-oriented economy was possible only with the guaranteed availability of the U.S. and Japanese markets. This was but one aspect of how Cold War concerns influenced the trajectory of the Korean development experience.

For the export-oriented strategy to work, Korea had to build capacity in targeted sectors to mass produce products for export. Given the low level of technology in Korea at the time, its range of choices was rather narrow. The initial sectors chosen were the light industries such as textiles, garments and food processing, in which the late-developing country could take advantage of abundant cheap labor. The factories required to turn out large quantities of products were imported through "turnkey" arrangements, or wholesale importation of entire plants, mostly from the U.S. and Japan. The influx of foreign investment was facilitated by the enactment of the Foreign Capital Inducement Act in 1960, which streamlined the process of foreign investment. The technical personnel and expertise required at this stage were low-skilled labor and plant management capabilities, both of which the Korean secondary and tertiary educational system could furnish. To expedite exports, which was the ultimate goal, the Korean government established an export industrial park in the Guro-dong neighborhood of western Seoul. The items produced at the park were eligible for substantial tax breaks.

〈Table 1-5〉 Korean Development & Role of STI Policies

Period	1960s	1970s	1980s	1990s	2000s	2010~
Era	Export-Oriented	Export-led	Economic Liberalization	Democrat.	Advancement	Global Leading
Focused Industries	Light Industries	Heavy Industries	Assembly & Processing Industries	ICT	Knowledge Intensive Industries	Knowl. Service/New Converging Green Ind.
Compet. Factor	Cheap Labor	Skilled Labor	Capital Investment	Technologies	S&T Innovation	Advanced S&T Innovation
Demanding S&T HR	Skilled HR	Technical HR	Higher S&E	High Calibre S&E	Creative S&E	Creative & Converging S&E
Demanding Tech.	Plant Mgt.	Facility M&O	Mfg.	Core Tech.	Endogenous Tech.	Source Tech.
S&T Policy	Turn-key Capital Import/Tech. Learning	Internalizing Imported Tech./Reserve Eng.	Modify Imported Tech./Develop Domestic Tech.	Advancing Tech. Catch-up/ Large Gov. R&D Prog.	Focus on endogenous Tech./ Systemize Prog.	Globalize S&T Focus on Convergence

Source: Jang (2011).

The strategy of export-oriented development focused on the light industries led to moderate success by the mid-1960s. But the Korean government continued to push for further transformation. A major opportunity to upgrade the national innovation system came in 1966, when the U.S. offered assistance for establishing an applied think tank for industrial technology in Korea. In part, this gift from U.S. President Lyndon B. Johnson was thanks for Korea's deployment of combat troops to Vietnam that started in 1964. For the conflict in Southeast Asia, Korea had sent two combat infantry divisions and one marine brigade to assist the U.S. military. Korean participation in the Vietnam War had also resulted in significant implications for the burgeoning Korean economy, including the influx of foreign currency through combat allowance. Korea's setup of the think tank, however, had a major long-term impact on the country. At any rate, this was yet another instance of the Cold War influencing the evolution of NIS in Korea.

When American advisers arrived in Korea, they made it clear that the new institute should be an applied think tank for industrial technologies, as opposed to being a basic research institute like those prevalent in the U.S. in the postwar years. The idea was that the new institute would make direct contributions to industrial development, which would then help Korea identify and nurture new industries.

As such, Korea Institute of Science and Technology (KIST), founded in 1966, was intended to be a state-funded industrial research laboratory to provide technical services to the burgeoning industrial sector. The modus operandi of KIST was to be “contract research” paid for by companies needing technical assistance in their product development and production processes. In the initial stages, however, Korean industry lacked the motivation to conduct R&D. So KIST researchers, the majority of whom had been recruited from American universities and companies, had to serve the dual role of educating domestic industry on the necessity of further investment in technical research and conducting R&D (Moon, 2010).

The establishment of KIST was perhaps one of the most significant events in Korean STI history. First, the institute played a critical role in reversing the brain drain, which was perceived as a huge problem at the time. A significant proportion of the top Korean students were going abroad (mostly to America) to pursue advanced education in science and engineering, and most were not coming back. This was understandable given the dire state of the Korean economy in the 1960s with little opportunities for highly trained personnel. KIST reversed this trend not only by offering higher salaries and benefits than other peer institutions, but by also appealing to the sense of patriotism and national duty of skilled expatriates, stressing that their services would make great contributions to Korea’s national development. Second, KIST stimulated national interest in the role science and technology could play. Its researchers were instrumental not only in educating those in industry but also raising the bar for those in academia. Domestic universities around this time were largely teaching oriented, with little capacity and support for research. The establishment of KIST provided much needed stimuli for scientists and engineers at universities, though the full-fledged transformation of the academic sector would take time.

The late 1960s was a transformative period for the Korean NIS. The creation of KIST and subsequent influx of talented researchers from abroad sparked further developments in setting up a framework for science and technology policy. Two events that followed were particularly important in this regard. First, the enactment of the Science and Technology Promotion Law in 1967 signaled the beginning of a systematic and coordinated STI policy in Korea. The law included the creation of committees for promotion of science and technology, human resource and R&D policies, importation of technology and a science and technology promotion fund. Enactment of the law did not necessarily ensure implementation of the tasks covered in its provisions, but did provide the legal framework to coordinate STI policy at the national level. Second, the Ministry of Science and Technology (MOST) was established around the same time as KIST to serve as the control tower for national STI policy. One of the first things MOST did was to develop the “Long-Term Plan for the Comprehensive Development of Science and Technology, 1967-1986.” More than

400 experts from the government, universities, industry and PRIs participated in the deliberation leading up to the plan. So by the end of the 1960s, the institutional and legal frameworks for further development of science and technology were put in place within the government bureaucracy (Hong, 2010).

5.1.2. From Light to Heavy & Chemical Industries

The next inflection point came in the early 1970s. The late 1960s and early 1970s were a tumultuous time for the Korean peninsula. North Korean military provocations were on the rise. On January 21, 1968, a group of North Korean commandos waged a surprise attack on the South Korean presidential mansion of Cheong Wa Dae (Blue House). Two days later on January 23, the North attacked and captured the U.S. Navy intelligence ship USS Pueblo off the east coast, claiming that the vessel had been spying. Despite direct attacks on U.S. servicemen, Washington's main concern was to bring the Vietnam War to a close. This led U.S. President Richard M. Nixon to announce the Nixon Doctrine in late 1969, which said the U.S. would not "undertake all the defense of the free nations of the world." The doctrine meant that South Korea, as well as other American allies, was responsible for its own security.

The shift in the regional geopolitical environment led President Park to place added emphasis on national security, now that U.S. involvement in the event of another conflict on the Korean peninsula could not be automatically assumed. In response, Seoul readjusted its strategy to place equal emphasis on economic development and national security. The history of science and technology is flush with examples in which national security concerns serve as drivers for scientific and technological innovation. International conflicts provided incentives for national governments to invest in the development of advanced weaponry, from chemical to atomic weapons. These wartime advances sometimes, but by no means always, trickled back to the civilian sector in times of peace. The case of South Korea in the early 1970s was certainly one in which national security fears led to increased investment in STI. In 1973, Park made an official announcement that the government would pursue development of the heavy and chemical industries (HCI), with the catchphrase "We work as we fight; we fight as we work." The targeted industries were iron and steel, shipbuilding, petrochemicals and automobiles, with many of these being critical strategic industries for supporting military capabilities (Kim, 2004).

Making a policy shift was easier said than done. Promoting these industries required a completely different set of skilled technical personnel, compared to what was required a decade earlier. If the light industries required low-level, low-cost labor, the new HCI demanded skilled technicians and engineers. To meet its new labor needs, the Korean government began to restructure the nation's education system.

In 1973, the Ministry of Science and Technology conducted a nationwide survey on the supply and demand of technical personnel. The results are listed in <Table 1-3>.

<Table 1-6> Supply & Demand Plan for Sci-tech Personnel

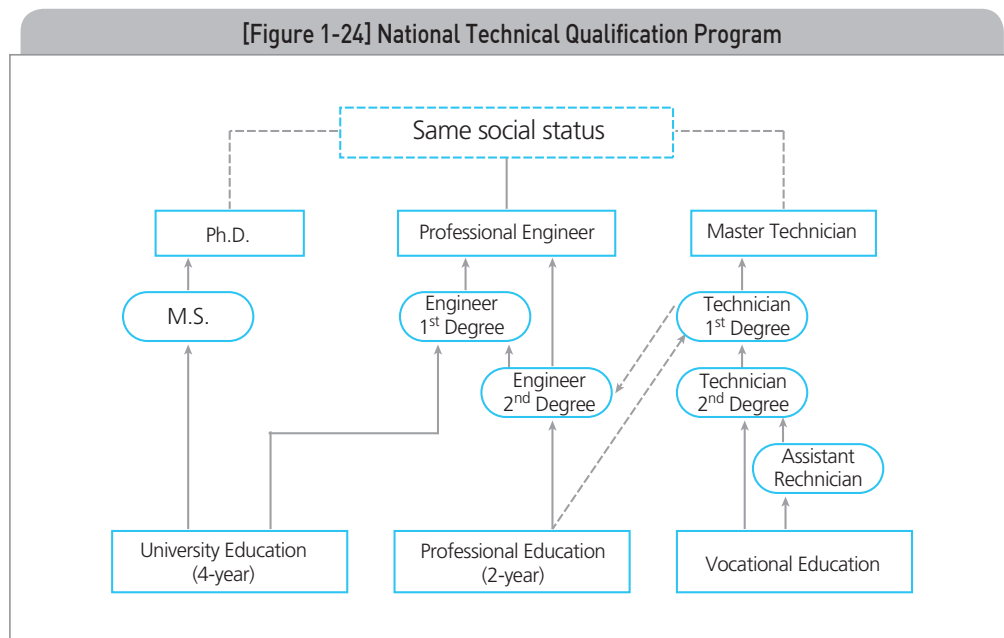
		1972	1981	1973-81
Scientists & Engineering Researchers	All Fields	27,400	126,700	+54,900 (surplus)
	HCI	10,300	66,300	-8,000 (shortage)
Engineers	All Fields	71,800	258,200	-160,300 (shortage)
	HCI	25,500	121,500	-89,700 (shortage)
Technicians	All Fields	451,400	1,991,800	-1,340,800 (shortage)
	HCI	177,000	906,000	-648,200 (shortage)
Total	All Fields	550,600	2,376,700	-1,446,200 (shortage)
	HCI	212,800	1,098,800	-745,900 (shortage)

If HCI policy was to be pursued for the remainder of the 1970s, the survey showed, serious shortages of technicians and engineers were predicted. The problem was less pronounced for researchers, with certain fields expected to see a glut of researchers. The numbers clearly showed that Korea's priorities lay in vocational technical education at the secondary level, which produced skilled technicians who would find employment in the emerging heavy and chemical industries. Most high school students in Korea at the time, however, were preparing for the college entrance exam, even at technical high schools. This was partly a legacy of the long cultural tradition of Confucianism in Korean society emphasizing scholarly learning as the central goal of education (Seth, 2002).

The Korean government attempted to shake that tendency through founding an elite technical high school lavishly supported by public funds. Kumoh Technical High School in Gumi, North Gyeongsang Province, was a shining example of the campaign. Opened in 1973 with partial funding from Japanese developmental aid, Kumoh was to be a benchmark for elite technical education in Korea. Students received full scholarships and monthly stipends, or incentives to talented students from relatively meager financial backgrounds to gain technical expertise. The state-of-the-art curriculum was designed by Japanese experts with considerable experience in technical education. Korean teachers received extensive training in Japan, and several Japanese teachers were dispatched to Gumi to make sure that their advice was properly implemented. Kumoh was also equipped with the most advanced facilities at the time so that students could gain hands-on experience on what they learned in the classroom. According to contemporary observers, the facilities at

Kumoh went far beyond what the best Korean universities at the time possessed, so university students and faculty frequently visited the high school to use equipment. In return for their privileged treatment, Kumoh graduates had to serve in the military for five years as “technical sergeants.” This service obligation reflected the underlying motivation behind the HCI policy goal of modernizing the military (Lim, 2015).

Kumoh was but one example among many technical high schools established and strengthened in Korea throughout the early 1970s. The goal, of course, was to meet the growing demand for skilled technicians in the pursuit of HCI policy. Growing numbers of technicians were expected to find employment in rapidly expanding steel mills, petrochemical plants and heavy industry (automobiles and shipbuilding) factories. To induce talented students toward these career trajectories, the Korean government enacted the National Technical Qualifications Act in 1973 to provide certification for qualified personnel. The purpose of the law was to ensure that technicians would receive “equal social recognition” as those taking the college education route. In other words, if a technician continued to accumulate experience in an area of expertise, he or she would receive the title of “master mechanic,” which was to equal to a Ph.D. in rank (See [Figure 1-24]) In retrospect, this project failed to achieve its original purpose but reflected the strong government intent to enhance the social status of skilled technicians in the early 1970s.



Vocational and technical education was Korea's top priority in the early 1970s, but empowering the university sector was also important to ensure the long-term growth of domestic technical capability. Eventually, the Korean government came around on this issue. In 1977, the Korea Science and Engineering Foundation (now the National Research Foundation of Korea) was established with the purpose of providing research grants and fellowships to faculty and graduate students. The measure reflected the need to integrate universities into the national program for economic development. Until the late 1970s, even the country's best universities were largely teaching oriented and lacked the capacity and facilities to conduct R&D. Revamping of the academic sector was a complex task that took a long time to accomplish. Simply pumping more funds into the sector would not solve the problem. The task would have to entail significant turnover in faculty as well as a conceptual change in a university's role in society. While the foundation's establishment represented the first step in this direction, more than a decade would pass before Korean universities would play significant roles within the emerging innovation system.

The NIS of Korea in the 1970s also saw the vast expansion of its PRI system. Partially to cope with renewed demand for technologies to support HCI policy, the Korean government established scores of specialized PRIs in this period.

〈Table 1-7〉 List of Sci-tech PRIs Established in Latter 1970s

Institution	Year
Korea Atomic Energy Research Institute	1959
Korea Standards Research Institute	1975
Korea Test Institute of Machinery & Metals	1976
Korea Research Institute of Chemical Technology	1976
Korea Research Institute of Geoscience & Mineral Resources	1976
Korea Institute of Electronics Technology	1976
Korea Electric Testing Institute	1978
Korea Telecommunications Research Institute	1977
Korea Electrotechnology Research Institute	1978
Korea Ocean Research and Development Institute	1978

Most of the new specialized PRIs founded in the latter 1970s were located in Daedok, a district about two hours south of Seoul designed to be a hub of science and technology. The purpose of the cluster strategy was to maximize the synergy effect among these institutions. The rapid—almost instantaneous—expansion of

the PRI system in 1975-78 was possible thanks to the success of KIST that began in the late 1960s. The think tank served as the breeding ground for the proliferation of specialized PRIs in the early 1970s. Many KIST researchers played instrumental roles in setting up these new institutes. And the KIST model was considered the template for designing the governance structure of the new PRIs. As with KIST, PRIs were closely connected to industrial activities in their respective fields and expected to support the technological needs of industry (Moon, 2010).

By the late 1970s, the basic components of the Korean NIS were in place. Industrial development, first in the light industries and later in the heavy and chemical industries, was guided by government policy and eventually led to the emergence of a strong private sector. Accompanying industrial development was the proliferation of PRIs to support the technological needs of specialized sectors. Academia was the last NIS component to receive interest. A few elite universities transformed their orientations from teaching to research at a modest rate, but only starting in the 1980s. All these sound like a linear success story in retrospect but establishing and strengthening the basic “triple helix” structure was not enough. The success of the Korean experience owed as much to designing the necessary legal and regulatory frameworks (i.e., the Foreign Capital Inducement Act of 1960 and the National Technical Qualification Act of 1973) and planning for the supply and demand of technical staff needed for the evolving stage of industrial development. As the critics of the NIS approach said, history and culture matter when applying the NIS framework to developing countries.

5.1.3. Imbuing the Scientific Spirit: The Saemaul Movement

For countries late to industrialization, focusing on science and technology institutions would certainly be critical to the development of national technical capability. Perhaps just as important was nurturing a scientific culture throughout society to inspire the innovative spirit to blossom. For this reason, this section will take a brief detour to discuss Korea’s “Saemaul Undong (New Community Movement)” of the early 1970s. On October 17, 1972, President Park said in a special announcement that he would treat the movement as the priority task to “cultivate a social spirit of eliminating irrationalities and absurdities” within Korean society. The goal of the movement was to transform Korean culture into one that embraces rational thinking and continuous improvement of its way of life. Seen from the NIS approach, the culture of a society comprises the broad environment within which the components are embedded. Thus, imbuing society with a culture that values science and technology is an important precondition for the long-term success of STI.

[Figure 1-25] New Community Movement



Source: JOONGANG PHOTO, <http://news.join.com/article/11183811#none>

The New Community Movement focused primarily on rural communities. It mobilized a large number of “village leaders” to educate farmers, who continued to use traditional farming methods and lifestyles. Partly based on the movement, the average annual income for farm households increased drastically from 255,800 KRW in 1970 to 1,531,300 KRW in 1979, a nearly six-fold jump. The improved standard of living was due to increased productivity in farming and enhancement of rural infrastructure such as paved roads, electrification and better housing. Based on the slogan of “diligence, self-help and collaboration,” the movement helped overcome the backwardness of Korean rural communities over this period.

For the purposes of this discussion, the important component of the New Community Movement was the National Scientization Campaign. The two goals of the campaign were the “scientization of everyday life” and “scientization of agriculture.” The idea was to imbue a scientific and rational way of thinking to agricultural practices and everyday life to raise productivity in rural areas. On the one hand, the campaign emphasized education for farmers, stressing the scientization of their lifestyles and cultivating respect for science and technology. For this purpose, the government used all available media to disseminate propaganda materials: newspapers, magazines, radio, TV, public talks, movies and exhibitions (Song, 2008).

The campaign also mobilized scientists and engineers to provide expert guidance to rural communities in a program dubbed “Saemaul Technology Guidance.” For this campaign, the Korean government mobilized a large number of scientists and engineers through the Korean Federation of Science and Technology Societies

(KOFST), an organization representing the scientific and engineering community in Korea. A combined 118 member societies of KOFST volunteered for the campaign. The Technology Volunteer Corps published booklets and produced radio programs on technology guidance, as well as directly visiting rural communities to interact with farmers and assess their technological needs. This was aligned with the government's intent to make a symbolic gesture to rural communities that intellectuals were taking the initiative in the modernization of agriculture. The slogan of "scientization of agriculture" was made visible in the field as large groups of professional scientists and engineers roamed the countryside (Lee, 2009).

The guidance provided by the Technology Volunteer Corps covered a broad spectrum, from lifestyle improvement (public baths, sanitary wells and more modern kitchens) to advice on agricultural technology (cultivation method, management of infectious diseases and prevention of parasites), in addition to ways to raise household income in rural communities. In short, the corps aimed to provide an encyclopedia for the improvement and modernization of rural communities. The professionals served as consultants for the education and welfare of individual villages by visiting the communities, as well as responding to their questions and requests. Once the initial stages of the Saemaul Technology Guidance were complete, the Technology Volunteer Corps morphed into a kind of agricultural extension service station based in provincial national universities, usually in the agricultural colleges. The dean of a college served as the director of the stations.

The New Community Movement and Technology Volunteer Corps were characteristic of 1970s Korea. They mobilized community leaders and professional scientists and engineers for the national campaign to modernize and "scientize" the broader society. Korean leaders considered backward and irrational thinking as a critical hindrance in the path toward continued economic development. First, the campaign created a new social environment placing high value on a rational and scientific way of life, and this further enhanced the social status of science and technology within Korean society. Without the widespread belief in the power of science and technology, all efforts to achieve national development through technological innovation would have experienced difficulty. Second, the campaign aimed to fully utilize the technological capabilities available within Korea and directed them to the most urgent problems that needed solving. Such problems were most acute in rural communities, and professional scientists and engineers were urged to mingle with farmers to arrive at possible solutions.

5.1.4. The 1980s and Beyond

By 1980, the Korean NIS had evolved into a substantial network of institutions, embedded in a bureaucratic and legal framework to foster STI for national

development. Also, the social and cultural environment had changed dramatically into one that valued science and technology. Based on this foundation, the economy made impressive strides in the 1980s and beyond. External conditions, however, were a mixed bag. The global economy was largely favorable, with oil prices, the value of the U.S. dollar and domestic interest rates all hitting lows. Economic indicators steadily improved throughout the decade. The problem was the worldwide trend toward protecting advanced technologies. With a more globalized economy, advanced industrial countries were incentivized to hold on more tightly to their technological knowledge that enabled innovation. A prime example of this competitive behavior was the high-tech trade war between the U.S. and Japan, especially in semiconductors. The new regime of global technology trade posed significant challenges for Korea, still a developing nation heavily reliant on technology transfers from the two economic superpowers. Amid the disputes between the two tech giants, Korea found it increasingly difficult to procure advanced technologies from abroad. To cope with this problem, Seoul began to pursue an aggressive program of state-led technological development through the “technology drive policy.” This approach emphasized the importance of directed basic research, signaling an influx of meaningful financial support for research activities at leading universities (Choi, 2014).

Through the 1990s, the roles of each component of the Korean NIS began to take shape. The government continued to pursue state-led technological development programs for targeted technologies such as computers and semiconductors. PRIs coordinated these projects and bridged the gap between the academic and industrial sectors. Universities produced high-caliber scientists and engineers at the postgraduate level, and conducted long-term basic research. Meanwhile, the share of the industrial R&D budget ballooned to peak at more than 80 percent of national R&D spending. Thus, the division of labor within the NIS was more or less established in this period. Under this framework, the Korean economy excelled over the next two decades in manufacturing semiconductors, cars and mobile devices.

The established structure proved to be quite effective, but only for a while. The government played a critical role in coordinating STI activities across the industrial, academic and PRI sectors and directing them toward economic development. The success of the Korean economy over the last 20 years depended largely on the effective operation of the NIS. Depending on national needs (export-oriented, heavy and chemical industrialization, and technology drive policies), the government made plans for the appropriate division of labor among the triple helix. While state planning was not always optimal, the coordinating function the public sector played served as one of Korea’s competitive advantages in this period. In the 21st century, however, there emerged a new need for a creative, post-catch-up innovation system. With the rise of China, the competitive terrain for Korea’s leading edge sectors

had changed. It was no longer sufficient to follow the technological trajectories of advanced economies. Korea now had to figure out its next move without guidance.

Korea's development experience can serve as a useful guide for late developing and newly industrializing countries. The country's experience between 1960 and 2000 illustrate the importance of government policy in fostering the STI system, through creation of technology demand in burgeoning industry, strengthening the education system and establishing new institutions. Will this experience, during which Korea established a successful NIS, be an advantage or disadvantage for the nation in the new world? The results are not yet in but the cards look stacked against Korea. Seoul is desperately groping in the dark for a renewed NIS. The process of searching for a new structure itself should also provide important lessons to benchmark for countries late to industrialization.

5.1.5. The Korean Development Model

Based on the previous discussions of the Korean development experience, this is a sketch of the basic outlines of the Korean model of STI-based development strategy.

〈Table 1-8〉 Policy Cases for Fostering Korean NIS

		Roles & Functions	Policy Cases
1	Vision	• Provide overall direction	• Export-oriented economic development • STI as major driver
2	Methodology	• Framework for prioritization	• "Steeple building" strategy in stage • Light industries (1960s), Heavy and chemical industries (1970s), ICT industries (1980s-)
3	Governance and legal framework	• Framework for policy coordination	• Science and Technology Promotion Law (1967) • Ministry of Science and Technology (1967)
4	Investment	• Financial stability for STI activities	• Foreign Capital Inducement Act (1960) • Foreign aid (United State, Japan) • Science and Technology Promotion Fund (1967)
5	Human Resources	• Attracting and educating STI manpower	• Attract foreign talent • Strengthen vocational technical education • Natrure university sector as component of NIS
6	Technology	• Meet private sector needs in STI	• Korea Institute of Science and Technology (1966) • GRI System (1975~1978) • Technology Importation
7	Society	• Create mood that values STI	• New Community Movement • Saemaul Technology Guidance

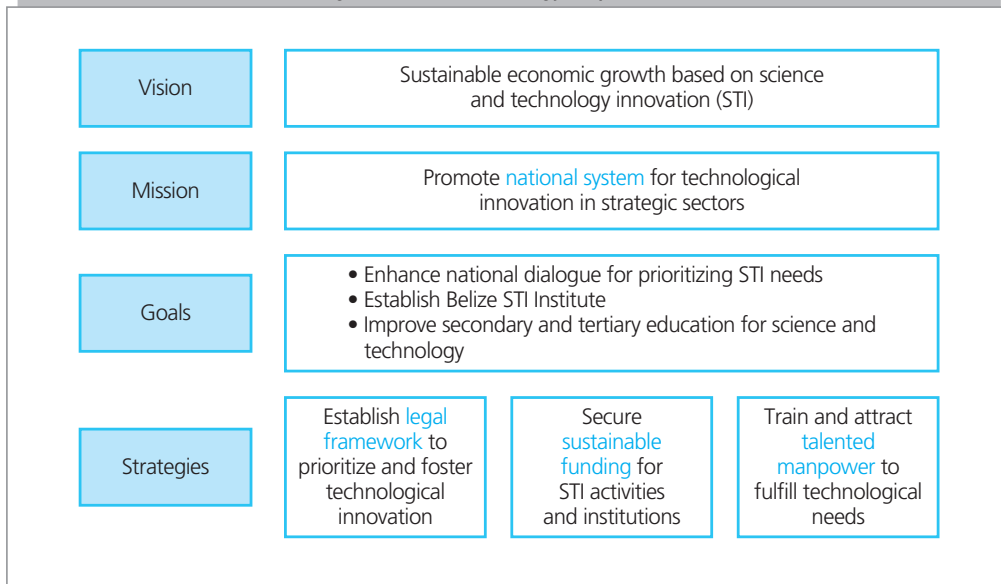
<Table 1-8> shows the various components of Korean STI policy connected to the policy cases discussed above. The overall direction and scope of the policy is guided by a broad vision promulgated by political leadership. In the Korea of the 1960s, that vision was export-led growth driven by science and technology. To achieve this vision, the government adopted a “steeple building” strategy of selecting key industrial sectors for focused development, beginning with the light industries and later moving onto HCI and ICT. To pursue this strategy, the governance and legal frameworks were put in place by the late 1960s. In 1967, the Ministry of Science and Technology was established and the Science and Technology Promotion Law enacted. These institutions provided the basic legal and bureaucratic frameworks to cope with emerging STI needs. In particular, the ministry served as a control tower of STI policy by coordinating policy options with other public agencies, universities and industries.

Implementing the policy vision required substantial resources in finance, human capital and technology. Korea secured the needed financial resources through a combination of foreign investment and aid and domestic tax income. Human resources were initially attracted from abroad, with adequate preparation to produce homegrown talent through both the secondary and university education systems. Technology needs were met initially through technology importation, primarily from the U.S. and Japan, but this was gradually replaced by domestic R&D at KIST, other GRIs, and later universities and corporate laboratories. Though the three resources discussed above served as the basic ingredients for fostering STI, the social component was perhaps equally important for Korean development. The New Community Movement allowed the country to foster a widespread social attitude of valuing science and technology as a driver for national development.

6. STI Strategy & Roadmap for Belize

This section will propose a basic vision for Belize’s STI strategy and outline a step-by-step action plan to realize it. Based on analysis of theoretical considerations, Belize’s economic status and existing plans for STI activities, and Korea’s development experience, the following strategy map is presented in [Figure 1-26].

[Figure 1-26] STI Strategy Map for Belize



6.1. Vision & Three Main Goals

The ultimate goal or vision of the STI strategic plan is to achieve *sustainable economic growth based on STI*. As seen above, Belize’s economy has significant unmet needs but can attain better performance when integrated with higher levels of STI activities. In particular, we strongly believe that the agricultural sector should be designated the initial “steeple” to be upgraded with knowledge and skills in biotechnology, food technology and agronomy. This, in turn, will allow Belize to gain a higher share of the export market. In pursuing this goal, enhanced productivity of agricultural products should not be the only aim but be complemented by targeted efforts to meet the needs of the international market. This will necessitate careful analysis of the regional and global market for agricultural products, to which a streamlined Belizean agricultural sector could cater.

To realize this vision, we suggest that Belize needs to *nurture the national system for technological innovation, with a focus on strategic sectors*. This strategy requires two distinct approaches. The first is a “systems approach” that considers the diverse sectors impinging on the nation’s technological capability, including the government, private sector and academia. To nurture the system, Belize should aim for simultaneous growth and active participation of all three sectors. The second is a “selection and prioritization approach” that brings into focus the most urgent areas to which STI activities should be targeted. Selection and prioritization require in-depth discussions among the numerous agencies within the central government, as

well as institutions in the education system and private industry. Hence, a national dialogue for prioritizing STI needs is required, which, in turn, needs assessment and reconfiguration of the legal and governance frameworks for STI.

Though we advocate a systems approach in the development of Belize's technological capability, focusing on the new Belizean STI institute, at least in the initial stages, will be prudent. The think tank is expected to serve as a catalyst for promoting STI activities focused on prioritized strategic sectors. Also, given the relative paucity of technical education in Belize, the institute should also play a role in training technical personnel in the initial stages of development. The process for setting up the institute will be covered later in the report.

Finally, for the medium- to long-term growth of STI in Belize, the national education system needs reconfiguration and reform. This streamlining should be tailored to the strategic prioritization made within the legal and governance frameworks for STI. Reform measures should include curricular reform, new units within universities and networking activities between academia and industry. Educational reform should not be limited to higher education, however; technical education within the vocational school system is also in urgent need of strengthening. The secondary level of technical education should go well beyond the scope of producing workers in the service sector, and must foster those who can find employment in agro-technology, a sector that will expand in scale and scope with the integration of STI.

The following sections will explain detailed strategic steps to realize the goals outlined above. The three strategic directions are: 1) establish legal and governance framework; 2) secure sustainable funding for STI; and 3) train and attract talented personnel.

6.2. Establish Legal and Governance Framework

Establishing appropriate legal and governance frameworks for planning and implementing STI policy is the most important step. Unlike Korea circa 1960, Belize has a ministry responsible for STI, namely MESTPU. The ministry thus should play a central role in coordinating the divergent needs arising from various sectors of the Belizean economy. The following action plans are proposed to achieve this strategic goal:

6.2.1. Form National STI Committee

Selection and prioritization are political acts that involve a number of stakeholders. The first thing Belize should do is to implement a framework for

national dialogue to allow discussion of prioritizing STI needs. Though we suggest agriculture as the initial strategic area for integrating STI activities, ratification of the choice at the national level is needed. To formalize the decision-making process, creating an institutional framework for prioritizing national STI needs is critical, perhaps in the form of a National STI Committee answerable to the Office of the Prime Minister. Ideally, the committee chairman should be a recognizable and authoritative member of the STI community, but in the meantime, a high-ranking government official can assume the post. The committee will serve as an umbrella organization comprising core actors of the Belizean NIS, namely staff from MESTPU, the ministries of economic development, agriculture, tourism and Trade, and other government offices related to STI. Representatives from the University of Belize and ITVET should also be present, as well as those from industry. The mission of the committee is to prioritize Belize's STI needs and determine the strategic sectors on which to focus over a timeframe of five, ten and 20 years. Also, a national body discussing STI issues will enhance Belizean science and technology and ensure the sustainable drive required for STI-led growth.

6.2.2. Enact Basic Law for Promotion of STI

The Basic Law for the Promotion of STI will serve as a de facto constitution for STI activities in Belize. The National STI Committee should offer a discussion framework for drafting the law, with input from ministries, universities and private industry. The government should also make clear that the law has priority over other existing statutes, so that its covenants can be implemented without hindrance.

The STI promotion law must also outline basic procedures for decision making and designate responsibilities in the implementation of STI policy. It should also make provisions for ensuring stable revenue for STI-related activities, including basic operation funds for the new STI institute. Finally, the law should support the formulation of comprehensive STI plans over the short, medium- and long- term.

6.3. Secure Sustainable Funding for STI

Once in place, the National STI Committee can help steer national discussion on STI investment. Belize currently spends about 0.06 percent of gross domestic expenditure on R&D (GERD). Though this seems higher than other countries in the region (i.e., 0.03 percent for El Salvador in 2011), the absolute size of the Belizean population needs consideration to get a realistic sense of what this means. Calculating from the 2013 GDP figure of 1.624 billion USD, the rate of 0.06 percent comes out to a meager 974,400 USD. The country's GERD needs to be raised to at least 0.5 percent (approximately a ten-fold increase) over the short term (2020) and eventually to 1 percent (20-fold increase) over the medium term (2025) if a sizable STI

institute is to be built. Both ODA funding and domestic funding sources should be sought to meet this goal.

6.3.1. Seek ODA Funding for STI Institute

The Belizean government, spearheaded by MESTPU, should actively look into applying for ODA opportunities for use as initial funding sources to kick-start Belize's STI activities. These efforts to secure initial funding will lay the foundation for the establishment of the new STI institute, which will require substantial sunk costs in infrastructure. Potential funding sources include the Economic Development Cooperation Fund (EDCF) of Korea, programs of the Japan International Cooperation Agency and other international aid and loan programs that support economic advancement in developing countries.

In Korea, KIST was established in part through funds made available by the U.S. Granted, the motivation for development assistance in the 1960s came largely from the Cold War, but in the 21st century, comparable opportunities for assistance are available.

6.3.2. Secure Domestic Funding Sources for STI Activities

Though KIST was partly funded by U.S. aid, its operating costs were covered by the Korean government. So even if ODA funding is available and forthcoming, Belize must secure domestic sources of funding to ensure the sustainability of STI activities. The National STI Committee should hold in-depth negotiations to secure a portion of tax revenue, such as higher tourist tax for those disembarking from cruise ships, for allocation for STI activities. Procedures for this kind of decision making should be outlined in the Basic Law for the Promotion of STI. Securing a sustainable funding source is critical for initiating STI in Belize, at least in the short term.

Over the medium term, a different approach is needed to ensure a steady flow of funds for STI. Once STI-led growth is in orbit, industrial sectors that benefit from STI integration should contribute a portion of their profits toward the development of Belizean STI activities. Such funds should be reinvested in the next set of strategic sectors selected by the National STI Committee.

Finally, STI activities should not rely solely on public funds, though this could be inevitable in the initial stages. Over time, however, private sector investment in R&D must be encouraged, either in the form of industrial R&D contracts with the STI institute or industrial enterprises choosing to set up their own R&D organizations to meet their technical needs. Either way, the government could consider providing tax breaks to induce the private sector to invest in STI activities.

6.4. Train & Attract Talented Workers

Education is the foundation of STI-led development. The Belizean academic sector, however, is underutilized, to say the least. Universities in Belize should cope with the national need to produce talented professionals in science and technology. To make that possible, the focus and curriculum of universities need significant reconfiguration and revision to reflect STI needs identified through selection and prioritization exercised at the national level.

6.4.1. Attract Talent from Overseas

At least in the initial stages, Belize should seek to attract technical talent from abroad while its education system is brought up to speed and acquires the capability to produce the necessary personnel. As in the case of KIST, attracting foreign-based talent requires not only proper financial compensation but also the participation of technical personnel in national development. This brings back to mind the broader vision of STI-led economic development, which needs to take concrete shape to be persuasive enough to attract foreign-based talent.

The pool of foreign-based talent could be identified in two areas. First, Belize should keep close track of its nationals who choose to pursue further education abroad — what they studied and where — especially for those in technical fields. This group of people has the potential to return home if offered the right mix of incentives and motivation. Even if they decline to return, they could be utilized as useful contacts to form global networks with domestic institutions. Second, Belize could attract foreigners on a short-term basis to work on specific projects, though not necessarily in the form of employment. For example, the new STI institute might engage GRIs in Korea to arrange for short-term visiting scientists or engineers. Crucial to making these arrangements is concerted efforts to fully absorb the knowledge of the visitors. Meiji Japan is a case in point. In the late 19th century, the Japanese government invited a number of foreign teachers (oyatoi, or hired hands) in science and engineering to train an initial batch of Japanese students. Within a few decades, Japan had replaced the foreign teachers with Japanese.

6.4.2. Strengthen Technical Education at both Secondary & Tertiary Levels

In the meantime, strengthening the domestic education system is absolutely critical for sustained STI development in Belize. The more immediate action should be taken for secondary-level vocational training in areas with the most urgent STI needs. At least over the medium term (10 years), more attention should go toward development of the university sector, which should serve as both a training ground

for a new generation of technical talent and a research institution producing technical knowledge. This also entails a revised curriculum to meet STI needs based on strategic selection and prioritization.

Over the longer term, however, Belize could consider establishing a separate university modeled after agriculture and mechanical arts (A&M) universities in the U.S. state university system, such as Texas A&M University. If the University of Belize remains the premier educational institution that produces the nation's political leaders, lawyers and businessmen, the new A&M university will aim to produce engineers and scientific farmers. The latter school could also serve the economy through extension services for agriculture and engineering.

The short-, medium- and long-term roadmap for the strategies and action plans outlined above are summarized in <Table 1-9>.

<Table 1-9> Brief Roadmap of STI Action Plan for Belize

	Short-Term (5 yrs)	Medium-Term (10 yrs)	Long-Term (20 yrs)
Goals	• 0.5% GERD/GDP • 25 R&D personnel	• 1% GERD/GDP • 50 R&D personnel	-
Strategic Industries	• Agriculture	• Biotechnology (tentative) • Biofuel (tentative)	Three additional strategic industries
STI Governance and Legal Framework	• National STI Committee	• ST Basic Law	
STI Investment	• Foreign assistance (ODA) • Tourist Tax	• Feedback from beneficiary sector • Tax break for industrial R&D contracts with SRI Institute	• STI Institute self-sufficient • R&D investment from private industry
STI Human Resources	• Attract foreign-based talent • STI Institute serve training function	• Revise university curriculum on ST and agriculture to meet STI needs	• A&M(Agriculture and Mechanical Art) university
STI System	• Establish STI Institute in a small scale	• Expand and diversify STI Institute	• International networks • Innovation clusters

6.5. Short-term Tasks

Between 2015 and 2020, Belize will establish the National STI Committee comprising major stakeholders as a framework for national dialogue for STI-related issues. The committee's role is to select and prioritize the country's STI needs. Assuming agriculture is selected as an initial strategic sector, the new STI institute will focus on agricultural science and technology. Initial investment in the institute will come from a mix of foreign aid and domestic sources. If the GERD/GDP can be raised to 0.5 percent, the annual funding flow can be assumed to be maintained at approximately 10 million USD. The money could pay for the institute's personnel and operating costs, as well as partial matching funds for infrastructure construction.

6.6. Medium-term Tasks

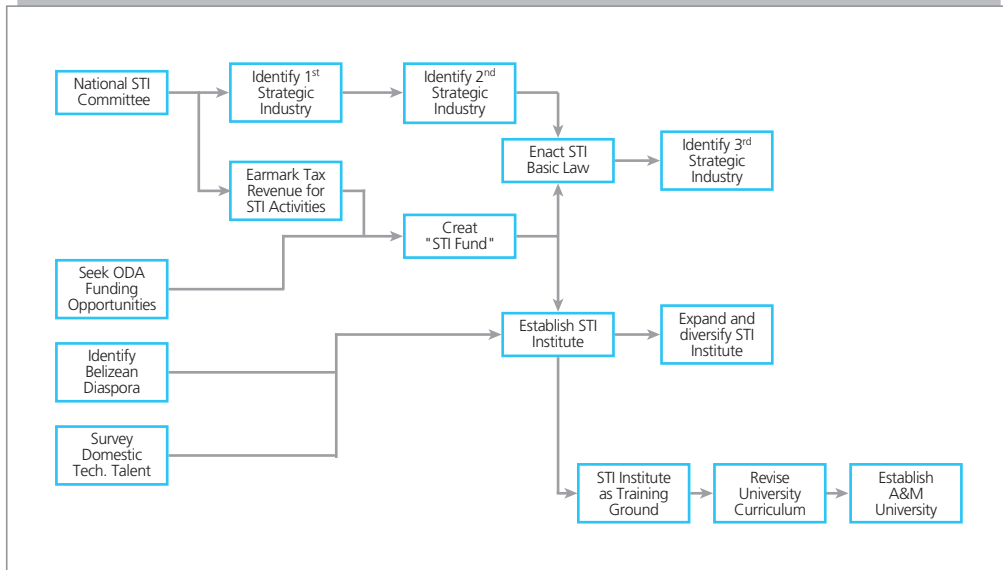
Between 2020 and 2025, the National STI Committee will take the initiative in enacting the Basic Law for the Promotion of STI, which will outline the basic procedures for decision making. Based on the law, the committee will also start devising a comprehensive STI plan, including a guarantee for stable funding of STI activities. Successful integration of STI in agricultural industries will yield an additional source of revenue, and this will raise the level of GERD/GDP to 1 percent, or more than 20 million USD per annum. Thanks to the funding level, the STI institute will gradually diversify its scope to the next set of strategic industries such as biotechnology and biofuel. This will require hiring more skilled personnel. By this time, efforts should begin to revise the university curriculum to meet new STI needs. The STI institute will make concerted efforts to form links with global peer institutions to create a network of professionals Belize can tap.

6.7. Long-term Tasks

By 2030, the STI institute will further diversify to comprise five to six divisions. It will closely interact with industry to perform R&D projects on a contractual basis. The academic sector has matured enough for agriculture and engineering to be taught separately at an A&M university, with the mission of training skilled personnel in strategic industries, providing extension services and performing basic R&D. The STI institute, A&M university and private industry will form clusters within which diverse players can collaborate.

The tasks listed above can be summarized by the roadmap below:

[Figure 1-27] Roadmap of Short, Medium and Long Term Tasks



7. Conclusion

The recommendations made in this report are based on the South Korean development experience between the 1960s and today. During this period, South Korea has successfully built up a cadre of institutions committed to perform research and development, revamped its technical education system, and forged close links among the government, university, and industrial sectors. The financial resources to support STI activities have been secured through a variety of means, including foreign aid, targeted taxation, and, in later years, the lottery fund. Since the 1980s, private industry has provided the lion's share of the national R&D expenditure, gradually releasing the public sector from solely bearing the burden.

A brief analysis of Belize's current status makes it abundantly clear that the country will not find a sufficient domestic market to ensure the benefits of economies of scale and scope. We believe, and most would agree, that the only feasible strategy for STI-based economic growth in Belize would be in identifying and capturing an export market in strategic areas of excellence. As in the South Korean experience, the initial investment for STI activities will have to come from the public sector, through foreign aid and targeted taxation. Utilizing the initial influx of secured funds, it will be critical to set up an adequate governance framework to prioritize the most urgent STI activity to jump-start the operation. The driving force of Belizean STI should be vested in the new STI Institute, which will be something

entirely new in the ecosystem of Belize's institutional environment. Based on the idea of global excellence in targeted areas, the Belize STI Institute will spearhead STI activities and align governmental goals with those of private industry.

Although we have noted at the outset that we will base our recommendations upon the South Korean development experience, one cannot overemphasize the fact that the Korean experience cannot be directly applied to the Belizean context. There are some major differences between the two countries that make it difficult to identify compatible patterns. First, one can immediately see the asymmetry in scale. The population of South Korea has always been in the order of tens of millions, compared to hundreds of thousands in Belize, a 1:100 difference. Therefore, Belize should not expect to build up the full spectrum of publicly-funded research institutes, as South Korea have done in the 1970s. Rather, Belize would benefit from focusing more intensively on those select few areas that are chosen through careful deliberation within the framework of Believe STI Committee.

Second, the geopolitical situation surrounding the two countries are disparate. As noted above, South Korea in the 1960s was at the frontier of the Cold War struggle between the United States (and partly Japan) and the Communist bloc. Admittedly, this had its merits and demerits, but from an economic standpoint it is difficult to deny that South Korea benefited greatly from economic and technical aid from its Cold War allies throughout the decades of rapid economic growth. Seen in this light, the geopolitical condition surrounding Belize is completely different. In the era of worldwide economic depression, it may not be realistic for Belize to expect the level of foreign assistance South Korea enjoyed. Therefore, it would be prudent for Belize to make keen efforts develop its long-term vision, so as to make the country an attractive candidate for official development aid (ODA). Why would any country provide assistance, economic or otherwise, to Belize? What could Belize, as a country armed with STI capabilities, contribute to the region and the world? While the answer to this question can only be produced through Belizean consensus, we see glimpses of the answer in Belize's emphasis in environment preservation. This will be a particularly important angle in the age of global climate change.

Taking these qualifications into account, we sincerely hope that Belize would come up with a truly actionable plan to transform its NIS into a higher level, based on the crude recommendations that we managed to deliver within the limited time allotted to us. It would make us happy if this report could spark further discussions within the Belizean society on the importance of STI.

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2014/15 Knowledge Sharing Program with Belize:
Formulation of a National STI Strategy and Action
Plan and Establishment of a Belize-Korea Science,
Technology and Innovation (STI) Institute

Chapter 2

Establishment of a Belize-Korea STI Institute

1. Introduction
2. Need for STI Institute in Belize
3. Analysis of Success of KIST, Korea's STI Institute
4. Policy Actions for Establishing STI Institute in Belize
5. Conclusion

Establishment of a Belize-Korea STI Institute

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Summary

This chapter's primary objective is to promote greater understanding of the Korean experience focusing on best practices in the development of STI to help Belize improve its agencies responsible for strategic planning and implementation of program activities within STI.

The Belizean STI institute will be a key player in the implementation of The STI Strategy and Action Plan for Belize. The plan identifies the institute as the principal actor in the implementation of actions in several of the think tank's strategic objectives. The institute aims to contribute to Belize's national and provincial development by serving as a strategic center for research knowledge, a hub for related activities, a center for STI talent and a site for building infrastructure for research and education.

This report proposes a feasible model for the institute. For this, we conducted a thorough assessment of the country's STI capability and related needs; evaluated the feasibility of establishing an STI institute in Belize; and examined the legal and institutional mechanisms for the institute. Next we proposed three approaches to provide a general framework for the institute. Our three models, which are tailored for our assessment of Belize's STI capability and needs, are based on a greater understanding of the Korean experience focused on best practices in the development of STI, especially those from KIST. Through these three models, we

propose key policy recommendations for the proper establishment and sustainability of the Belizean STI institute.

A necessity-based approach is crucial in planning the STI institute in Belize, and for that, we had three approaches (that are not exclusive to each other). First, we determined if Belize needs a strategic focus in developing an STI institute. Second, we wanted to study the feasibility of Belize having a training center that produces the talent needed for the country. Third, we thought ICT infrastructure would have a transformative impact on various sectors in Belize such as education, tourism and other private industries.

1. Introduction

The Belizean economy is small, open and based on commodities, as well as dominated by non-technology-intensive activities. The service industry is the biggest contributor to GDP and employment, but agriculture accounts for most of the exports. Output of agricultural commodities and tourism accounts for almost 60 percents of exported goods and services. Realizing its heavy reliance on outdated views of socioeconomic development and production and management systems, the Belizean government devised a strategy to provide a level playing field for all businesses and entrepreneurs, marked by the use of appropriate technology to increase productivity and competitiveness in an environmentally sustainable way.²⁷⁾

The Global Competitiveness Report (2011-12), or GCR, analyzed the competitiveness of Belize for the first time. The analysis considered basic requirements (institutions, infrastructure, macro-economic environment, health and primary education); efficiency enhancers (higher education, efficiency of the goods and labor markets, technological readiness and market size), and innovation and sophistication (business sophistication and innovation). [Figure 2-1] compares Belize with other efficiency-driven economies.

The country ranked 123rd out of 142 countries surveyed due to weak institutional bases (120th), especially in security (122nd), poor infrastructure and insufficient competition that hinders efficient allocation of resources in the goods market. Moreover, the country scored well in primary education enrolment with virtually universal schooling, but the rate of higher education attendance was low (105th), as was the quality of the country's educational system overall (126th). Participation in tertiary education remained low compared to the regional average, and despite significant investment in technical and vocational education, participation rates there

27) Government of Belize. Horizon 2030 Belize. Retrieved from http://www.nationalplanningcycles.org/sites/default/files/country_docs/Belize/ndp_belize.pdf

remained dismal. Females outnumbered males in the ratio of 62:38 at the tertiary level. Of 7,008 students at the tertiary level, 3,006 were enrolled at junior colleges in 2009/10, among which females numbered 1,702 and males 1,304. Policies to tackle these weaknesses will help boost the country's competitive edge (GCR, 2012).

[Figure 2-1] Belize's Technological Readiness and Innovation Scores/Rank

9th pillar: Technological readiness			12th pillar: Innovation		
9.01	Availability of latest technologies	4.6 92	12.01	Capacity for innovation	2.3 131
9.02	Firm-level technology absorption	3.9 129	12.02	Quality of scientific research institutions	2.4 128
9.03	FDI and technology transfer	3.4 136	12.03	Company spending on R&D	1.9 140
9.04	Internet users/100 pop.*	14.0 101	12.04	University-industry collaboration in R&D	2.5 132
9.05	Broadband Internet subscriptions/100 pop.*	2.9 84	12.05	Gov't procurement of advanced tech products	2.8 125
9.06	Internet bandwidth, kb/s/capita*	2.6 79	12.06	Availability of scientists and engineers	2.4 140
			12.07	Utility patents granted/million pop.*	0.0 90

Source: *Doing Business 2012: Doing Business in a More Transparent World – Belize Profile*; Co-publication of The World Bank and the International Finance Corp. (IFC), Washington, D.C.

Development of sci-tech initiatives applied to strategic economic areas can serve as the driver of a resilient Belizean economy. The Ministry of Energy, Science and Technology and Public Utilities' (MESTPU) Strategic Plan 2012-17 describe a "framework that supports the economic growth and competitiveness of Belize in the global market through the promotion of energy and food security, improved access to potable water, and utilization of science and technology to advance the wellness and livelihoods of the population to lead longer, wealthier and more productive lives."²⁸⁾ (MESTPU Strategic Plan 2012-17)

The report's primary objective is to promote greater understanding of the Korean experience focusing on best practices in the development of STI to help Belize improve its agencies responsible for strategic planning and implementation of program activities within STI areas. To accomplish this objective, the report first analyzed STI's status in Belize covering its capacities in research, education and innovation. This provided a background on which this report built the foundation for the Korean-Belizean STI institute. Second, greater understanding of the Korean experience focused on best practices in development of STI would help Belize improve its agencies responsible for strategic planning and implementation of program activities within STI areas. This report examines the success of KIST, Korea's first government-funded research institute (GRI) that led the country's STI-based economic development, and explores several factors crucial to KIST's development

28) Government of Belize. *Ministry of Energy, Science and Technology and Public Utilities Strategic Plan 2012-2017*. Retrieved from <http://estpu.gov.bz/images/media/mestpu%20strategic%20plan%20-%202012-2017.pdf>

as a prominent STI think tank in Korea. The conclusion will be used as a model for the proposed Belizean STI institute based on an analysis of Belize's STI capability and related needs.

The Belizean STI institute will play a key role in the implementation of The STI Strategy and Action Plan for Belize. The plan identifies the institute as the principal actor in the implementation of actions in several strategic objectives. The institute aims to contribute to the country's national and provincial development by serving as a strategic center of research knowledge, a hub for research activities, a center for STI talent and a site for building infrastructure for research and education.

The institute will serve: (1) as a policy and planning institute that runs the operations of MESTPU's STI Strategic Options for Implementation, as described in the ministry's Strategic Plan; (2) as an incubator for fostering partnerships between industry and academia to conduct joint research synchronized with national growth objectives²⁹⁾ (Mokhele, 2007).

The development of a national STI institute has positive implications for the development of a knowledge-based society and innovation-based economy. So a necessity-based approach in planning institute in Belize is crucial. We took three approaches (that are not exclusive to each other). First, we determined if Belize needs a strategic focus in developing an STI institute. Second, we assessed the feasibility of Belize having a training center to help the country get the talent needed. Third, we considered ICT infrastructure as having a transformative impact on various sectors in Belize such as education, tourism and other private industries.

2. Need for STI Institute in Belize

Belize has a small, open and commodity-based economy dominated by non-technology-intensive activities. The service industry is the biggest contributor to GDP and employment. Agriculture accounts for most exports, which include sugar and bananas to the European Union under the CARIFORUM-EU Economic Partnership and citrus to the U.S. under the Caribbean Basin Initiative. Additionally, agriculture serves as an input to Belize's agro-processing industries, which comprise the bulk of the country's manufacturing base (World Trade Organization, 2010).³⁰⁾ Agricultural commodities and tourism account for almost 60 percent of goods and service exports, indicating high dependence on a narrow range of sectors and products.

29) Mokhele, K. (2007) *Using Science, Technology and Innovation to Change the Fortunes of the Caribbean Region* ("The Mokhele Report"). UNESCO and the CARICOM Steering Committee on Science and Technology. UNESCO, Paris: www.unesco.org/science/psd/thm_innov/cariscience.shtml

30) World Trade Organization (2010). *Trade Policy Review*.

The economy's heavy reliance on preferential market treatment has also prevented adequate responses to market competition.

Recognizing its reliance on outdated views of socioeconomic development and production and management systems, Belize has established the following high-level goal to enhance its economic resilience:

- A. resilient economy with a level playing field for all businesses and entrepreneurs using appropriate technology to increase productivity and competitiveness in an environmentally sustainable way.³¹⁾ (Government of Belize)
- B. Priority economic goals for 2030 include higher agricultural output with strong added value through industrial/agro-processing development, well-performing micro, small and medium enterprises (MSME), and a strong workforce and corps of entrepreneurs fostered by an education system that nurtures innovation. In addition, the government will invest in key economic infrastructure.
- C. Development of sci-tech initiatives applied to strategic economic areas can serve as the vehicle for developing a more resilient Belizean economy. By generating employment and well-being through design and commercialization of new products and services, these initiatives are widely considered underlying factors to promote economic and industrial growth, wealth creation and higher quality of life for the people³²⁾ (Planning Institute of Jamaica, 2009).

The Ministry of Energy, Science and Technology and Public Utilities' (MESTPU) Strategic Plan 2012-17 aims to build "framework that supports the economic growth and competitiveness of Belize in the global market through the promotion of energy and food security, improved access to potable water and utilization of science and technology to advance the wellness and livelihood of the population to lead longer, wealthier and more productive lives."³³⁾ (MESTPU' Strategic Plan 2012-17)

A science, technology and innovation (STI) institute in Belize could serve the following dual purposes:

- A. Serve as a policy and planning think tank that puts into operation MESTPU's STI Strategic Options for Implementation, as described in the ministry's Strategic

31) Government of Belize. Horizon 2030 Belize. Retrieved from http://www.nationalplanningcycles.org/sites/default/files/country_docs/Belize/ndp_belize.pdf

32) Planning Institute of Jamaica (2009). Vision 2030 Jamaica: National Development Plan. Retrieved from <http://www.vision2030.gov.jm/National-Development-Plan>

33) Government of Belize. *Ministry of Energy, Science & Technology and Public Utilities' Strategic Plan 2012-17*. Retrieved from <http://estpu.gov.bz/images/media/mestpu%20strategic%20plan%20-%202012-2017.pdf>

Plan. This can mean development and coordination of a national action plan for the coordination of various government initiatives as conducted through various ministries, non-government organizational initiatives and private sector efforts in science and technology.

In particular, the institute could take on the important project of development, implementation and oversight of a national sci-tech education program, coupled with adequate ICT support in collaboration with the Ministry of Education and other relevant stakeholders to meet labor demands and industrial needs. Such a program can tackle the following issues as described in the data:

- 1) Development of technical and vocational skills at the post-secondary level of education, including entrepreneurial skills, linked to key industries and meet the needs of the private sector
 - 2) Higher scholastic performance, particularly in mathematics, on national and regional assessments
 - 3) Need for incentive programs for inclusion of a larger proportion of women in science and technology education
 - 4) Promotion of incentive programs for inclusion of a larger proportion of women to enter the workforce, especially in science and technology.
- B. Serves as an incubator to foster partnerships between industry and academia for research synchronized with national growth objectives³⁴⁾ (Mokhele, 2007). The institute can build links with other think tanks in the industrial/private sector, including CARDI, the Sugar Industry Research and Development Institute (SIRDI) and the Citrus Research and Education Institute (CREI). The STI institute can provide a discussion forum for policymakers and scientists on national and regional sci-tech objectives. In addition, it can serve as a center of research knowledge and a hub for research activities conducted by visiting scientists from the Belizean diaspora. Such activities will have positive effects on technology transfers and knowledge spillover³⁵⁾ (UNESCO, 2010).

The development of a national STI institute has positive implications for the development of a knowledge-based society and innovation-based economy.

What is crucial is a necessity-based approach in planning an STI institute in Belize, so we took three approaches (that are not exclusive to each other). First, we determined if Belize needs a strategic focus in developing an STI institute. Second,

34) Mokhele, K. (2007) *Using Science, Technology and Innovation to Change the Fortunes of the Caribbean Region*. ("The Mokhele Report"). UNESCO and the CARICOM Steering Committee on Science and Technology. UNESCO, Paris: www.unesco.org/science/psd/thm_innov/cariscience.shtml

35) UNESCO (2010). *UNESCO Science Report 2010: The Current Status of Science Around the World*. Retrieved from <http://www.uis.unesco.org/Library/Documents/UNNESCOSR10-eng.pdf>

we studied the feasibility of Belize having a training center for producing talent needed for the country. And third, we concluded that ICT infrastructure will have a transformative impact on various sectors in Belize such as education, tourism and other private industries.

3. Analysis of Success of KIST, Korea's STI Institute

3.1. Introduction

Korea Institute of Science and Technology (KIST) was founded in 1966 as the nation's first government-funded research institute (GRI). KIST has played a leading role in national development, and many other GRIs in Korea have been spun off from KIST or modeled after it. KIST has set the benchmark for a national think tank of science and technology. This chapter, through a historical examination of KIST, aims to identify five factors behind the institute's success: KIST's vision and political leadership for STI-led economic development; efforts toward sustainable funding; unique institutional and legal implementation; formation of a KIST-industry nexus; and policy toward sci-tech personnel.

Since its founding in 1966, KIST has led the country's STI-based economic development. Its focus on the application of science and technology to industrial innovation contributed to the growth of Korea's STI capability, and its formation of a KIST-industry nexus played a key role in sci-tech-based innovation. In the 1970s KIST led Korea's industrialization through inflow of technological knowhow and learning, and its major contribution was in heavy industry, which needed strong sci-tech capability. Korea is now a world leader in chemicals, steel and shipbuilding.

As Korea's first sci-tech research institute, KIST also contributed to the development of a national innovation system. GRIs were spun off from KIST through diversification, and KIST also contributed to the growth of high-quality sci-tech personnel by importing talented staff from overseas and via educational programs.

Greater understanding of the Korean experience, specifically best practices in STI development, will help Belize improve its agencies responsible for strategic planning and implementation of program activities within STI areas. This chapter aims to examine KIST's success and identify several factors crucial to KIST emerging as Korea's most prominent STI institute. The conclusion is to serve as a model for the proposed Belizean STI institute based on analysis of Belize's STI capability and needs.

3.2. KIST: A Brief History – Outline

A. Stage 1: Establishment (1967) ~ early 1970s

- 1) KIST's most important goal was to provide momentum for industry to recognize the necessity for and merits of R&D.
- 2) KIST's main tasks were technology training and understanding and improving imported technologies.

B. Stage 2: Mid-1970s

- 1) Increase in number of contracted studies from industrial corporations
- 2) Establishment of K-TAC for commercializing more studies conducted by KIST
- 3) Research institute incubator: Many branch institutes of KIST were separated and many veteran KIST researchers were appointed to head newly rising institutes
- 4) Prototype platform for policies and laws supporting think tanks

C. Stage 3: Late-1970s: National Lab Systems & Diversification

- 1) The government founded a number of strategic industrial think tanks whose structures and operating systems were modeled after those of KIST.
- 2) Granting a subsidy, gratuitous transfer of national property and autonomous operation were guaranteed under the Act for the Support of Specific Research Institutes.

3.3. KIST: Vision and Leadership

The establishment of KIST was a risky venture. Korea was a developing country whose major industry was agriculture, and thus demand for sci-tech capability was not high enough to justify the founding of a think tank dedicated to science and technology. Yet strong political leadership by then President Park Chung-hee, along with the opportunity from Korea's participation in the Vietnam War, made it possible to envision a think tank that would fundamentally transform Korea's potential for industrial development.

In 1965, the presidents of Korea and the U.S. (Park and Lyndon B. Johnson) signed a joint statement on "Establishment of a Korean Industrial and Applied Science R&D Institute," and KIST was subsequently founded the next year. Its primary aim was to support national industrial development within the framework of Rostow's stage theory of development, which postulates that economic growth occurs in five basic stages that ultimately lead to modernization. Accordingly, Korea needed to drive its industry to maturity, and KIST's role was to support this transition by fostering and diversifying the industrial base. In other words, its main focus was to support Korea's industrial development as opposed to studying basic science.

In 1966, KIST, under the leadership of first director Choi Hyung-sub, began operations as an autonomous think tank. In its first two years, it focused on the recruitment of sci-tech personnel in addition to planning and construction of the institute headquarters. KIST had its own governance structure, thus shielding it from government and academic intervention. Director Choi believed that KIST's independence was critical for its own survival as a high-quality sci-tech institute. He was also fortunately a man of political dexterity and had the full backing of then President Park.

Choi implemented a few organizational innovations to maintain KIST's autonomy as a research institute. First came the adoption of a self-supporting accounting system for each laboratory as a way to respect the autonomy of KIST scientists. To recruit and retain high-quality scientists and engineers from overseas, a flexible wage structure allowed salaries higher than those of Korean government and academic researchers. The relaxation of rigid administrative rules also enabled scientists and engineers to enjoy certain privileges such as exemption from military service.

The so-called package deal contract research was also a clever administrative measure that ensured KIST's autonomy. This was meant to provide government grants as support for KIST without state intervention. Every year, a package of research proposals from various areas was subject to government review. The sum was added to the government's annual budget and granted to KIST. Then the Council for Research distributed it to labs and the rest could be used to supplement operational expenses.

Foreign aid agents also played a great role in protecting KIST's autonomy. In the process of establishing and developing KIST, other key ministries failed to understand that KIST's autonomy was sorely needed. National leaders could also disturb KIST's well-knit plan for their own political ambitions. Foreign aid agents were always worried over government intervention, irrational pressure and misuse of investments. So the scientific workforce of KIST often depended on their authority.

Battelle Memorial Institute, the "sister institute" of KIST, offered to dispatch its well-trained personnel to help early KIST projects and conduct staff education for lead KIST researchers for more than six months at its headquarter labs. For an STI institute set up between Belize and KIST, KIST needs close contact with the institute and offer its expertise.

3.4. KIST: Institutional & Legal Implementation

3.4.1. KIST: Legal Framework

The Korean government needed to legislate new mandates to support science and technology in Korea. First, Seoul in 1967 ordered the setup of a ministry exclusively in charge of science and technology.

A. Launch of Ministry of Science and Technology (1967)

The result was the formal launch of the Ministry of Science and Technology (MOST) to consolidate sci-tech activities in other government agencies. The ministry's primary mandates were:

- 1) Devise collective and foundational policies for sci-tech promotion
- 2) Organize testing, investigation and research tasks
- 3) Supervise operations of facilities for testing and research
- 4) Promote international cooperation for sci-tech promotion
- 5) Formulate plans for researching and utilizing nuclear science

B. Support for Korea Institute of Science and Technology Act (1966)

In addition to creating a public agency dedicated to science and technology, the government also enacted a law to authorize the creation of and support for KIST as an autonomous think tank. In 1966, the Ministry of Science and Technology prepared a proposal for KIST's establishment.

The contents of the Support for KIST Act are summarized below:

- 1) The government can contribute to construction and operating expenses and operating funds of KIST.
- 2) The government can gratuitously rent or convey national property KIST needs.
- 3) Though KIST receives financial aid from the Korean and U.S governments, it is free from financial audit and requires no approval for its operating plans. KIST just has to report its annual plan to the proper Cabinet minister and submit a balance sheet audited by a certified and government-designated auditor.

When the Support for KIST Act came out in 1966, objectors, especially members of the Finance Ministry, claimed that KIST should not be free from financial audit and must get approval for its operating plans. But KIST head Choi Hyung-sub and leading members waged an aggressive movement against such opposition, and even mobilized the U.S. Embassy in Seoul and U.S. Operations Mission. Finally, KIST won and earned even greater autonomy.

The Korean government also enacted a series of laws on science, technology and innovation (STI), establishing a legal system for STI through the Professional Engineering Act, Science and Technology Promotion Act, Technology Development Promotion Act and Support for Specific Research Institutes Act.

C. Science and Technology Promotion Act (1967)

The Science and Technology Promotion Act aimed to 1) systematize the administration of human resource development, R&D and technology cooperation; 2) organize the Science and Technology Promotion Council; and 3) allocate more than 0.1 percent of the government budget to the Science and Technology Promotion Fund. The government expected that urgent needs for promoting science and technology could be fulfilled by enforcing the Act.

The contents of the Science and Technology Promotion Act are summarized below:

- 1) Give coherence to formerly dispersed and unsystematized policies
- 2) Define the details of comprehensive basic plans for promoting sci-tech and systematizing administration for making and executing plans.
- 3) Organize a Science and Technology Promotion Council to respond to the sci-tech minister on basic plans for promoting sci-tech related budget for promotion and other important matters.
- 4) Suggest a legal basis to establish policies and administrative systems for effective development of sci-tech human resources.

The Ministry of Science and Technology eventually formulated along-term comprehensive plan with a 20-year time frame based on a nationwide study of sci-tech research activities. The plan intended to:

- 1) Facilitate introduction of advanced technology / imitate and learn for fostering creativity through expanding independent R&D
- 2) Develop and maximize utilization of sci-tech personnel
- 3) Foster private R&D
- 4) Technological development in consideration of specialized and international division of labor (such as metal, mechanical and chemical engineering, electronic materials, ceramic engineering, breeding science for agriculture and fisheries, and oceanography).

3.4.2. KIST: Legal Framework

KIST laid out its basic governance principles to structure its status as an autonomous think tank dedicated to STI promotion. KIST's ideals were intended to sustain an ideal environment for creative research.

KIST's Three Ideals:

- 1) Autonomy of research: KIST should be free from administrative surveillance. The government blamed the inefficiency of former think tanks on rigid and unfit regulations of administrative bureaucracy.
- 2) Financial stability: For an innovative research structure, especially at the start and early stage, huge state investment is needed at any cost.
- 3) Rational and dynamic working atmosphere: Veteran researchers who founded KIST were generally from more advanced economies overseas. The first director of KIST, Choi Hyung-sub, was concerned whether they would not work at KIST because of the gap in atmosphere and culture. And this slogan was one way of persuading them.

Basic principles aimed to structure KIST's operational philosophy:

- 1) Develop and accumulate sci-tech knowhow for national development.
- 2) Balance development of applied science with basic research.
- 3) Form ties with industry for practical use of research.
- 4) Hold broad exchanges and technical cooperation with domestic and international think tanks.
- 5) Ensure adequate employment of researchers for a free and lively research climate.
- 6) Offer careful employment contracts on the basis of performance and suitability of a project.
- 7) Set responsibility accounting based on accurate cost estimation.

KIST Board of Directors:

The KIST Board of Directors consisted of 13 members: five ex officio directors, six directors appointed by industry and the sci-tech community, an auditor and secretary-general.

KIST's five ex officio director members are:

- 1) Vice minister of Economic Planning Board,
- 2) Vice minister of commerce and industry
- 3) Vice minister of science and technology
- 4) Ex officio member appointed by USAID
- 5) Ex officio member appointed by KIST's sister organization Battelle Memorial Institute

KIST's six directors are:

- 1) Three appointed by private industry
- 2) Three appointed by the sci-tech community

KIST Board of Directors shall vote on the following matters:

- 1) Annual plan and budget
- 2) Appointment of directors
- 3) Acquisition and Sale of KIST properties
- 4) Change in KIST's articles of association
- 5) Changes in KIST's administrative rules
- 6) Additional building of research facilities
- 7) Other matters related to KIST operations and policies

The KIST Board of Directors was primarily a deliberative body that voted on an agenda item. This was basically a board that set KIST's policy agenda, and was separated from the implementation of the institute's administrative functions.

KIST director:

The KIST director was a member of and appointed by the Board of Directors with a term of three years.

While the board appointed the director, he or she traditionally enjoyed absolute authority over the institute. The first director, Choi Hyung-sub, played a central role in turning KIST's vision into reality. After his retirement from KIST, he was appointed to head the Ministry of Science and Technology.

Council for Research and Administration:

KIST established a Council for Research to review the following upon request from its head.

- 1) Matters on research projects and contract research
- 2) Employment of researchers
- 3) Audit for research spending
- 4) Other administrative matters

3.4.3. KIST: Organizational Growth & Diversification

Due to lack of industrial need, KIST initially functioned as a think tank for government and industry. Its intent was to educate industry on the significance of sci-tech research in advancing competitiveness. KIST did indeed conduct types of technology transfers and thus introduced advanced foreign technologies to domestic industries through technical imports, guidance and education.

〈Table 2-1〉 Changes in Research Departments, Centers and Affiliated Institutes of KIST

	1972. 2	1974. 6	1976. 12	1980. 8
Main Depts.	Mechanics & metal Electric & electronic Food & feed Chemistry & chemical engineering Industrial economy	Mechanics Metal Material engineering Electric & electronic Food & bioengineering Chemistry & chemical engineering Organic chemistry Industrial economy Computing	Mechanics Metal Material engineering Electronic engineering Food & bioengineering Chemical engineering Resource engineering Organic chemistry Industrialization testing Industrial economy Computing organization	Mechanics Metal Material engineering Electronic engineering Food & Feed engineering Bioengineering Chemical engineering Polymer Industrialization Applied chemistry Environment system Industrial economy Development of technical kow-how Development of machinery
Centers		Center for technology of precision machine Center for technology of casting	Center for technology of precision machine Center for technology of casting Center for consultation about technology introduction Center for Semiconductor technology development	Center for technology of precision machine Center for technology of casting Center for consultation about technology introduction Center for information system development
Affiliated institutes		Research institute of ships Research institute of ocean development	Research institute of electronics and telecommunications	Research institute of ocean development Research institute of regional development
Total No.	5 depts.	9 depts. 2 centers 2 attached institute	11 depts. 4 centers 1 attached institute	14 depts. 4 centers 2 attached institute

Source: Changes in KIST Research Departments, Centers & Affiliated Institutes Moon, Man Yong. Formation of the Science and Technology System in Korea., Seon-In, 2010.

From the mid-1970s due to KIST's dedication, private industry began to appreciate the usefulness of sci-tech research. Certain private companies arranged contracted research with KIST. In addition, KIST tried to do developmental research and commercialization through ancillary organizations like K-TAC (Korean Technology Advancement Corp.). K-TAC's purpose was to commercialize more research conducted within KIST. Since KIST was a nonprofit corporation and tax exempt,

another ancillary organization like K-TAC was needed for business ventures. KIST's commercialization efforts pioneered the development of technology transfers in Korea.

The government founded a number of strategic industrial think tanks whose structures and operating systems were modeled after those of KIST. Granting a subsidy, gratuitous transfer of national property and autonomous operation were guaranteed under the Support for Specific Research Institutes Act.

Under the same law, Korea Research Institute of Ships, Korea Electronics Technology Institute, Korea Machinery and Metal Testing Research Institute, Korea Research Institute of Chemical Technology and Ocean Research and Development Research Institute were founded under the KIST umbrella.

More think tanks founded under the law were Korea Nuclear Fuel Corp., Resource Development Institute, Korea Electric Device Testing Institute, Korea Telecommunication Technology Research Institute, Energy Research Institute, Korean Ginseng Research Institute, Korea Tobacco Research Institute, Korea Atomic Energy Research Institute and Korea Research Institute of Standards.

Before the emergence of KIST, most think tanks were intended for experimental study of public needs and mired in red tape. KIST gave birth to many other institutes and was the prototype for the modern state-run think tank in the form of a nonprofit corporation.

3.5. Sustainable Funding Strategies

KIST heavily depended on foreign aid, especially from the U.S., in financing its establishment. The Korean government initially hoped that the U.S. would supply most of the funds necessary to set up KIST, including for facility construction and initial operations. From 1966 to 1970, however, Washington funded only about a third of the funds KIST sought. The U.S. gave 7.18 million USD for construction, laboratory instrumentation and staffing. About a quarter of the money went to contracted research by KIST. The Korean government, contrary to its initial expectation, had to foot the remaining two-thirds of KIST's construction and operating budget, amounting to 14.1 million USD from 1966 to 1970.

〈Table 2-2〉 Estimated and Actual Expenses for KIST

Expected expense suggested by Battelle (1966~1970)

(Unit: 10,000 USD)

Total	By source		By purpose				
	Korea Gov.	U.S. Gov.	Construction		Research facilities	Operating expenses	Advisory fees for Battelle
1,208	532*	676	Korea 275	U.S. 200	U.S. 300	Korea 256	U.S. 176

Actual expense for KIST (1966~1970)

(Unit: 10,000 USD)

Total	By source			By purpose							
	Korea Gov.	U.S. Gov.	ETC	Construction			Research facilities			Operating expenses	Advisory fees for Battelle
2,341	1,410**	718	AID*** 190 JAPAN *** 20	Korea 635	U.S. 256	AID 172	Korea 6	U.S. 145	AID 17 Japan 22	Korea 768***	U.S. 317

Note: * Not including land cost of 70

** Not including land cost of 140

*** Loan from AID

**** Fund of the claim to Japan

***** Including money for books

***** Including 633 from operational fund of KIST wholly consisted of counterpart fund from USA at that time.

Source: Kim, Keun Bae, in Korean History of Science Society 12.1, 1990: 44–69.

KIST had four major sources of operating funds in its initial stage.

1. Research contracts from corporations: revenue from industrial projects via R&D or technical support
2. Research contracts from government ministries: revenue from state projects via R&D or technical support.
3. Government subsidies for research via Package Deal Contract Research.
4. Utilization of other monetary aid: Similar fund in U.S. and the fund of the claim to Japan (provided by the 1965 Claims Agreement between Japan and Korea)

KIST's survival was heavily dependent on getting a sustainable operating budget, a daunting task. First, KIST was the outcome of an ambitious government plan, and as such, its scale and scope were big compared to the nation's economic and industrial needs. Its primary mission was national industrial development, but Korea lacked key industrial sectors that would lead to modernization in STI. So more often than not, KIST heavily relied on government research contracts. Substantial parts of its operating budget came from counterpart funds from U.S. and other foreign

sources as well. From 1966 to 1970, funds from the Korean government and other foreign sources, including counterpart funds, amounted to 6.33 million USD from its overall operating budget of 7.68 million USD.

〈Table 2-3〉 KIST's Technology Transfer Contracts 1969-96

	No. of contracts	Percentage	Total Income (Unit: million won)	Average income for each contract (Unit: million won)
Chemistry & Chem engineering	95	48.7	3980	41.9
Food & Bio	27	13.8	466	17.3
Electric & electronic	22	11.3	351	16.0
Metal	28	14.4	925	33.0
Mechanics	19	9.7	415	21.8
ETC	4	2.1	29	7.3
Total	195	100	6166	31.6

Source: Korea Institute of Science and Technology, Thirty-Year History of KIST, Korea Institute of Science and Technology.

KIST tried to get industrial contracts for research as a way to obtain sustainable funds. Its initial strategy was to foster a relationship with industry, focusing on meeting industrial demand for scientific and engineering needs. KIST's contracted research from private industry, however, remained a small fraction of its operating budget partially because of KIST's large institutional size. So more often than not, KIST relied on government contract funds. Without the ability to get sustainable funding, KIST ultimately became a government-owned think tank in the late 1970s, with most of its operating budget coming from public coffers. Meanwhile, as a result of its growth and diversification in the government-led rapid industrialization of the 1970s and beyond, KIST evolved into the backbone of the national research system with many specialized state-run think tanks under its wing.

3.6. KIST: Government-Industry Nexus

In the early stage of KIST, the major part of KIST research was in the form of research contracts from government ministries.

- A. Investigation and technology training
- B. Applied R&D
- C. Computerization of administrative works

The modernization of the government system conducted by the KIST Electrical and Computer Laboratory stands out in the institute's history.

The lab computerized the administrative operations of ministries and built data systems for them. It also played a leading role in introducing OCR (optical character reader) technology.

The lab was selected for many crucial government tasks because of its high competence and accumulated public trust.

The main difference between five-year R&D plans in 1969 and 1971 was the attitude toward technology transfer from overseas. This change was also largely related to KIST.

The importance of understanding and improving newly introduced technologies was recognized in KIST's early contracted works.

The five-year R&D plan set in 1969 was rather focused on "development of original technologies" befitting Korea's underdeveloped status at that time.

KIST also had huge influence over the conversion of the plan's content through its experience in technology training and the government audit of KIST on the real condition of introduced technologies.

3.6.1. Survey on Status of Industry

From November 1966 for eight months, 600 companies and related government agencies and educational institutions were visited by 80 researchers (57 from Korea, 23 from Battelle Research Institute of the U.S.).

Corporations were classified into 16 fields. The results were individually analyzed both on technological and economic bases by assessment teams with economists for each field.

3.6.2. General Technical Support Project

In the early stage of Korea's industrial development, SMEs needed technical support rather than R&D because they lacked researchers or technicians who could solve their technical problems. So KIST established the General Technical Support Center.

If SMEs paid a deposit of 100,000 won and registered at the KIST R&D office, the

latter connected each SME to the proper laboratory. The researchers and technicians in that lab studied the SME's problems and offered solutions using KIST facilities. The expenses were then deducted from the deposit.

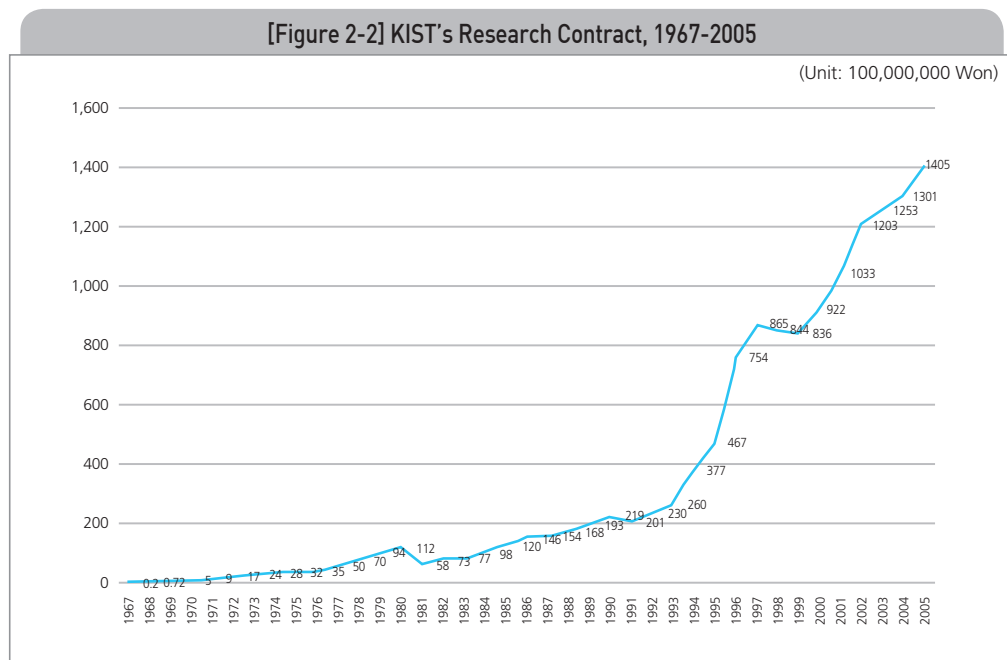
3.6.3. Hosting Social Gathering of Industrial Executives

Industrial executives and KIST researchers meet regularly and exchange opinions about technical development.

3.6.4. Foundation of K-TAC (Korean Technology Advancement Corp.)

K-TAC was established to commercialize more research conducted by KIST, a task that had been impossible because KIST was a nonprofit corporation and tax exempt.

Basically clients decided research subjects, namely the government and companies. But under Package Deal Contract Research, researchers could set their own topics with prospects of success for both the research itself and its commercialization. When researchers could not find a client for the conducted research, K-TAC organized a new corporate client, gathered funds and built factories for it; it later sold both the technology and facilities to investors.



Source: Korea Institute of Science and Technology, *Forty-Year History of KIST*, Korea Institute of Science and Technology, 2006.

K-TAC can be considered a bridge between KIST and industry and encouraged autonomous research not to be buried. It also contributed to KIST by supporting the institute's autonomous financial stability.

K-TAC commercialized many studies conducted by KIST and was Korea's first venture capital.

3.7. KIST: Staffing

In the late 1960s, sci-tech personnel needed to be recruited to work at KIST. Scientists and engineers in Korea in that era were severely lacking in number and quality. In addition, the nation's higher education system was primarily focused on education rather than research, making it difficult for the Korean government to train qualified scientists and engineers within a short time frame.

3.7.1. Korean Researchers from Overseas

In staffing KIST, the Korean government tried to attract ethnic Korean researchers from overseas. In addition to appealing to a sense of patriotism and national pride, KIST offered the following incentives for ethnic Korean researchers considering moving back to Korea:

A. Higher Rewards for Ethnic Korean Researchers from Overseas

- 1) Higher Salary: The monthly wage of leading researchers ranged from 60,000 to 90,000 won (a third or fourth of their previous wages in the U.S.). At that time, the average monthly wage of a full-time professor was 30,000 won and that of a senior researcher at KIST 45,000 won.
- 2) Affordable Housing
- 3) Sabbatical Year for Research

B. Research Autonomy

Each KIST laboratory, as a core functioning unit of KIST, was self-financing, that is, independently responsible for its own budget. Each principle researcher also had the full authority and responsibility for his or her laboratory. Thus each laboratory could maintain autonomy when pursuing its research agenda. The assurance of research autonomy in KIST's institutional design significantly improved the prospect of attracting ethnic Korean researchers from overseas.

C. Foreign Citizenship

KIST, unlike other state-run institutions in Korea at that time, allowed its researchers to retain foreign citizenship to attract ethnic Korean researchers from overseas, many of whom had a foreign passport. Many such researcher wanted to

retain their foreign citizenship given their anxiety over moving back to Korea or convenience when traveling abroad. KIST allowed this with no penalty.

- 1) Overseas recruitment conducted based on need of industry at that time
- 2) Considerable percentage of ethnic Korean researchers from abroad remained in Korea to contribute to the virtuous cycle of “reverse brain drain”

〈Table 2-4〉 No. of Ethnic Korean Scientists & Engineers from Overseas, 1966-80, by Research Field

	1966~1969	1970~1975	1976~1980.7	Total	Remaining (1980.7)
Chemistry & Chem engineering	6	15	17	37	23
Food & Feed & Bio	5	2	6	13	11
Mechanics	4	6	7	17	4
Metal & material	5	6	9	20	11
Electric & electronic	2	8	4	14	5
Industrial economics	2	3	2	7	5
Computing	0	1	4	5	4
ETC	1	0	4	5	5
Total	25	41	53	119	68

Source: Moon, Man Yong. Formation of the Science and Technology System in Korea., Seon-In, 2010.

3.7.2. Korean Researchers from Overseas

In addition to the design of an incentive system for attracting ethnic Korean researchers from overseas, KIST actively promoted its vision abroad by distributing promotional materials to about 500 foreign think tanks. Initially, about 800 ethnic Korean scientists and economists applied for research positions at KIST. The institute then screened them based on application documents, and about 150 were selected. By the second round of screening, 75 were selected according to the priorities of each field.

A recurring emphasis for KIST was the need to recruit researchers who could meet the needs of domestic industry, as opposed to offering cutting-edge or rather basic science. So KIST used a survey on the status of industry (November 1966~June 1967) in screening its candidates. Industrial challenges and needs drawn from the survey were shown to potential candidates, who then were requested to submit proposals

for what kind of research they would do and how they would run their own laboratory to meet these challenges and needs. Sixteen researchers were selected in the end after interviews to head each KIST laboratory.

3.7.3. Recruiting Domestic Research Staff at Early Stage of KIST

When KIST was established, the number of well-qualified sci-tech personnel at domestic universities and institutes was quite small. As such, a problem could have arisen had KIST tried to attract them in an ambitious way. So domestic recruitment was done in a silent and individual manner, not through a large-scale open campaign.

At the end of 1969, the number of faculty at the senior level or higher in Korea was 30 (seven lead researchers, three lead technicians, 20 senior researchers, three senior technicians). In 1968, KIST held its first large-scale open recruitment, and 50 of the 700 applicants were selected (three as senior researchers and 47 as researchers). Most of them were fresh graduates or had little experience after finishing university.

3.7.4. KIST's Contribution to Training Sci-tech Researchers

While KIST kept on recruiting veteran scientists abroad as lead researchers, it played an increasingly significant role in training sci-tech personnel. Careful selection of lead researchers and giving them full responsibility of their own laboratories helped to foster overall academic excellence. Those working and trained in this excellent research environment often wanted to pursue higher education. KIST thus provided an opportunity for its staff to attend graduate school.

KIST also provided the opportunity for post-doctoral training, mainly overseas. Its staff often underwent further technical and science training at foreign research institutions, and its faculty also enjoyed a generous sabbatical (every three years initially) so that they could catch up with advanced research trends.

KIST offered a graduate course on cooperation between academia and research institutes. Students attending research universities in Korea could take advanced lectures at the KIST academy and participate in key research projects performed at KIST. They could also get practical and hands-on training at KIST while working on their dissertations. Thus they could cultivate and acquire technical sophistication with applications to practical problems in an industry. This program greatly contributed to the training of sci-tech personnel who could be employed by domestic industry while meeting technical needs.

3.8. Implications: Assessment of KIST's Experience

[Figure 2-3] Implications

Vision & Leadership	Autonomous STI research institute
	Lab leaders: Self-supporting accounting system for each lab
Sustainable Funding	Initially foreign aids + government support
	Strive for industry funding + tech transfer operations
Legal and Institutional Structure	Legislation for STI's independent status & central planning for STI policy
	Governance structure for research autonomy, sustainable funding and rational research environment
KIST-Industry Nexus	Technology Transfer to Industry: foreign technology
	Efforts to survey and meet industry needs
	K-TAC: Tech Transfer and Commercializing Institutions
S&T Personnel	Reverse Brain Drain: recruit overseas S&T Personnel
	KIST: Evolved into a training institution for industry personnel needs for S&T
	Diversification: Backbone for Korea's STI system

4. Policy Actions for Establishing STI Institute in Belize

This chapter aims to propose a feasible model for the Belizean STI institute. For this, we conducted a thorough assessment of Belize's STI capability and needs, evaluated the feasibility of establishing such an institute in Belize, and examined legal and institutional mechanisms for it. Next, we proposed three approaches that could provide a general framework for the institute. Our three models are tailored for our assessment of Belize's STI capability and needs and based on a greater understanding of the Korean experience focused on best practices in STI development, especially the success of KIST. Through these three models, we proposed key policy recommendations for the establishment and sustainability of the Belizean STI institute.

4.1. Introduction

The Belizean STI institute will be a key player in the implementation of The STI Strategy and Action Plan for Belize. The plan identifies the institute as the principal actor in the implementation of actions in several of its strategic objectives. The aim is to contribute to national and provincial development by accomplishing the following objectives through research, education and service:

- A. Create a national think tank
- B. Build research and service partnerships with industry
- C. Establish STI talent hub and infrastructure
- D. Enhance STI capability through education

4.2. Feasibility of STI Institute in Belize: Educational and Economic Needs

The Belizean government is responsible for establishing and maintaining a functional sci-tech system. This is due to weak investment in R&D by the private sector given the disconnect between R&D efforts and industry needs. Few or no incentives exist for researchers to study the problems of industry. An integrated environment in which academia, especially personnel with the capacity to conduct R&D, the government and the private sector work together on sci-tech efforts will impact economic growth and job creation.

Despite the government's role as the primary supporter of sci-tech initiatives in Belize, only 0.06 percent of GDP was invested in the Ministry of Energy, Science and Technology, and Public Utilities, according to the Approved Estimates of Revenue and Expenditure for Fiscal 2013/2014 (Ministry of Finance, 2015).³⁶⁾ This funding level is inadequate for the establishment of a dynamic research culture, falling below the target of 1 percent of gross domestic expenditure on R&D many developing countries agree they should strive for³⁷⁾ (UNESCO, 2010).

Other needs are risk capital funding, an incentive system for R&D and the promotion of advanced training in innovation. These can be used to bolster working relations among the government, researchers, and the private sector for sci-tech development and application. In particular, other needs include risk and venture capital funds for technology-led projects and pilot plant facilities and higher assessment capabilities of banks to estimate risk in a knowledge economy.

36) Ministry of Finance (2015). Retrieved from <http://www.cdn.gov.bz/mof.gov.bz/images/stories/docs/APPROVED%20BUDGET%202013-2014.pdf>

37) UNESCO (2010). UNESCO Science Report 2010: The Current Status of Science around the World. Retrieved from <http://www.uis.unesco.org/Library/Documents/UNNECOSR10-eng.pdf>

4.3. Feasibility of STI Institute in Belize: Legal and Institutional Framework

4.3.1. Legal Framework

The Substantive Laws of Belize Revised Edition 2000 make provisions for the setup of statutory bodies in Belize. Existing statutory bodies under various governmental ministries include the following:

A. Ministry of Agriculture & Fisheries

- 1) The Belize Agricultural Health Authority (BAHA), established under the Laws of Belize Belize Agricultural Health Authority Act, Chapter 211, is designed to modernize agricultural health services.³⁸⁾
- 2) The Pesticides Control Board (PCB), established under the Pesticides Control Act, Chapter 181B of the Laws of Belize, is responsible for the implementation of the provisions of the aforementioned act.³⁹⁾
- 3) The Coastal Zone Management (CZM), established under the Coastal Zone Management Act, Chapter 329 of the Laws of Belize, is responsible for defining the policies, strategies and guidelines for management and conservation of Belize's coastal resources.⁴⁰⁾

B. Ministry of Tourism

The Belize Tourism Board (BTB), established under the Belize Tourism Board Act, Chapter 275 of the Laws of Belize, is responsible for the development of domestic tourism in the interest of national socioeconomic growth through prudent, transparent and effective governance.⁴¹⁾

C. Ministry of Trade and Investment

The purpose of the Belize Trade and Investment Development Service (BELTRAIDE), established under the Belize Trade and Investment Services Act, Chapter 282 of the Laws of Belize, is to “promote and facilitate local and foreign investment, and unfetter the export potential of local businesses to increase foreign exchange, employment, skills development or transfer technology and government revenue.”⁴²⁾

D. Ministry of Economic Development

The Belize Social Investment Fund (BSIF), established under the Belize Social

38) Retrieved from <http://baha.bz/about.html>

39) Retrieved from http://www.agriculture.gov.bz/Statutory_Bodies.html

40) Retrieved from http://www.coastalzonebelize.org/?page_id=38

41) Retrieved from <http://faolex.fao.org/docs/pdf/blz118275.pdf>

42) Retrieved from <http://belizeinvest.net/about/>

Investment Fund Act, Chapter 53 of the Laws of Belize, shall “subject to the availability of resources, approve projects and programs and provide, either wholly or partially, financial and technical assistance to community groups with development goals, and local government organizations, for the execution of such projects or programs that will serve to provide basic services to the most severely affected groups in the country.”⁴³⁾

E. Ministry of Education

The University of Belize (UB), established under the University of Belize Act, Chapter 37 of the Laws of Belize, is a public university designed to meet national needs, encourage sustainable development, increase self-sustainability and prepare Belize to meet global challenges.⁴⁴⁾

F. Ministry of Finance

- 1) The Statistical Institute of Belize (SIB), established under the Statistical Institute of Belize Act, 2006, of the Laws of Belize, is the autonomous national statistical agency of Belize.
- 2) The Central Bank of Belize, established under the Central Bank Act, Chapter 262 of the Laws of Belize, regulates Belize’s financial system, provides economic data and publications, offers services to the government and financial institutions, and issues Belize’s currency.⁴⁵⁾

The establishment of the proposed STI institute can be done in either of two forms:

- A. An autonomous statutory body per the Laws of Belize under the direction of the Ministry of Energy, Science & Technology, and Public Utilities (MESTPU). As the Belizean government is the primary promoter of sci-tech initiatives in the country, it plays a key role in formulating policy and facilitating related discussions among stakeholders from the public and private sectors.
- B. An autonomous institution under the University of Belize, with articles of association (see Annex A) and memorandum of association (Annex B) similar to those of the Environmental Research Institute (ERI).

4.3.2. Institutional Framework: Key Stakeholders and Their Roles

The following are potential key stakeholders in addition to the Ministry of Energy, Science & Technology and Public Utilities, and their roles in the establishment of the STI institute.

43) Retrieved from <http://www.sifbelize.org/>

44) Retrieved from http://www.ub.edu.bz/media/misc/UB_Act-Chapt37-RevisedEdition2000.pdf

45) Retrieved from http://www.incorporatebelize.com/downloads/Banks-And-Financial-Institutions/central_bank_of_belize_act_cap262.pdf

〈Table 2-5〉 Primary Stakeholders and Potential Roles

Primary Stakeholders	Potential Roles
Prime Minister's Council of Science Advisors (PM-CSA)	Formulation and implementation of an effective strategy for Belize's economic advancement through STI by employing practical and environmentally sustainable applications of science, technology, engineering, and mathematics (STEM)
Ministry of Education, Youth and Sports	Review of national mathematics and science curriculum at the primary, secondary and tertiary levels of education to align with national policy on science and education; Promotion of technical and vocational education for the development of a skilled workforce;
University of Belize Faculty of Science and Technology (FST) Department of Agriculture Environmental Research Institute (ERI)	Review of mathematics, IT and science academic programs to include practical field experience for graduates ready to enter the workforce; Promotion of science education programs at the primary and secondary levels of education, especially among women, to provide a base for student enrolment in STEM programs at the tertiary level; Provision of adequate funding in partnership with the Belizean government, University of Belize, and private sector and industry to encourage faculty research in their areas of expertise in collaboration with the private sector
Central Information Technology Office (CITO)	Creation of centers of excellence to facilitate R&D and innovation with emphasis on indigenous technology Proliferation and promotion of the delivery of e-government services Enhance connectivity among ICT infrastructure systems across the public sector Development of standards for the procurement of ICT hardware and software in the public sector
Ministry of Natural Resources and Agriculture BAHA	Identification of key agricultural areas for R&D (Sanitary and Phyto-sanitary (SPS) Certification)
Ministry of Tourism & Culture	Identification of key sectors of the tourism industry for R&D
Ministry of Trade, Investment Promotion, Private Sector Development & Consumer Protection Belize Trade and Investment Development Services (BELTRAIDE) Belize Bureau of Standards (BBS)	Promotion of micro, small, and medium enterprise (MSME) goods, services and processes; Implementation of standards in relation to goods, services and processes
R&D Institutes Caribbean Agricultural Research and Development Institute (CARDI) Sugar Industry Research & Development Institute (SIRDI) Citrus Research & Education Institute	In conjunction with the government and the University of Belize, development of research initiatives in agriculture that align with national policies for development and growth

〈Table 2-5〉 continued

Primary Stakeholders	Potential Roles
Caribbean Community Climate Change Centre (CCCCC)	Provision of policy advice and guidelines for effective management of climate change; Access to regional database on regional climate change data. The Caribbean Community Climate Change Centre tries to find effective solutions for the coordination of the region's response to climate change and global warming.
Belize Chamber of Commerce and Industry (BCCI)	Private sector input on policy toward research: agricultural, production, service and industrial sectors
Ministry of Finance Economic Development Belize Social Investment Fund (BSIF)	Attract ICT-related foreign direct investment Approval of budget
Financial institutions Domestic banks & credit unions Development Finance Corp. (DFC)	Provision of risk and venture capital funds for technology-led projects and pilot plant facilities

4.4. STI Institute: Three Models

A necessity-based approach is crucial for planning an STI institute in Belize. So we took three approaches (that are not exclusive to each other). First, we wanted to know if Belize needs a strategic focus in developing the institute. Second, we studied the feasibility of Belize having a training center that would help attract needed talent for the country. Third, our recommendation is to build ICT infrastructure given its transformative impact on various sectors in Belize such as education, tourism, and other private industries.

4.4.1. Steeples of Excellence Approach: Strategic Focus

It is crucial to design a STI Institute that would focus on a set of priorities the Belizean government, academia, and industry agree on. This is partly because Belize lacks sufficient academic, financial and institutional resources to build a full-fledged R&D institute. So the key is to focus on a few strategic fields the institute can contribute to in scientific research, technical assistance and personnel training.

A. Short term - You can't do everything well!

Identify strategic areas and build them into visible centers of national and international prominence

1) What are the potential “steeples” in Belize?

In our initial survey, potential steeples in Belize are identified in agriculture and tourism. In the Belize STI Roadmap, the following key areas are discussed as the potential strategic objectives of the STI institute:

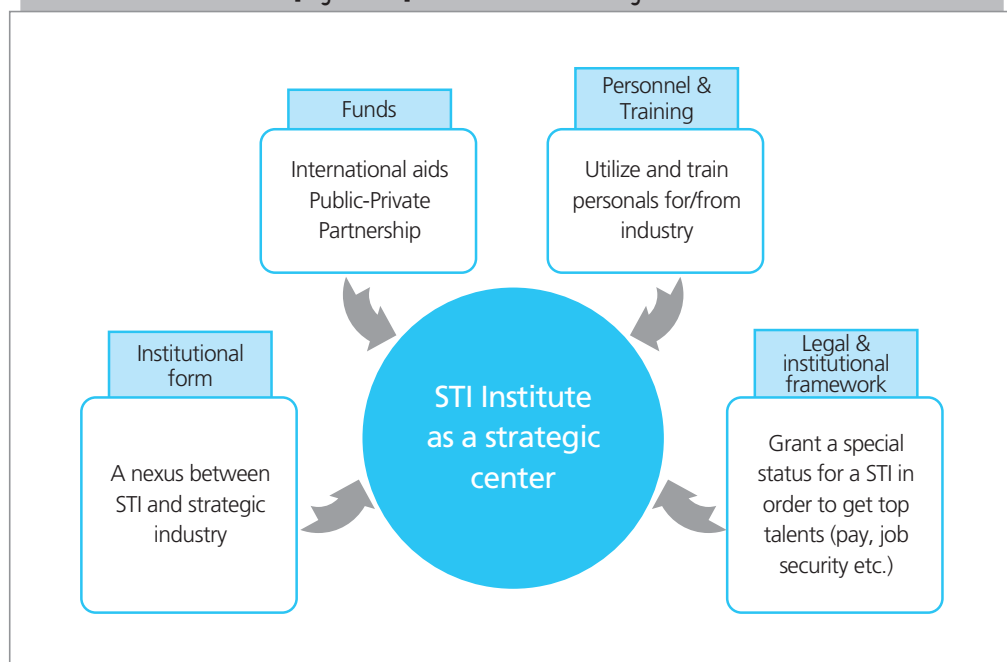
Food processing, ICT or technological resources for tourism

- Candidates include food technology (related to food safety), accreditation laboratory, processing technology (automation, electronics control) and soil science (including fertilizer research and irrigation).

2) Institutional form: STI institute

The institute should be located at the University of Belize but needs to attract sci-tech personnel from overseas. Ultimately the institute should build key steeples in the areas of its strategic focus(es), for which high-quality sci-tech research, development and innovation are carried out. By building a recipe for steeples in a few strategic areas, the institute can further diversify into other areas for development, evolving into a comprehensive STI institute that can contribute to the development of STI capability through research, education and innovation.

[Figure 2-4] STI Institute as Strategic Center



3) Strategies for Strategic Center

a. Design STI Institute to Focus on Set of Priorities

It is crucial to focus on a few strategic fields that the Belize STI institute can contribute to in scientific research, technical assistance and personnel training.

b. Building Recipe for Steeples for Few Strategic Areas

Ultimately the institute needs to build key steeples in the areas of its strategic focus(es) for which high-quality sci-tech R&D and innovation are carried out.

c. Diversification

Based on a successful recipe for steeples, the institute can further diversify into other areas for development, evolving into a comprehensive STI institute that can contribute to the development of STI capability through research, education and innovation.

4.4.2. Brain Circulation Approach: STI as Hub for Sci-tech Talent

In the age of globalization, a brain drain can be leveraged toward regional and national advantage if a channel through which STI talent can circulate is built. The Belizean STI institute should be positioned as a strategic site for “brain circulation.” Belize has a few advantages for this: English as a primary language, modest initial investment needed for building an STI institute and the country’s potential to become a regional sci-tech talent hub.

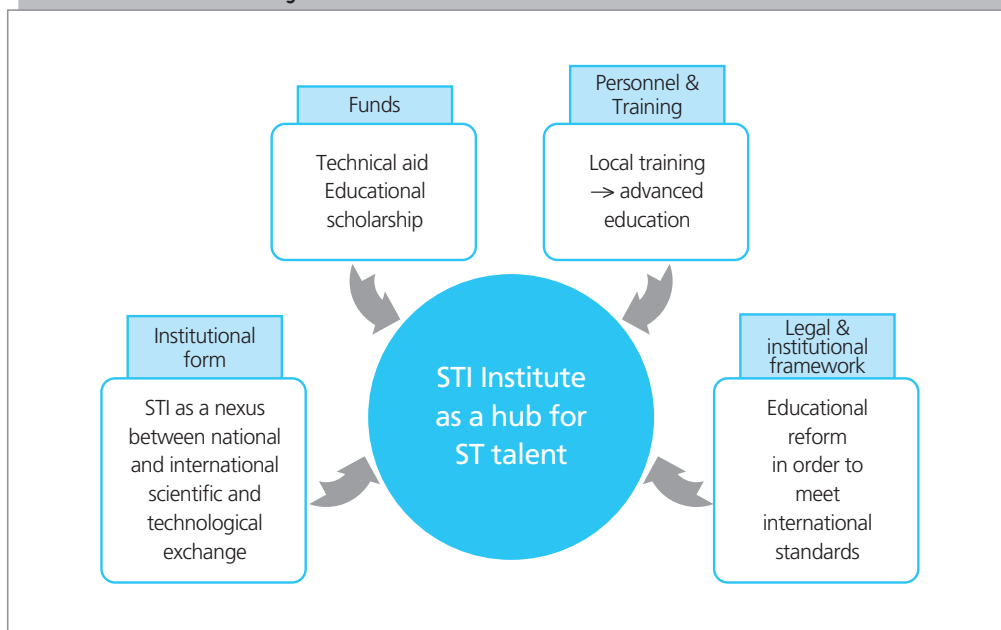
A. Short-term Stage

- 1) Recruit Sci-tech personnel from overseas
- 2) Begin foreign exchange/technical education

B. Mid-term Goal

- 1) Expand STI institute and empower university sector
- 2) Reform educational system toward technical and vocational training
- 3) STI as a sci-tech talent hub: The Belizean STI Institute can serve as a hub for the relevant technical community worldwide.

[Figure 2-5] STI Institute as Hub for Sci-tech Talent



C. Strategies: Toward Brain Circulation

1) Incentives

The Belizean government should offer incentives to alleviate the brain drain and attract to Belize those who are trained and have been working overseas, under a structure similar to KIST's incentive system.

2) Clear Sci-tech Personnel Policy

To maintain the high-quality of its sci-tech personnel, the STI institute needs a clear policy on the recruitment and management of research staff.

3) Training Next Generation of Sci-tech Personnel

- a. The STI institute will need to cultivate its function as a sci-tech talent hub by utilizing its own incentive system and other funding sources.
- b. The institute, through collaboration with the University of Belize, will also need to offer formal training and education programs (e.g., KIST's training program with academy and industry).

A critical problem for the Belizean STI institute is the lack of high-quality sci-tech personnel who can contribute to R&D activities in the areas of its strategic focuses. While the empowerment of sci-tech educational institutions would be the first step to train the personnel necessary for the STI institute, the status of Belizean R&D institutions (both academic and industrial alike) is rather unsuitable for rapid implementation of sci-tech training programs due to the institute's own lack of advanced personnel.

Recognition of the significance of human capital in Belize is the first crucial step toward the success of the STI institute. For this, the Belizean government is advised to implement incentives to alleviate the brain drain and attract to Belize those who are trained and have been working overseas. In the case of KIST, the Korean government decided to grant the think tank special legal status to allow an incentive system that differed from those of other government, educational and research institutions. This meant higher wages and more benefits such as housing, ample sabbatical leave and even exemption from military service. Since the Belizean STI institute's location could be at the University of Belize, the institute must have a clear policy on the recruitment and management of its own sci-tech personnel to maintain staff quality.

At another level, the STI institute should aim to educate Belize's next generation of scientists and engineers. In the initial stage, the institute would need to recruit personnel from overseas and then cultivate its ability to function as a sci-tech talent hub through its own incentive systems and other funding. The institute, through collaboration with the University of Belize, will also need to offer formal training and educational programs. In the case of KIST, only those who collaborated directly with private industry were eligible for such a program, making it easier for KIST to defray the educational expenses through cost sharing with industry for such education and training. So the recommendation is to initially limit the number of students participating in SIT training programs to those who collaborate with private industry on joint research projects with the STI institute.

4.4.3. Infrastructure Approach: Transformative Impact through ICT

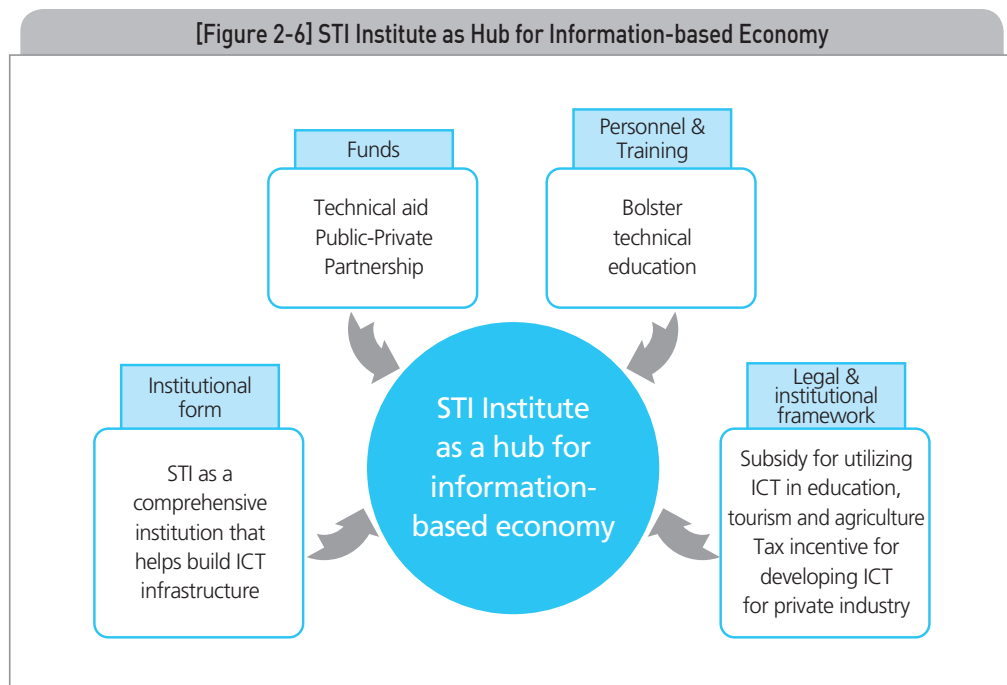
ICT (information communication technology) is an infrastructure with broad and transformative impact on education, R&D and innovation. As a latecomer in industrial development, Belize should heavily invest in ICT initially for use in education, government and a few strategic industries, such as agriculture and tourism.

A. Infrastructure: Building a National Innovation System

Forming a cluster: STI institute & University of Belize
→ NIS cluster

B. STI Institute as a Hub

Given the size, location and personnel of Belize, the institute should serve as the hub of national and regional information rather than attempt to do everything in-house.



C. Strong ICT Connectivity

ICT can have a transformative effect on the public and private sectors such as in education, tourism, trade and other private industries.

D. Strategies for ICT approach

1) ICT (Internet and communication technologies) as infrastructure

- ICT can have a broad and transformative impact on education, R&D and innovation.
- As a latecomer in industrial development, Belize should take advantage of ICT by investing heavily in ICT initially for use in education, government and a few strategic industries such as agriculture and tourism.

2) Funding: PPP (Public and Private Partnership)

- a. ICT is a public utility, so the government should take the initiative. The Belizean government can fund the construction and technical implementation of ICT infrastructure. It can also revise its tax policy and design investment incentives for private industry to participate in the building of ICT infrastructure.
- b. ICT also offers rapid and profitable business opportunities, so private industry should not hesitate to invest in ICT infrastructure in Belize.
- c. We recommend that the Belizean government form a PPP in its funding operations for successful building of ICT infrastructure.

3) Belize STI Institute: Center for ICT Infrastructure

- a. The Belizean STI institute could play a key role in building ICT infrastructure by recruiting and training sci-tech personnel crucial to implementing and spreading ICT infrastructure both in hardware and software.
- b. Government Modernization: The benefits of building ICT infrastructure are broad in scope. KIST, for example, computerized Korea's government administrative system to make such service and operations highly efficient.
- c. Transforming Education: The Belizean government could utilize ICT for making accessible online high-quality educational materials to more people.
- d. Advantage of Information-based Economy: Affordable ICT would attract many tourists by making it easier to navigate key tourist areas. The Belizean STI institute can ultimately play a central role in making the country a center for information economy.

Funding for building a broad ICT infrastructure needs to come from somewhere. Theoretically, ICT is a public utility, so the government should take the initiative. The government can fund construction and technical implementation of ICT infrastructure, but also revise its tax policy and design an investment incentive for private industry to help build ICT infrastructure. At another level, ICT is a rapidly and profitable business, so private industry should jump at investing in ICT infrastructure in Belize. The Belizean government is urged to form a PPP (public and private partnership) in its funding for successful building of national ICT infrastructure.

The Belizean STI institute could play a key role in building ICT infrastructure by recruiting and training sci-tech personnel crucial to implementing and spreading ICT infrastructure, both in hardware and software. The benefits of building ICT infrastructure are broad in scope. KIST, for example, computerized Korea's government administrative system to make the provision of administrative services such as tax matters and government work like economic planning more efficient. The government could also utilize ICT for making accessible online high-quality

educational materials to more Belizeans. Affordable ICT would attract many tourists by making it easier to navigate key tourist areas. Ultimately the Belizean STI institute can play a central role in turning Belize into a center for information economy.

5. Conclusion

5.1. Follow-up Action Plan for Belizean STI Institute

5.1.1. Legal Establishment

The proposed science, technology and innovation (STI) institute should be established as an autonomous statutory body under the Laws of Belize and through the supervision of the Ministry of Energy, Science and Technology, and Public Utilities (MESTPU).

The government of Belize, in consultation with MESTPU, should enact a special law to give MESTPU the authority to oversee a national sci-tech policy. MESTPU, under the legal authority bestowed upon it, should then provide a legal framework for the proposed institute including laws related to funding and personnel recruitment.

5.1.2. Task Force

- A. A task force is crucial for setting the objectives and priorities of the proposed STI institute. MESTPU should be in primary charge of forming the task force in consultation with primary stakeholders listed in Table 3-1. The leader of this task force should be an eminent administrator or senior researcher who is respected and familiar with STI research and education.
- B. The task force will design an STI institute based on three models suggested in this KSP report. It should have a clear organizational structure, and the scale and scope of the institute will be decided by the task force.
- C. The task force will appoint the first STI board of directors in consultation with MESTPU and the University of Belize. The board should then decide on administrative and operational rules similar to those of the Environmental Research Institute (ERI).
- D. The task force should devise a five-year strategy based on the STI Strategy and Action Plan for Belize, and decide on a five-year implementation budget for the institute (The five-year implementation budget for ERI was BZ\$27 million,

of which the University of Belize contributed BZ\$11.2 million).

- E. The task force will calculate the proposed budget for the setup of the Belizean STI institute

5.1.3. Governance

- A. The STI institute will be governed by a board of directors:
- B. The board of directors shall have a chairman and deputy chairman appointed by MESTPU. Both positions require extensive administrative and research skills and experience, and appointees should be respected and familiar with STI research and education.
- C. Two science directors will be appointed in consultation with MESTPU, private industry and the University of Belize. They will oversee overall scientific R&D projects at the institute in consultation with the board of directors.
- D. The institute can appoint other officers and staff deemed necessary by the board of directors.

5.1.4. Science Directors

- A. Science directors will decide program areas for the STI institute in consultation with private industry and academia.
- B. Science directors should find ways to encourage private sector participation in the proposed STI institute as an active partner in research projects, an enthusiastic funder of the institute and a contributor to STI capability.

5.1.5. Revenue & Sustainability

The income and property of the STI institute, whenever derived, shall be applied solely toward the promotion of the institute's objectives.

5.2. Sustainability of STI Institute: Obstacles and Tasks

Belize's infrastructure in education, information and communication has the foundation on which to build its capacity to harness science and technology more effectively for economic development.

Obstacles that could negatively affect the sustainability of the STI institute include but are not limited to the following:

- A. High cost of public utilities that negatively affect foreign investment in the country. To resolve this issue, MESTPU, as outlined in its Strategic Plan 2002-17, has offered initiatives to improve energy efficiency and conservation and increased the number of energy carriers derived from renewable energy sources.⁴⁶⁾
- B. Belize ranks low in the region in the technological readiness of its economy to adopt to existing technologies to enhance industrial productivity. Recent initiatives to expand Internet access include the doubling of Web speed while maintaining the same cost before the increase and establishment of a domestic Internet exchange point to route online traffic at no additional cost to Internet service providers⁴⁷⁾.

Given the huge potential for transform the Belizean economy through sci-tech initiatives, greater funding is needed to foster R&D via public and private sector partnerships. One of the greatest challenges for Belize, a developing country, lies in the unintentional allocation of much needed funding to promote and develop sci-tech initiatives to other areas of pressing need.

46) Government of Belize. *Ministry of Energy, Science & Technology and Public Utilities Strategic Plan 2012 - 2017*. Retrieved from <http://estpu.gov.bz/images/media/mestpu%20strategic%20plan%20-%202012-2017.pdf>

47) Retrieved from <http://www.7newsbelize.com/sstroy.php?nid=32172>

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